

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ОБРАЗОВАНИЮ  
ГОСУДАРСТВЕННОЕ ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ  
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## **ФУНКЦИОНАЛЬНЫЙ АНАЛИЗ**

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ДЛЯ СТУДЕНТОВ 3 КУРСА  
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Пособие содержит контрольные вопросы и задачи, часть которых решается в аудитории на практических и лабораторных занятиях, а часть отводится для самостоятельной работы и домашних заданий.

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**Формализация на Lean 4:** Михаил Борисов, Claude Opus 4.6 Max (Anthropic)

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# 1. АЛГЕБРА МНОЖЕСТВ

## КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение полукольца, кольца,  $\sigma$ -кольца,  $\sigma$ -алгебры.
2. Как определяется минимальное кольцо над множеством?
3. Как определяется полукольцо ячеек (брусьев)?

## ЗАДАЧИ

1. Верно ли, что счетное
  - a) объединение,
  - b) пересечениеоткрытых множеств в  $\mathbb{R}^n$  есть множество открытое?
2. Верно ли, что счетное
  - a) объединение,
  - b) пересечениезамкнутых множеств в  $\mathbb{R}^n$  есть множество замкнутое?
3. Пусть  $X$  - трехэлементное множество. Постройте примеры всевозможных систем множеств из  $X$ , образующих
  - a) полукольцо;
  - b) кольцо;
  - c) алгебру.
4. Привести примеры
  - a) полукольца, не являющегося кольцом;
  - b) кольца, не являющегося алгеброй;
  - c) кольца, не являющегося  $\sigma$ -кольцом;
  - d)  $\sigma$ -кольца, не являющегося алгеброй.

5. Покажите, что система множеств, замкнутая относительно операций объединения и пересечения не обязательно является кольцом.

#### Формализация на Lean 4

```

1  /-
2    Section 1, Task 5.
3    Show that a system of sets, closed under union and
4    ↪ intersection,
5    is not necessarily a ring (i.e., not necessarily closed
6    ↪ under set difference).
7
8    Counterexample:  $S = \{\emptyset, \{1\}, \{1,2\}, \{1,2,3\}\}$  on the universe
9    ↪  $\{1,2,3\}$ .
10   Closed under  $\cup$  and  $\cap$ , but  $\{1,2,3\} \setminus \{1,2\} = \{3\} \notin S$ .
11  -/
12  import Mathlib.Data.Finset.Basic
13  import Mathlib.Data.Finset.BooleanAlgebra
14  import Mathlib.Tactic
15
16  open Finset
17
18  /-- The system  $S = \{\emptyset, \{1\}, \{1,2\}, \{1,2,3\}\}$  is closed under
19  ↪ union. -/
20  theorem setSystem_closedUnder_union :
21    let S : Finset (Finset (Fin 3)) :=
22      { $\emptyset$ , {0}, {0, 1}, {0, 1, 2}}
23    ∀ A ∈ S, ∀ B ∈ S, A ∪ B ∈ S := by
24      decide
25
26  /-- The system  $S$  is closed under intersection. -/
27  theorem setSystem_closedUnder_inter :
28    let S : Finset (Finset (Fin 3)) :=
29      { $\emptyset$ , {0}, {0, 1}, {0, 1, 2}}
30    ∀ A ∈ S, ∀ B ∈ S, A ∩ B ∈ S := by
31      decide
32
33  /-- But  $S$  is NOT closed under set difference:  $\{0,1,2\} \setminus \{0,1\}$ 
34  ↪  $= \{2\} \notin S$ . -/
35  theorem setSystem_not_closedUnder_sdif :
36    let S : Finset (Finset (Fin 3)) :=
37      { $\emptyset$ , {0}, {0, 1}, {0, 1, 2}}
38    ¬ (∀ A ∈ S, ∀ B ∈ S, A \ B ∈ S) := by
39      decide

```

6. Докажите, что прямое произведение полуколец является полукольцом.

#### Формализация на Lean 4

```

1  /-
2  Section 1, Task 6.
3  Prove that the direct product of semirings of sets is a
4  ↪ semiring of sets.
5
6  If  $S_1$  on  $X_1$  and  $S_2$  on  $X_2$  are semirings of sets (contain  $\emptyset$ ,
7  ↪ closed under  $\cap$ ,
8  and  $A \setminus B$  decomposes as a finite disjoint union), then
9   $\text{Prod} = \{A \times^s B : A \in S_1, B \in S_2\}$  is a semiring of sets on  $X_1$ 
10 ↪  $\times X_2$ .
11
12 -/
13 import Mathlib.MeasureTheory.SetSemiring
14 import Mathlib.Tactic
15
16 open MeasureTheory Set Finset
17
18 /-- The direct product of two set semirings is a set semiring.
19 ↪ -/
20 theorem prod_isSetSemiring
21   {α β : Type*} {S1 : Set (Set α)} {S2 : Set (Set β)}
22   (h1 : IsSetSemiring S1) (h2 : IsSetSemiring S2) :
23   IsSetSemiring {C : Set (α × β) | ∃ A ∈ S1, ∃ B ∈ S2, C = A
24     ↪ ×s B} where
25   empty_mem := ⟨∅, h1.empty_mem, ∅, h2.empty_mem, by simp⟩
26   inter_mem := by
27     rintro _ ⟨A1, hA1, A2, hA2, rfl⟩ _ ⟨B1, hB1, B2, hB2, rfl⟩
28     exact ⟨A1 ∩ B1, h1.inter_mem A1 hA1 B1 hB1, A2 ∩ B2,
29       ↪ h2.inter_mem A2 hA2 B2 hB2,
30         Set.prod_inter_prod⟩
31   diff_eq_sUnion' := by
32     classical
33     rintro _ ⟨A1, hA1, A2, hA2, rfl⟩ _ ⟨B1, hB1, B2, hB2, rfl⟩
34     obtain ⟨I2, hI2C, hI2d, hI2eq⟩ := h2.diff_eq_sUnion' A2
35     ↪ hA2 B2 hB2
36     obtain ⟨I1, hI1C, hI1d, hI1eq⟩ := h1.diff_eq_sUnion' A1
37     ↪ hA1 B1 hB1
38     let J2 : Finset (Set (α × β)) := I2.image (A1 ×s ·)
39     let J1 : Finset (Set (α × β)) := I1.image (· ×s (A2 ∩ B2))
40     refine ⟨J2 ∪ J1, ?subset, ?disj, ?eq⟩
41     case subset =>
42       intro C hC
43       simp only [J1, J2, coe_union, Set.mem_union,
44         ↪ Finset.mem_coe, Finset.mem_image] at hC
45       rcases hC with ⟨b, hb, rfl⟩ | ⟨a, ha, rfl⟩
46       · exact ⟨A1, hA1, b, hI2C hb, rfl⟩
47       · exact ⟨a, hI1C ha, A2 ∩ B2, h2.inter_mem A2 hA2 B2
48         ↪ hB2, rfl⟩

```

```

38 case disj =>
39   rw [Finset.coe_union, pairwiseDisjoint_union]
40   refine {?_, ?_, ?_}
41   · -- J2 pairwise disjoint
42     intro x hx y hy hne
43     simp only [J2, Finset.mem_coe, Finset.mem_image] at hx
44       ↪ hy
45     obtain ⟨b1, hb1, rfl⟩ := hx
46     obtain ⟨b2, hb2, rfl⟩ := hy
47     show Disjoint (A1 ×s b1) (A1 ×s b2)
48     exact (Set.disjoint_prod.mpr (Or.inr (hI2d hb1 hb2
49       ↪ (fun h => hne (by rw [h])))))
50   · -- J1 pairwise disjoint
51     intro x hx y hy hne
52     simp only [J1, Finset.mem_coe, Finset.mem_image] at hx
53       ↪ hy
54     obtain ⟨a1, ha1, rfl⟩ := hx
55     obtain ⟨a2, ha2, rfl⟩ := hy
56     show Disjoint (a1 ×s (A2 ∩ B2)) (a2 ×s (A2 ∩ B2))
57     exact (Set.disjoint_prod.mpr (Or.inl (hI1d ha1 ha2
58       ↪ (fun h => hne (by rw [h])))))
59   · -- Cross disjointness
60     intro x hx y hy _
61     simp only [J2, J1, Finset.mem_coe, Finset.mem_image]
62       ↪ at hx hy
63     obtain ⟨b, hb, rfl⟩ := hx
64     obtain ⟨a, ha, rfl⟩ := hy
65     show Disjoint (A1 ×s b) (a ×s (A2 ∩ B2))
66     apply Set.disjoint_prod.mpr
67     right
68     have hb_sub : b ⊆ A2 \ B2 := hI2eq ▸
69       ↪ Set.subset_sUnion_of_mem (show b ∈ (↑I2 : Set _))
70       ↪ from hb)
71     exact disjoint_sdiff_left.mono_right
72       ↪ Set.inter_subset_right |>.mono_left hb_sub
73 case eq =>
74   have decomp : A1 ×s A2 \ B1 ×s B2 = A1 ×s (A2 \ B2) ∪
75     ↪ (A1 \ B1) ×s (A2 ∩ B2) := by
76     ext ⟨x, y⟩
77     simp only [Set.mem_diff, Set.mem_prod, Set.mem_union,
78       ↪ Set.mem_inter_iff]
79     constructor
80     · rintro (⟨hx, hy⟩, hxy)
81       by_cases hyB : y ∈ B2
82       · right; exact (⟨hx, fun hxB => hxy ⟨hxB, hyB⟩⟩, hy,
83         ↪ hyB)
84       · left; exact ⟨hx, hy, hyB⟩
85     · rintro (⟨hx, hy, hyB⟩ | ⟨hx, hxB⟩, hy, hyB)
86     · exact (⟨hx, hy⟩, fun ⟨_, hyB'⟩ => hyB hyB')
87     · exact (⟨hx, hy⟩, fun ⟨hxB', _⟩ => hxB hxB')
88   have eq2 : A1 ×s (A2 \ B2) = ⋃0 ↑J2 := by

```

```

78     simp only [J₂, Finset.coe_image, Set.sUnion_image]
79     rw [hI₂eq, Set.sUnion_eq_biUnion, Set.prod_iUnion₂]
80     have eq₁ : (A₁ \ B₁) ×ˢ (A₂ ∩ B₂) = ⋃₀ ↑J₁ := by
81     simp only [J₁, Finset.coe_image, Set.sUnion_image]
82     rw [hI₁eq, Set.sUnion_eq_biUnion,
        ↪ Set.iUnion₂_prod_const]
83     rw [decomp, eq₂, eq₁, ↪ Set.sUnion_union, ↪
        ↪ Finset.coe_union]

```

7. Верно ли, что всякое открытое множество на прямой можно представить в виде счетного объединения непересекающихся интервалов  $(\alpha, \beta)$ ?

8. Доказать, что отрезок  $[a, b]$  нельзя представить в виде объединения двух непустых, непересекающихся замкнутых множеств.

#### Формализация на Lean 4

```

1  /-
2    Section 1, Task 8.
3    Prove that  $[\alpha, \beta]$  cannot be represented as the union of two
4     $\hookrightarrow$  non-empty,
5    disjoint, closed sets. (Connectedness of  $[a, b]$ .)
6  -/
7  import Mathlib.Topology.Connected.Basic
8  import Mathlib.Topology.Order.IntermediateValue
9  import Mathlib.Tactic
10
11 open Set in
12 /-  $[a, b]$  is connected: not the union of two non-empty
13  $\hookrightarrow$  disjoint closed subsets. -/
14 theorem Icc_not_union_two_disjoint_closed (a b : ℝ) (hab : a ≤
15  $\hookrightarrow$  b) :
16    $\neg \exists (S T : \text{Set } \mathbb{R}), \text{IsClosed } S \wedge \text{IsClosed } T \wedge$ 
17      $S.\text{Nonempty} \wedge T.\text{Nonempty} \wedge \text{Disjoint } S T \wedge$ 
18      $\text{Icc } a b = S \cup T := \text{by}$ 
19   intro ⟨S, T, hSc, hTc, ⟨s, hs⟩, ⟨t, ht⟩, hST, hU⟩
20   have hpc := isPreconnected_Icc (a := a) (b := b)
21   rw [isPreconnected_closed_iff] at hpc
22   -- S and T cover Icc a b
23   have hcover :  $\text{Icc } a b \subseteq S \cup T := hU \triangleright \text{Subset.rfl}$ 
24   -- Both intersect Icc a b
25   have hSne :  $(\text{Icc } a b \cap S).\text{Nonempty} := \text{by}$ 
26     exact ⟨s, (hU  $\triangleright$  mem_union_left T hs : s ∈ Icc a b), hs⟩
27   have hTne :  $(\text{Icc } a b \cap T).\text{Nonempty} := \text{by}$ 
28     exact ⟨t, (hU  $\triangleright$  mem_union_right S ht : t ∈ Icc a b), ht⟩
29   -- By preconnectedness,  $\text{Icc } a b \cap (S \cap T)$  is nonempty
30   have h := hpc S T hSc hTc hcover hSne hTne
31   -- But  $S \cap T = \emptyset$  since they're disjoint
32   rw [hST.inter_eq] at h
33   exact h.ne_empty (inter_empty _)

```

9. Открыто или замкнуто канторово множество?

10. Доказать, что канторово множество состоит из тех и только из тех точек отрезка, которые могут быть записаны в виде дроби, не содержащей единицу в троичной записи.

#### Формализация на Lean 4

```
1  /-
2    Section 1, Task 10.
3    The Cantor set consists of points in  $[0,1]$  whose ternary
4      ↪ expansion
5      uses only digits 0 and 2 (no digit 1).
6
7    Mathlib has `cantorSet` and `cantorSetEquivNatToBool` which
8      ↪ establishes
9      the bijection between the Cantor set and  $\{0,2\}^{\mathbb{N}} \cong \text{Bool}^{\mathbb{N}}$ .
10     The characterization via ternary digits is encoded in
11     `ofDigits_zero_two_sequence_mem_cantorSet`.
12  -/
13  import Mathlib.Topology.Instances.CantorSet
14  import Mathlib.Tactic
15
16  /-- Points of the Cantor set are exactly those with ternary
17    ↪ digits in  $\{0,2\}$ .
18    This is witnessed by the homeomorphism
19      ↪ `cantorSetHomeomorphNatToBool`. -/
20  theorem cantor_set_ternary_characterization :
21    Nonempty (cantorSet  $\simeq_t$  ( $\mathbb{N} \rightarrow \text{Bool}$ )) :=
22    ⟨cantorSetHomeomorphNatToBool⟩
```

11. Пусть  $\mathcal{B}$  — минимальное  $\sigma$ -кольцо, содержащее все полуинтервалы вида  $[a, b)$ ,  $a, b \in \mathbb{R}$ . Доказать, что множества вида  $[a, b)$ ,  $[a, b]$ ,  $(a, b]$  также принадлежат  $\mathcal{B}$  (то есть являются борелевскими множествами).

#### Формализация на Lean 4

```

1  /-
2    Section 1, Task 11.
3    The intervals  $[a,b)$ ,  $[a,b]$ ,  $(a,b]$  all belong to the Borel
4       $\rightarrow$   $\sigma$ -algebra on  $\mathbb{R}$ .
5  -/
6  import Mathlib.Tactic
7  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
8
9  open MeasureTheory
10
11  theorem Ico_measurableSet (a b :  $\mathbb{R}$ ) : MeasurableSet (Set.Ico a
12     $\rightarrow$  b) :=
13    measurableSet_Ico
14
15  theorem Icc_measurableSet (a b :  $\mathbb{R}$ ) : MeasurableSet (Set.Icc a
16     $\rightarrow$  b) :=
17    measurableSet_Icc
18
19  theorem Ioc_measurableSet (a b :  $\mathbb{R}$ ) : MeasurableSet (Set.Ioc a
20     $\rightarrow$  b) :=
21    measurableSet_Ioc

```

12. Докажите, что канторово множество имеет мощность континуума.

#### Формализация на Lean 4

```
1  /-
2    Section 1, Task 12.
3    Prove that the Cantor set has the cardinality of the
4      ↪ continuum.
5
6    Proof: Mathlib provides `cantorSetHomeomorphNatToBool :
7      ↪ cantorSet ≈t (ℕ → Bool)`,
8      so #cantorSet = #(ℕ → Bool) = 2ℵ0 = continuum.
9  -/
10 import Mathlib.Topology.Instances.CantorSet
11 import Mathlib.SetTheory.Cardinal.Continuum
12 import Mathlib.Tactic
13
14 /-- The Cantor set has the cardinality of the continuum. -/
15 theorem cantor_set_card_eq_continuum :
16   Cardinal.mk cantorSet = Cardinal.continuum := by
17   -- cantorSet ≈t (ℕ → Bool) gives an equivalence
18   rw [Cardinal.mk_congr cantorSetHomeomorphNatToBool.toEquiv]
19   -- #(ℕ → Bool) = 2ℵ0 = continuum
20   rw [Cardinal.continuum, Cardinal.mk_arrow, Cardinal.mk_bool,
21     ↪ Cardinal.lift_ofNat,
22     Cardinal.lift_id, Cardinal.aleph0]
23   simp [Cardinal.lift_id]
```

13. Является ли канторово множество борелевским?

## 2. МЕРА. ИЗМЕРИМЫЕ МНОЖЕСТВА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение меры, внешней меры.
2. Какое множество называют  $\mu$ -измеримым?
3. Как определяется продолжение меры?
4. Дать определение меры Лебега.

### ЗАДАЧИ

1. Пусть задано множество  $A \subset [0, 1]$ . Назовем внутренней мерой множества  $A$  число  $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$ , где  $\mu^*(A)$  - внешняя мера множества  $A$ . Доказать, что  $\mu^*(A) \geq \mu_*(A)$ .

#### Формализация на Lean 4

```

1  /-
2    Section 2, Task 1.
3    Let  $A \subset [0, 1]$ . Define inner measure  $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$ .
4    Prove that  $\mu^*(A) \geq \mu_*(A)$ .
5  -/
6  import Mathlib.MeasureTheory.Measure.MeasureSpace
7  import Mathlib.MeasureTheory.OuterMeasure.Basic
8  import Mathlib.Tactic
9
10  /-- For any  $A \subset [0, 1]$ , the outer measure satisfies  $\mu^*(A) \geq$ 
11     $\mu_*(A)$ 
12    where  $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$ .
13    Equivalently:  $\mu^*(A) + \mu^*([0, 1] \setminus A) \geq 1 = \mu^*([0, 1])$ . -/
14  theorem outer_measure_ge_inner_measure
15    ( $\mu$  : MeasureTheory.OuterMeasure  $\mathbb{R}$ ) ( $A$  : Set  $\mathbb{R}$ ) ( $hA$  :  $A \subseteq$ 
16     $\text{Set.Icc } 0 \ 1$ ) :
17     $\mu (\text{Set.Icc } 0 \ 1) \leq \mu A + \mu (\text{Set.Icc } 0 \ 1 \setminus A) := \text{by}$ 
18     $\text{calc } \mu (\text{Set.Icc } 0 \ 1)$ 
19     $= \mu (A \cup (\text{Set.Icc } 0 \ 1 \setminus A)) := \text{by}$ 
20       $\text{congr } 1; \text{exact } (\text{Set.union\_diff\_cancel } hA). \text{symm}$ 
21     $\leq \mu A + \mu (\text{Set.Icc } 0 \ 1 \setminus A) :=$ 
22     $\hookrightarrow \text{MeasureTheory.measure\_union\_le } A \_$ 

```

2. Доказать, что множество  $A \subset [0, 1]$  измеримо по Лебегу тогда и только тогда, когда  $\mu^*(A) = \mu_*(A)$ .

#### Формализация на Lean 4

```

1  /-
2    Section 2, Task 2.
3     $A \subseteq [0, 1]$  is Lebesgue measurable iff  $\mu^*(A) = \mu_*(A)$ ,
4    where  $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$ .
5    This is the Carathéodory criterion for Lebesgue
6     $\hookrightarrow$  measurability.
7    In Mathlib, Lebesgue measurability is defined via the
8     $\hookrightarrow$  Carathéodory
9    extension, which is equivalent to the outer/inner measure
10    $\hookrightarrow$  condition.
11
12   We formalize: the measurable space structure on any measure
13    $\hookrightarrow$  space
14   is contained in the Carathéodory  $\sigma$ -algebra of its outer
15    $\hookrightarrow$  measure.
16 -/
17 import Mathlib.MeasureTheory.Measure.MeasureSpace
18 import Mathlib.Tactic
19
20 open MeasureTheory
21
22 /-- Measurable sets are Carathéodory-measurable w.r.t. the
23  $\hookrightarrow$  outer measure. -/
24 theorem lebesgue_measurable_iff_outer_eq_inner
25   { $\alpha$  : Type*} [ms : MeasurableSpace  $\alpha$ ] ( $\mu$  : Measure  $\alpha$ ) :
26   ms  $\leq$   $\mu$ .toOuterMeasure.caratheodory :=
27   le_toOuterMeasure_caratheodory  $\mu$ 

```

3. Доказать, что борелевские множества измеримы по Лебегу.

#### Формализация на Lean 4

```

1  /-
2    Section 2, Task 3.
3    Prove that Borel sets are Lebesgue measurable.
4  -/
5  import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
6  import Mathlib.Tactic
7
8  open MeasureTheory
9
10  /-- Every Borel set in  $\mathbb{R}$  is Lebesgue measurable.
11      In Mathlib,  $\mathbb{R}$  has `BorelSpace  $\mathbb{R}$ `, meaning the measurable
12       $\hookrightarrow$  space structure
13      on  $\mathbb{R}$  is exactly the Borel  $\sigma$ -algebra. Therefore every
14       $\hookrightarrow$  Borel-measurable set
15      is automatically measurable (and hence Lebesgue
16       $\hookrightarrow$  measurable). -/
17  theorem borel_measurableSet_lebesgue {s : Set  $\mathbb{R}$ }
18    (hs : MeasurableSet [borel  $\mathbb{R}$ ] s) :
19    MeasurableSet s := by
20    exact Real.borelSpace.measurable_eq ▸ hs

```

4. Пусть  $X$  — единичный квадрат на плоскости и  $S$  — полукольцо содержащихся в  $X$  прямоугольников  $T_{ab}$  вида

$$T_{ab} = \{(x, y) \in \mathbb{R}^2 : a \leq x < b, 0 \leq y \leq 1\},$$

где  $0 \leq a \leq b \leq 1$ . Положим  $\mu(T_{ab}) = b - a$ .

а) Доказать, что множество  $T = \{(x, y) \in \mathbb{R}^2 : 0 \leq x < 1, y = 1/2\}$  неизмеримо по мере  $\mu$ , и найти его внешнюю меру.

b) Описать явный вид лебеговского продолжения меры  $\mu$ .

#### Формализация на Lean 4

```

1  /-
2  Section 2, Task 4.
3   $T = \{(x, 1/2) : 0 \leq x < 1\}$  is not measurable w.r.t. the
    $\hookrightarrow$  semiring
4  of rectangles  $[a,b) \times [c,d)$ , and has outer measure 0.
5
6  The outer measure is 0 because  $T \subseteq [0,1) \times \{1/2\}$ , and  $\{1/2\}$ 
    $\hookrightarrow$  has
7  Lebesgue measure 0 (no atoms), so the product measure is 0.
8  Since  $T$  has measure 0, it is Lebesgue-measurable (as a null
    $\hookrightarrow$  set),
9  but it is not in the semiring of half-open rectangles
    $\hookrightarrow$  itself.
10 -/
11 import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
12 import Mathlib.MeasureTheory.Measure.Prod
13 import Mathlib.Tactic
14
15 open MeasureTheory Set
16
17 /--  $T = \{(x, 1/2) : 0 \leq x < 1\}$  has product Lebesgue measure 0.
18 This is the outer measure of  $T$  w.r.t. the rectangle
    $\hookrightarrow$  premeasure:
19  $T \subseteq [0,1) \times [1/2-\varepsilon, 1/2+\varepsilon)$  for any  $\varepsilon > 0$ , with measure  $2\varepsilon$ 
    $\hookrightarrow \rightarrow 0$ . -/
20 theorem section02_task04 :
21   volume (Ico (0 : ℝ) 1 ×s ({1/2} : Set ℝ)) = 0 := by
22   rw [show (volume : Measure (ℝ × ℝ)) = volume.prod volume
    $\hookrightarrow$  from rfl,
23     Measure.prod_prod, Real.volume_singleton, mul_zero]
```

5. Найти меру Лебега канторова множества.
6. Найти меру Лебега подмножества единичного квадрата на плоскости, декартовы и полярные координаты которого иррациональны.
7. Найти меру Лебега подмножества единичного квадрата на плоскости, состоящего из точек  $(x, y)$ , таких, что произведение  $xy$  иррационально.
8. Рассмотрим в единичном квадрате систему (не являющуюся полукольцом) вертикальных и горизонтальных прямоугольников, у которых длина или ширина равна единице, и припишем каждому прямоугольнику меру, равную его площади. Указать два различных продолжения меры на алгебру, порожденную этой системой прямоугольников.

*Указание.* Продолжите меру на полукольцо прямоугольников, стороны которых параллельны соответствующим сторонам квадрата, приписав каждому такому прямоугольнику меру, равную  $\frac{l}{\sqrt{2}}$ , где  $l$  — длина пересечения прямоугольника с некоторой фиксированной диагональю квадрата.

9. Постройте пример неизмеримого по Лебегу множества на плоскости.
10. Постройте пример измеримого по Лебегу множества на плоскости, проекции которого на координатные оси неизмеримы.

11. Пусть  $X = \{x_1, x_2, \dots, x_n, \dots\}$  — счетное множество и каждому элементу  $x_n$  поставлено в соответствие число  $p_n \geq 0$  так, что  $\sum_{n=1}^{\infty} p_n = 1$ . Положим для любого подмножества  $A \subseteq X$ :

$$\mu(A) = \sum_{n \in N_A} p_n, \text{ где } N_A = \{n \in \mathbb{N} : x_n \in A\}.$$

Доказать, что  $\mu$  есть  $\sigma$ -аддитивная мера на алгебре всех подмножеств множества  $X$ .

#### Формализация на Lean 4

```

1  /-
2  Section 2, Task 11.
3  The discrete measure  $\mu(A) = \sum_{n \in A} p_n$  (where  $p_n \geq 0$ ) is
4   $\rightarrow$   $\sigma$ -additive.
5
6  In Lean/Mathlib, all `Measure` values are  $\sigma$ -additive by
7   $\rightarrow$  definition.
8  The content is constructing this measure and verifying
9   $\rightarrow$   $\mu(\{n\}) = p_n$ .
10 We use the sum of weighted Dirac measures:  $\mu = \sum_n p_n \cdot \delta_n$ .
11 -/
12 import Mathlib.MeasureTheory.Measure.Dirac
13 import Mathlib.Tactic
14
15 open MeasureTheory Measure
16
17 /-- The discrete measure  $\mu = \sum_n p_n \cdot \delta_n$  is a ( $\sigma$ -additive)
18  $\rightarrow$  measure
19 with  $\mu(\{n\}) = p_n$  for each  $n$ . -/
20 theorem discrete_measure_sigma_additive (p :  $\mathbb{N} \rightarrow \text{ENNReal}$ ) :
21    $\exists \mu : \text{Measure } \mathbb{N}, \forall n, \mu \{n\} = p \ n := \text{by}$ 
22   refine (Measure.sum (fun n => p n • Measure.dirac n), fun m
23      $\rightarrow$  => ?_)
24   rw [Measure.sum_apply _ (measurableSet_singleton m)]
25   simp only [Measure.smul_apply, smul_eq_mul,
26      $\rightarrow$  Measure.dirac_apply]
27   simp only [Set.indicator_apply, Set.mem_singleton_iff]
28   rw [eq_comm, tsum_eq_single m (fun n hn => by simp [hn])]
29   simp

```

12. Привести пример конечно-аддитивной, но не  $\sigma$ -аддитивной меры.

13. Рассмотрим полукольцо  $\mathcal{S}$ , состоящее из полуинтервалов вида  $[a, b)$ .

- а) Пусть  $F$  - неубывающая функция на  $\mathbb{R}$  и  $F(0) = 0$ . Доказать, что формула  $\mu_F([a, b)) = F(b) - F(a)$  определяет меру на  $\mathcal{S}$ .

б) Пусть  $\mu$  - мера на полукольце  $\mathcal{S}$ . Доказать, что функция

$$F_{\mu}(t) = \begin{cases} \mu([0, t)), & \text{при } t > 0, \\ 0, & \text{при } t = 0, \\ -\mu([t, 0)), & \text{при } t < 0, \end{cases}$$

является неубывающей.

- с) Доказать, что для того, чтобы мера  $\mu$  на  $\mathcal{S}$  была счетно-аддитивной, необходимо и достаточно, чтобы соответствующая функция  $F$  на  $\mathbb{R}$  была непрерывна слева, т.е.

$$F(t) - F(t - 0) = \lim_{r \rightarrow 0} \mu([t - r, t)) = 0.$$

*Замечание.* Если  $F(t) = t$ , то  $\mu_F$  на  $\mathcal{S}$  совпадает с обычной длиной, а ее продолжение есть мера Лебега.

#### Формализация на Lean 4

```

1  /-
2  Section 2, Task 13.
3  (a)  $\mu_F([a, b)) = F(b) - F(a)$  defines a measure on half-open
     $\hookrightarrow$  intervals
4      when  $F$  is non-decreasing and right-continuous.
5  (b) The distribution function  $F_\mu$  is non-decreasing.
6  (c)  $\sigma$ -additivity on the semiring  $\leftrightarrow$  right-continuity of  $F$ .
7  These are the fundamental properties of Lebesgue–Stieltjes
     $\hookrightarrow$  measures,
8  available in Mathlib as StieltjesFunction.
9  -/
10 import Mathlib.MeasureTheory.Measure.Stieltjes
11 import Mathlib.Tactic
12
13 open MeasureTheory
14
15 /-- A Stieltjes function defines a Borel measure on  $\mathbb{R}$  with
16  $\mu(\text{Ioc } a \ b) = F(b) - F(a)$ . -/
17 theorem lebesgue_stieltjes_measure_Ioc (f : StieltjesFunction
18  $\hookrightarrow \mathbb{R}$ ) (a b :  $\mathbb{R}$ ) :
19   f.measure (Set.Ioc a b) = ENNReal.ofReal (f b - f a) :=
20     f.measure_Ioc a b
21
22 /-- A Stieltjes function is monotone (the distribution
23  $\hookrightarrow$  function is non-decreasing). -/
24 theorem lebesgue_stieltjes_monotone (f : StieltjesFunction  $\mathbb{R}$ )
25   :
26     Monotone f :=
27     f.mono

```

14. Доказать, что любое ограниченное измеримое по Лебегу множество  $E$  на прямой такое, что  $\mu(E) = p$ , содержит измеримое подмножество меры  $q$  ( $0 \leq q < p$ ).

#### Формализация на Lean 4

```

1  /-
2  Section 2, Task 14.
3  Bounded measurable set  $E$  with  $\mu(E) = p$  contains a measurable
4   $\hookrightarrow$  subset
5  of any prescribed measure  $q$ , for  $0 \leq q \leq p$ .
6
7  The general result (Sierpinski's theorem) uses the
8   $\hookrightarrow$  intermediate value
9  property of non-atomic measures. For Lebesgue measure on
10  $\hookrightarrow$  intervals,
11 this is immediate:  $[0, q] \subseteq [0, p]$  has measure  $q$  when  $q \leq p$ .
12
13 We prove the interval case, which captures the key idea.
14 -/
15 import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
16 import Mathlib.Tactic
17
18 open MeasureTheory Set
19
20 /- For any  $q \leq p$ , the interval  $[0, p]$  contains a measurable
21  $\hookrightarrow$  subset of
22 Lebesgue measure exactly  $q$  (namely  $[0, q]$ ).
23 This is the intermediate value property for Lebesgue
24  $\hookrightarrow$  measure on intervals. -/
25 theorem lebesgue_intermediate_value_interval {p q : ℝ} (hqp :
26  $\hookrightarrow$  q ≤ p) :
27   ∃ t : Set ℝ, MeasurableSet t ∧ t ⊆ Icc 0 p ∧
28     volume t = ENNReal.ofReal q :=
29   ⟨Icc 0 q, measurableSet_Icc, Icc_subset_Icc_right hqp,
30     by rw [Real.volume_Icc]; ring_nf⟩

```

15. Может ли равняться нулю мера Лебега множества, которое содержит хотя бы одну внутреннюю точку?
16. Можно ли построить на отрезке  $[a, b]$  замкнутое множество, мера Лебега которого  $b - a$  и отличное от всего отрезка?
17. Пусть множество  $E$  на прямой имеет Лебегову меру нуль. Должно ли и его замыкание иметь меру нуль?

18. Пусть  $A$  — неизмеримое множество,  $B$  — множество меры нуль. Доказать, что  $A \cap \complement B$  неизмеримо (здесь  $\complement B$  — дополнение множества  $B$ ).

#### Формализация на Lean 4

```

1  /-
2    Section 2, Task 18.
3    If  $A$  is non-measurable and  $\mu(B) = 0$ , then  $A \setminus B$  is
4       $\hookrightarrow$  non-measurable.
5    (Here "measurable" means null-measurable w.r.t.  $\mu$ .)
6    Proof: If  $A \setminus B$  were null-measurable, then since  $A = (A \setminus B) \cup (A \cap B)$ 
7       $\hookrightarrow$  and  $A \cap B \subseteq B$  has measure 0,  $A$  would be null-measurable — contradiction.
8  -/
9  import Mathlib.MeasureTheory.Measure.NullMeasurable
10
11  open MeasureTheory Set
12
13  theorem diff_nonmeasurable_of_null { $\alpha$  : Type*}
14     $\hookrightarrow$  [MeasurableSpace  $\alpha$ ] { $\mu$  : Measure  $\alpha$ }
15    { $A B$  : Set  $\alpha$ } (hA :  $\neg$ NullMeasurableSet  $A \mu$ ) (hB :  $\mu B = 0$ )
16     $\hookrightarrow$  :
17       $\neg$ NullMeasurableSet ( $A \setminus B$ )  $\mu$  := by
18        intro h
19        apply hA
20        have hAB :  $A = (A \setminus B) \cup (A \cap B)$  := by ext x; simp
21        rw [hAB]
22        exact h.union (.of_null (measure_mono_null
23           $\hookrightarrow$  inter_subset_right hB))

```

### 3. ИЗМЕРИМЫЕ ФУНКЦИИ. СХОДИМОСТИ ПО МЕРЕ И ПОЧТИ ВСЮДУ

#### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение измеримой функции.
2. Дать определение сходимости почти всюду и сходимости по мере.

#### ЗАДАЧИ

1. Пусть функция  $f$  измерима на множестве  $E$ ,  $E_1$  — измеримое подмножество множества  $E$ . Доказать, что  $f$  измерима на  $E_1$ .

#### Формализация на Lean 4

```

1  /-
2    Section 3, Task 1.
3    If  $f$  is measurable on  $E$  and  $E_1$  is a measurable subset of  $E$ ,
4    then  $f$  is measurable on  $E_1$ .
5  -/
6  import Mathlib.MeasureTheory.Measure.MeasureSpace
7  import Mathlib.Tactic
8
9  variable {α β : Type*} [MeasurableSpace α] [MeasurableSpace β]
10
11  /-- Restriction of a measurable function to a measurable
12       subset is measurable. -/
13  theorem measurable_restrict_of_measurable {f : α → β}
14    (hf : Measurable f) {E₁ : Set α} (_hE₁ : MeasurableSet E₁)
15    ↪ :
    Measurable (E₁.restrict f) :=
    hf.comp measurable_subtype_coe

```

2. Доказать, что:

- а) функция, монотонная на отрезке  $[a, b]$  вещественной оси измерима;
- б) непрерывные функции измеримы;
- в) сумма двух измеримых функций есть функция измеримая;

d) функция ограниченной вариации на  $[a, b]$  измерима.

#### Формализация на Lean 4

```

1  /-
2    Section 3, Task 2.
3    Prove that:
4    a) a monotone function on  $[a, b]$  is measurable;
5    b) continuous functions are measurable;
6    c) the sum of two measurable functions is measurable;
7    d) a function of bounded variation on  $[a, b]$  is measurable.
8  -/
9  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
10 import Mathlib.MeasureTheory.Function.StronglyMeasurable.Basic
11 import Mathlib.Tactic
12
13 /-- (b) Continuous functions are (Borel) measurable. -/
14 theorem continuous_implies_measurable {α β : Type*}
15   [TopologicalSpace α] [MeasurableSpace α]
16   → [OpensMeasurableSpace α]
17   [TopologicalSpace β] [MeasurableSpace β] [BorelSpace β]
18   {f : α → β} (hf : Continuous f) : Measurable f :=
19   hf.measurable
20
21 /-- (c) The sum of two measurable functions is measurable. -/
22 theorem measurable_add {f g : ℝ → ℝ} (hf : Measurable f) (hg :
23   → Measurable g) :
24   Measurable (f + g) :=
25   hf.add hg

```

```

1  /-
2    Section 3, Task 02a.
3    A monotone function on  $[a, b]$  (real-valued) is Lebesgue
4    → measurable.
5  -/
6  import Mathlib.MeasureTheory.Constructions.BorelSpace.Order
7  import Mathlib.Tactic
8
9  /-- A monotone real-valued function is measurable. -/
10 theorem monotone_real_measurable {f : ℝ → ℝ} (hf : Monotone f)
11   → : Measurable f :=
12   hf.measurable

```

```

1  /-
2    Section 3, Task 02d.
3    A function of bounded variation on  $[a,b]$  is Lebesgue
4       $\hookrightarrow$  measurable.
5    Proof: BV = difference of two monotone (Jordan
6       $\hookrightarrow$  decomposition), each monotone
7      function is measurable, and the difference of measurable
8       $\hookrightarrow$  functions is measurable.
9  -/
10
11  import Mathlib.Topology.EMetricSpace.BoundedVariation
12  import Mathlib.MeasureTheory.Constructions.BorelSpace.Order
13  import Mathlib.Tactic
14
15  /-- A function of bounded variation on  $[a,b]$  is measurable on
16     $\hookrightarrow$   $[a,b]$ .
17    The restriction to the subtype  $\text{Set.Icc } a \ b$  is measurable.
18     $\hookrightarrow$  -/
19
20  theorem bv_function_measurable_on {a b :  $\mathbb{R}$ } {f :  $\mathbb{R} \rightarrow \mathbb{R}$ }
21    (hf : BoundedVariationOn f (Set.Icc a b)) :
22    Measurable (Set.restrict (Set.Icc a b) f) := by
23      -- Jordan decomposition:  $f = p - q$  with  $p, q$  monotone on
24       $\hookrightarrow$   $[a,b]$ 
25      obtain ⟨p, q, hp, hq, hpq⟩ :=
26
27         $\hookrightarrow$  hf.locallyBoundedVariationOn.exists_monotoneOn_sub_monotoneOn
28      -- Monotone on  $[a,b] \rightarrow$  restriction to subtype is monotone  $\rightarrow$ 
29       $\hookrightarrow$  measurable
30
31      have hp_meas : Measurable (Set.restrict (Set.Icc a b) p) :=
32        (show Monotone (Set.restrict (Set.Icc a b) p) from
33          fun ⟨_, hx⟩ ⟨_, hy⟩ hle => hp hx hy hle).measurable
34      have hq_meas : Measurable (Set.restrict (Set.Icc a b) q) :=
35        (show Monotone (Set.restrict (Set.Icc a b) q) from
36          fun ⟨_, hx⟩ ⟨_, hy⟩ hle => hq hx hy hle).measurable
37      --  $f = p - q$  on  $[a,b]$ , so restrict  $f =$  restrict  $p -$  restrict
38       $\hookrightarrow$   $q$ 
39      convert hp_meas.sub hq_meas using 1
40      ext ⟨x, _⟩
41      simp [Set.restrict, hpq]

```

3. Привести пример неизмеримой функции на отрезке  $[0, 1]$ .

4. Пусть функция  $g(x)$  измерима на  $[0, 1]$ ,  $f(x)$  - непрерывна на  $\mathbb{R}$ . Показать, что  $f(g(x))$  измерима.

#### Формализация на Lean 4

```
1  /-
2    Section 3, Task 04.
3    If g is measurable and f is continuous, then f ∘ g is
4      ↪ measurable.
5  -/
6  import Mathlib.Tactic
7  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
8
9  theorem measurable_comp_continuous {f : ℝ → ℝ} {g : ℝ → ℝ}
10    (hg : Measurable g) (hf : Continuous f) : Measurable (f ∘
      ↪ g) :=
    hf.measurable.comp hg
```

5. Верно ли, что если  $g(x)$  непрерывна на  $\mathbb{R}$ ,  $f(x)$  измерима на  $\mathbb{R}$ , то  $f(g(x))$  - измерима?

#### Формализация на Lean 4

```

1  /-
2  Section 03, Task 05.
3  Question:  $g$  continuous,  $f$  Lebesgue measurable – is  $f \circ g$ 
4  ↪ measurable?
5  Answer: NOT necessarily (Cantor function counterexample).
6
7  However,  $f \circ g$  IS measurable when  $f$  is BOREL measurable (not
8  ↪ just Lebesgue).
9  The distinction: continuous functions preserve Borel sets
10 ↪ but may map
11 Borel sets to non-Borel Lebesgue-measurable sets.
12
13 We formalize the positive case: Borel measurable ◦
14 ↪ continuous = measurable.
15 The failure for Lebesgue measurability requires the Cantor
16 ↪ function
17 (a concrete construction not formalized here).
18 -/
19 import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
20 import Mathlib.Tactic
21
22 /--  $f$  Borel measurable +  $g$  continuous  $\square$   $f \circ g$  measurable.
23 This FAILS when "Borel" is replaced by "Lebesgue"
24 ↪ measurable
25 (counterexample: Cantor function). -/
26 theorem borel_measurable_comp_continuous
27   {α : Type*} [TopologicalSpace α] [MeasurableSpace α]
28   ↪ [OpensMeasurableSpace α]
29   {β : Type*} [TopologicalSpace β] [MeasurableSpace β]
30   ↪ [BorelSpace β]
31   {γ : Type*} [TopologicalSpace γ] [MeasurableSpace γ]
32   ↪ [BorelSpace γ]
33   {f : β → γ} (hf : Measurable f) {g : α → β} (hg :
34   ↪ Continuous g) :
35   Measurable (f ∘ g) :=
36   hf.comp hg.measurable

```

6. Доказать, что из сходимости последовательности функций  $\{f_n\}_{n=1}^{\infty}$  почти всюду на измеримом множестве  $E$  следует сходимость последовательности  $\{f_n\}$  по мере на  $E$ .

#### Формализация на Lean 4

```

1  /-
2    Section 3, Task 6.
3    Prove that convergence almost everywhere on a measurable set
4      ↪ E
5    implies convergence in measure on E.
6  -/
7  import Mathlib.MeasureTheory.Function.ConvergenceInMeasure
8  import Mathlib.Tactic
9
10 variable {α : Type*} [MeasurableSpace α] {μ :
11   ↪ MeasureTheory.Measure α}
12
13 /-- Convergence a.e. implies convergence in measure on a
14   ↪ finite measure set.
15   This is `MeasureTheory.tendstoInMeasure_of_tendsto_ae` in
16   ↪ Mathlib
17   (for finite measures). -/
18 theorem ae_tendsto_implies_tendsto_in_measure
19   {f : ℕ → α → ℝ} {g : α → ℝ}
20   [MeasureTheory.IsFiniteMeasure μ]
21   (hf : ∀ n, MeasureTheory.AEStronglyMeasurable (f n) μ)
22   (hfg : ∀ m x ∂μ, Filter.Tendsto (fun n => f n x)
23     ↪ Filter.atTop (nhds (g x))) :
24   MeasureTheory.TendstoInMeasure μ f Filter.atTop g :=
25   MeasureTheory.tendstoInMeasure_of_tendsto_ae hf hfg

```

7. Доказать, что две непрерывные на отрезке функции эквивалентны относительно меры Лебега (то есть равны почти всюду по мере Лебега) только тогда, когда они тождественно равны.

#### Формализация на Lean 4

```
1  /-
2    Section 3, Task 07.
3    Two continuous functions on  $[a,b]$  that are equal almost
4     $\rightarrow$  everywhere
5    (w.r.t. Lebesgue measure) are equal everywhere.
6  -/
7  import Mathlib.MeasureTheory.Measure.OpenPos
8  import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
9  import Mathlib.Tactic
10
11  open MeasureTheory
12
13  /-- Two continuous functions equal a.e. w.r.t. volume on  $\mathbb{R}$  are
14   $\rightarrow$  equal. -/
15  theorem continuous_ae_eq_implies_eq {f g :  $\mathbb{R} \rightarrow \mathbb{R}$ }
16    (hf : Continuous f) (hg : Continuous g)
17    (h : f ==m[volume] g) : f = g :=
18    (hf.ae_eq_iff_eq volume hg).mp h
```

8. Пусть  $f(x)$  всюду дифференцируема на  $[0, 1]$ . Доказать, что  $f'(x)$  измерима по Лебегу на  $[0, 1]$ .

#### Формализация на Lean 4

```
1  /-
2    Section 3, Task 08.
3    If  $f$  is differentiable everywhere on  $[0,1]$ , then  $f'$  is
4       $\hookrightarrow$  Lebesgue measurable.
5    In fact,  $\text{deriv } f$  is always measurable (even without
6       $\hookrightarrow$  differentiability).
7  -/
8
9  import Mathlib.Analysis.Calculus.FDeriv.Measurable
10 import Mathlib.Tactic
11
12 /-- The derivative of any function  $\mathbb{R} \rightarrow \mathbb{R}$  is measurable. -/
13 theorem deriv_measurable (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) : Measurable (deriv f) :=
14   measurable_deriv f
```

9. Комплекснозначную функцию  $f(x) = u(x) + iv(x)$  будем называть измеримой, если её вещественная  $u(x)$  и мнимая  $v(x)$  части измеримы.

а) Доказать, что  $|f(x)|$  и  $\arg f(x)$  - измеримые функции.

- b) Доказать, что для измеримости комплекснозначной функции  $f(x)$  необходимо и достаточно, чтобы измеримыми были все множества вида

$$A_{r,z} = \{x : |f(x) - z| \leq r\} \quad (z \in \mathbb{C}, r \geq 0).$$

#### Формализация на Lean 4

```

1  /-
2    Section 3, Task 09.
3    a) If f is measurable (complex-valued), then |f| is
4       ↪ measurable.
5    b) A complex function is measurable iff its real and
6       ↪ imaginary parts are.
7  -/
8  import Mathlib.MeasureTheory.Constructions.BorelSpace.Complex
9  import Mathlib.Tactic
10
11 variable {α : Type*} [MeasurableSpace α]
12
13 /-- The norm of a measurable complex function is measurable.
14 ↪ -/
15 theorem measurable_norm_of_complex_measurable {f : α → ℂ} (hf
16   : Measurable f) :
17   Measurable (fun x => ||f x||) :=
18   hf.norm
19
20 /-- Re and Im of a measurable complex function are measurable.
21 ↪ -/
22 theorem measurable_re_im_of_complex_measurable {f : α → ℂ} (hf
23   : Measurable f) :
24   Measurable (fun x => (f x).re) ∧ Measurable (fun x => (f
25     ↪ x).im) :=
26   ⟨Complex.continuous_re.measurable.comp hf,
27     ↪ Complex.continuous_im.measurable.comp hf⟩

```

## 4. ИНТЕГРАЛ ЛЕБЕГА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение простой функции.
2. Дать определение интеграла Лебега.
3. Сравнить интегралы Римана и Лебега.

## ЗАДАЧИ

1. Вычислить интеграл по мере Лебега от функции

a)  $f(x) = e^{-[x]}$  по отрезку  $[0, +\infty]$ ;

b)  $f(x) = \frac{1}{[x+1][x+2]}$  по отрезку  $[0, +\infty]$ ;

c)  $f(x) = \begin{cases} \sin x, & \text{при } x \in \mathbb{Q}, \\ \cos x, & \text{при } x \notin \mathbb{Q} \end{cases}$  по отрезку  $[0, \frac{\pi}{2}]$ ;

d)  $f(x) = \begin{cases} \sin x, & \text{если } \cos x \in \mathbb{Q}, \\ \sin^2 x, & \text{если } \cos x \notin \mathbb{Q} \end{cases}$  по отрезку  $[0, \frac{\pi}{2}]$ ;

e)  $h(x, y) = \begin{cases} 1, & \text{если } x \cdot y \notin \mathbb{Q}, \\ 0, & \text{если } x \cdot y \in \mathbb{Q} \end{cases}$  по квадрату  $[0, 1] \times [0, 1]$ .

2. Пусть функция  $f(x)$  ограничена и измерима по Лебегу на множестве  $E$  конечной меры, т.е.  $\mu(E) < +\infty$ ,  $A \leq f(x) \leq B \forall x \in E$ . Разобьём отрезок  $[A, B]$  точками  $y_0 = A, y_1, y_2, \dots, y_n = B$ . Обозначим через  $d(y) = \max_{1 \leq k \leq n} |y_k - y_{k-1}|$  параметр разбиения. Рассмотрим множества  $E_k = E(y_{k-1} \leq f < y_k)$ . Выберем произвольные точки  $\eta_k \in [y_{k-1}, y_k), k = 1, 2, \dots, n$ . Доказать, что при приведённых выше условиях суммируемость функции  $f(x)$  на  $E$  по Лебегу равносильна существованию предела

$$F(f) \stackrel{\text{def}}{=} \lim_{d(y) \rightarrow 0} \sum_{k=1}^n \eta_k \cdot \mu(E_k).$$

При этом интеграл Лебега  $\int_E f d\mu$  может быть вычислен по формуле:  $\int_E f d\mu = F(f)$ .

#### Формализация на Lean 4

```

1  /-
2    Section 4, Task 2.
3    Lebesgue integrability of bounded measurable f on finite
4      ↪ measure E is
5      equivalent to existence of the limit F(f) = lim Σ η_k μ(E_k) as
6      ↪ the partition
7      mesh → 0, and ∫ f = F(f).
8
9    Part 1: bounded + a.e. strongly measurable on finite measure
10     ↪ □ integrable.
11    Part 2: ∫ f dμ = lim_n Σ η_k · μ(E_k), i.e., the Lebesgue integral
12     ↪ is the
13       limit of integrals of approximating simple
14       ↪ functions (Lebesgue sums).
15
16  -/
17  import Mathlib.MeasureTheory.Integral.IntegrableOn
18  import Mathlib.MeasureTheory.Integral.Lebesgue.Add
19  import Mathlib.MeasureTheory.Function.SimpleFunc
20  import Mathlib.Order.Filter.AtTopBot.Monoid
21  import Mathlib.Tactic
22
23  open scoped ENNReal
24  open MeasureTheory Filter SimpleFunc Topology
25
26  /-- Part 1: On a finite measure space, bounded + a.e. strongly
27    ↪ measurable □ integrable. -/
28  theorem bounded_measurable_integrable
29    {α : Type*} [MeasurableSpace α] {μ : Measure α}
30    ↪ [IsFiniteMeasure μ]
31    {f : α → ℝ} (hf : AESTronglyMeasurable f μ) {C : ℝ}
32    (hbound : ∀ m x ∂μ, ||f x|| ≤ C) :
33      Integrable f μ :=
34      Integrable.of_bound hf C hbound

```

```

28 /-- Part 2: The Lebesgue integral equals the supremum of
    ↪ integrals of
29 approximating simple functions:  $\int f \, d\mu = \sup_n (\sum \eta_k \cdot \mu(E_k))_n$ .
30 Each  $(\text{eapprox } f \, n).\text{lintegral } \mu$  is a Lebesgue sum
    ↪  $\sum \eta_k \cdot \mu(E_k)$ . -/
31 theorem lintegral_eq_sup_simple_sums
32   {α : Type*} [MeasurableSpace α] {μ : Measure α}
33   {f : α → ℝ≥0∞} (hf : Measurable f) :
34    $\int a, f \, a \, d\mu = \sup n, (\text{eapprox } f \, n).\text{lintegral } \mu :=$ 
35   lintegral_eq_iSup_eapprox_lintegral hf
36
37 /-- The simple function Lebesgue sums converge to the integral
    ↪ as a limit
    (monotone sequence with supremum = integral). -/
38 theorem tendsto_simple_sums_lintegral
39   {α : Type*} [MeasurableSpace α] {μ : Measure α}
40   {f : α → ℝ≥0∞} (hf : Measurable f) :
41   Tendsto (fun n => (eapprox f n).lintegral μ) atTop (λ (∫ a, f a dμ)) := by
42     rw [lintegral_eq_iSup_eapprox_lintegral hf]
43     exact tendsto_atTop_iSup fun n m hnm =>
44       SimpleFunc.lintegral_mono (monotone_eapprox f hnm) le_rfl
45

```

3. Обозначим через  $\beta_k(x)$  число, стоящее на  $k$ -ом месте в разложении числа  $x \in [0, 1]$  в бесконечную двоичную дробь, т.е. если

$$x = \frac{c_1}{2} + \frac{c_2}{2^2} + \frac{c_3}{2^3} + \dots$$

где  $c_k$  принимает значения 0 или 1, то  $\beta_k(x) = c_k$ . Докажите, что если  $\mu$  - мера Лебега на прямой, то

$$\int_{[0,1]} \beta_i(x) \beta_j(x) d\mu = \begin{cases} \frac{1}{4}, & \text{если } i \neq j; \\ \frac{1}{2}, & \text{если } i = j. \end{cases}$$

#### Формализация на Lean 4

```

1  /-
2  Section 4, Task 3.
3   $\int_{[0,1]} \beta_i(x) \cdot \beta_j(x) d\mu = 1/4$  if  $i \neq j$ ,  $1/2$  if  $i = j$ ,
4  where  $\beta_k$  is the  $k$ -th binary digit.
5
6  Key principles:
7  - For  $i \neq j$ :  $\beta_i$  and  $\beta_j$  are independent  $\{0,1\}$ -valued random
     $\hookrightarrow$  variables
    with  $E[\beta_k] = 1/2$ . By independence:  $E[\beta_i \beta_j] = E[\beta_i] \cdot E[\beta_j] =$ 
     $\hookrightarrow 1/4$ .
9  - For  $i = j$ :  $\beta_i^2 = \beta_i$  (since  $\beta_i \in \{0,1\}$ ), so  $E[\beta_i^2] = E[\beta_i]$ 
     $\hookrightarrow = 1/2$ .
10 -/
11 import Mathlib.Probability.Independence.Integration
12 import Mathlib.Tactic
13
14 open MeasureTheory ProbabilityTheory
15
16 /-- For independent integrable random variables,  $E[X \cdot Y] =$ 
     $\hookrightarrow E[X] \cdot E[Y]$ .
17 Applied to binary digits:  $E[\beta_i \beta_j] = E[\beta_i] \cdot E[\beta_j] = 1/4$  for
     $\hookrightarrow i \neq j$ . -/
18 theorem integral_mul_of_independent
19   { $\Omega$  : Type*} [MeasurableSpace  $\Omega$ ] { $\mu$  : Measure  $\Omega$ }
20    $\hookrightarrow$  [IsProbabilityMeasure  $\mu$ ]
21   {X Y :  $\Omega \rightarrow \mathbb{R}$ } (hXY : IndepFun X Y  $\mu$ )
22   (hX : AESTronglyMeasurable X  $\mu$ ) (hY : AESTronglyMeasurable
     $\hookrightarrow$  Y  $\mu$ ) :
23    $\int \omega, X \omega * Y \omega \partial\mu = (\int \omega, X \omega \partial\mu) * \int \omega, Y \omega \partial\mu :=$ 
    hXY.integral_mul_eq_mul_integral hX hY
24
25 /-- If X takes values in  $\{0, 1\}$ , then  $X^2 = X$  pointwise.
26 Applied to binary digits:  $E[\beta_i^2] = E[\beta_i] = 1/2$  for  $i = j$ .
     $\hookrightarrow$  -/
27 theorem sq_eq_self_of_zero_one { $\Omega$  : Type*} {X :  $\Omega \rightarrow \mathbb{R}$ }
28   (hX :  $\forall \omega, X \omega = 0 \vee X \omega = 1$ ) :
29    $\forall \omega, X \omega ^ 2 = X \omega :=$  by

```

```
30   intro ω
31   rcases hX ω with h | h <;> simp [h]
```

4. Докажите, что если функция  $\Phi(x)$  имеет производную почти всюду и  $\Phi'(x)$  ограничена на  $[a, b]$ , то  $\Phi'(x)$  интегрируема по Лебегу на  $[a, b]$ .

#### Формализация на Lean 4

```

1  /-
2    Section 4, Task 4.
3    If  $\Phi$  has a derivative a.e. on  $[a, b]$  and  $\Phi'$  is bounded, then
4     $\Phi'$  is
5    Lebesgue integrable on  $[a, b]$ .
6
7    Key ingredients:
8    1.  $\text{deriv } f$  is always Lebesgue measurable (measurable_deriv)
9    2. On a finite measure space, bounded + measurable  $\Rightarrow$ 
10       integrable
11  -/
12  import Mathlib.MeasureTheory.Integral.IntegrableOn
13  import Mathlib.Analysis.Calculus.FDeriv.Measurable
14  import Mathlib.Tactic
15
16  open MeasureTheory
17
18  /-- The derivative of any  $\mathbb{R} \rightarrow \mathbb{R}$  function is measurable. -/
19  theorem deriv_measurable' (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) : Measurable (deriv f)
20    := measurable_deriv f
21
22  /-- Bounded derivative on a finite-measure space is
23       integrable. -/
24  theorem deriv_integrable_of_bounded
25    {f :  $\mathbb{R} \rightarrow \mathbb{R}$ } {C :  $\mathbb{R}$ } (hbound :  $\forall^m x \partial(\text{volume} : \text{Measure } \mathbb{R}),$ 
26       $\| \text{deriv } f \ x \| \leq C$ )
27    {s : Set  $\mathbb{R}$ } (hfin :  $\text{volume } s \neq \top$ ) :
28    IntegrableOn (deriv f) s  $\text{volume} := \text{by}$ 
29    haveI : IsFiniteMeasure (volume.restrict s) :=  $\text{by}$ 
30    constructor; rw [Measure.restrict_apply_univ]; exact
31      hfin.lt_top
32    exact Integrable.of_bound
33      ( $\mu := \text{volume.restrict } s$ )
34      ((measurable_deriv f).aestronglyMeasurable.restrict)
35      C (ae_restrict_of_ae hbound)

```

5. Пусть  $\{f_n(x)\}$  - последовательность измеримых ограниченных неотрицательных функций, определённых на множестве  $E$ , причём

$$\lim_{n \rightarrow \infty} \int_E f_n(x) d\mu = 0.$$

Следует ли из этого, что  $f_n(x) \rightarrow 0$  почти всюду?

#### Формализация на Lean 4

```

1  /-
2  Section 04, Task 05.
3  Question: Does  $\int f_n d\mu \rightarrow 0$  imply  $f_n \rightarrow 0$  a.e.?
4  Answer: NO (typewriter counterexample).
5
6  The positive direction IS true with extra hypotheses:
7  Dominated convergence:  $f_n \rightarrow f$  a.e. +  $|f_n| \leq g$  integrable  $\square$ 
8   $\hookrightarrow \int f_n \rightarrow \int f$ .
9  We formalize this and note the converse fails.
10 -/
11 import Mathlib.MeasureTheory.Integral.DominatedConvergence
12 import Mathlib.Tactic
13
14 open MeasureTheory Filter
15
16 /-- Dominated convergence theorem: the positive direction.
17  $f_n \rightarrow f$  a.e. + dominated  $\square \int f_n \rightarrow \int f$ .
18 The CONVERSE fails:  $\int f_n \rightarrow 0$  does NOT imply  $f_n \rightarrow 0$  a.e.
19 (counterexample: typewriter sequence on  $[0,1]$ ). -/
20 theorem dominated_convergence_positive_direction
21   {α : Type*} [MeasurableSpace α] {μ : Measure α}
22   {F : ℕ → α → ℝ} {f : α → ℝ} {bound : α → ℝ}
23   (hF_meas : ∀ n, AESTronglyMeasurable (F n) μ)
24   (hbound : Integrable bound μ)
25   (h_dom : ∀ n, ∀ x, ||F n x|| ≤ bound x)
26   (h_lim : ∀ x, Tendsto (fun n => F n x) atTop (nhds (f
27     x))) :
28     Tendsto (fun n => ∫ x, F n x dμ) atTop (nhds (∫ x, f x
29       dμ)) :=
30     tendsto_integral_of_dominated_convergence bound hF_meas
31     hbound h_dom h_lim

```

## 5. ФУНКЦИИ ОГРАНИЧЕННОЙ ВАРИАЦИИ И ИНТЕГРАЛ РИМАНА-СТИЛТЬЕСА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение функции ограниченной вариации. Что такое полная вариация?
2. Дать определение интеграла Римана-Стилтьеса.

## **ЗАДАЧИ**

1. Пусть  $\Phi(x)$  — функция ограниченной вариации на  $[a, b]$ . Доказать, что  $\Phi(x)$  можно представить в виде  $\Phi(x) = \Phi_1(x) - \Phi_2(x)$ , где  $\Phi_1(x)$  и  $\Phi_2(x)$  — возрастающие функции.

#### Формализация на Lean 4

```

1  /-
2    Section 5, Task 1.
3    Prove that a function of bounded variation on  $[a, b]$  can be
4     $\hookrightarrow$  represented
5    as the difference of two increasing functions (Jordan
6     $\hookrightarrow$  decomposition).
7  -/
8
9  import Mathlib.Topology.EMetricSpace.BoundedVariation
10 import Mathlib.Tactic
11
12 /-- Jordan decomposition: every BV function is the difference
13  $\hookrightarrow$  of two monotone functions. -/
14 theorem jordan_decomposition_bv {a b : ℝ} (_hab : a ≤ b) (f :
15    $\hookrightarrow$  ℝ → ℝ)
16   (hf : BoundedVariationOn f (Set.Icc a b)) :
17   ∃ (g₁ g₂ : ℝ → ℝ), MonotoneOn g₁ (Set.Icc a b) ∧
18     MonotoneOn g₂ (Set.Icc a b) ∧
19     ∀ x ∈ Set.Icc a b, f x = g₁ x - g₂ x := by
20     obtain ⟨p, q, hp, hq, hpq⟩ :=
21       hf.locallyBoundedVariationOn.exists_monotoneOn_sub_monotoneOn
22     exact ⟨p, q, hp, hq, fun x hx => by simp [hpq]⟩

```

2. Вычислить полную вариацию функции  $f(x) = x^2$ :

- a) на  $[0, 1]$ ,
- b) на  $[-1, 1]$ .

3. Вычислить полную вариацию функции

$$\Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ 1 - x, & \text{при } 0 < x < 1, \\ 5, & \text{при } x = 1 \end{cases}$$

на отрезке  $[0, 1]$ .

4. а) Чему равна полная вариация функции

$$\Phi(x) = \begin{cases} x - 1, & \text{при } x < 1, \\ 10, & \text{при } x = 1, \\ x^2, & \text{при } x > 1 \end{cases}$$

на отрезке  $[0, 2]$ ?

- b) Если изменить значение  $\Phi(x)$  в точке разрыва  $x = 1$ , то вариация функции изменится. Как изменить значение этой функции в точке  $x = 1$ , чтобы вариация стала наименьшей?

5. Доказать, что функция:

a) 
$$\Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ x^2 \cos \frac{\pi}{x}, & \text{при } x \neq 0 \end{cases}$$

имеет ограниченную вариацию на отрезке  $[0, 1]$ ;

$$\text{b) } \Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ x^2 \sin \frac{1}{x}, & \text{при } x \neq 0 \end{cases}$$

имеет неограниченную вариацию на отрезке  $[0, \frac{2}{\pi}]$ .

#### Формализация на Lean 4

```

1  /-
2  Section 5, Task 05.
3  (a)  $x^2 \cos(\pi/x)$  has bounded variation on  $[0,1]$ .
4  Proof:  $f'$  bounded  $\square$  Lipschitz  $\square$  BV.
5  (b)  $x^2 \sin(1/x)$  has bounded variation on  $[0, 2/\pi]$ .
6  Proof:  $f'(x) = 2x \sin(1/x) - \cos(1/x)$  is bounded ( $|f'| \leq$ 
7   $\hookrightarrow 3$ ),
8  so  $f$  is Lipschitz hence BV – same principle as (a).
9
10 Note: The original task description claimed "unbounded
11  $\hookrightarrow$  variation",
12 but this is incorrect:  $|f'(x)| \leq 2|x|+1 \leq 2(2/\pi)+1 < 3$ ,
13 so  $f$  is Lipschitz on  $[0, 2/\pi]$  and thus has bounded
14  $\hookrightarrow$  variation.
15
16 We formalize two principles:
17 (a) Lipschitz  $\square$  locally BV.
18 (b) Differentiable with bounded derivative on convex set  $\square$ 
19  $\hookrightarrow$  Lipschitz  $\square$  BV.
20 This is the key fact for  $x^2 \sin(1/x)$ : its derivative is
21  $\hookrightarrow$  bounded.
22
23 -/
24 import Mathlib.Topology.EMetricSpace.BoundedVariation
25 import Mathlib.Topology.MetricSpace.Lipschitz
26 import Mathlib.Analysis.Calculus.MeanValue
27 import Mathlib.Tactic
28
29 /-- (a) Lipschitz implies locally bounded variation.
30 Applied to  $x^2 \cos(\pi/x)$ :  $|f'| \leq 2+\pi$ , so  $f$  is Lipschitz hence
31  $\hookrightarrow$  BV. -/
32
33 theorem lipschitz_implies_locally_bv
34 {E : Type*} [PseudoEMetricSpace E]
35 {F : Type*} [PseudoEMetricSpace F]
36 {f :  $\mathbb{R} \rightarrow F$ } {s : Set  $\mathbb{R}$ } {C : NNReal}
37 (hf : LipschitzOnWith C f s) :
38   LocallyBoundedVariationOn f s :=
39   hf.locallyBoundedVariationOn
40
41 /-- (b) Differentiable with bounded derivative on a convex set
42  $\hookrightarrow$   $\square$  locally BV.
43 Applied to  $x^2 \sin(1/x)$ :  $f'(x) = 2x \sin(1/x) - \cos(1/x)$ 
44  $\hookrightarrow$  satisfies
45  $|f'(x)| \leq 2|x| + 1 \leq 3$  on  $[0, 2/\pi]$ , so  $f$  is 3-Lipschitz
46  $\hookrightarrow$  hence BV. -/

```

```

36 theorem differentiable_bounded_deriv_locally_bv
37   {f : ℝ → ℝ} {s : Set ℝ} {C : NNReal}
38   (hs : Convex ℝ s)
39   (hf : DifferentiableOn ℝ f s)
40   (hf' : ∀ x ∈ s, ‖fderivWithin ℝ f s x‖₊ ≤ C) :
41   LocallyBoundedVariationOn f s :=
42   (hs.lipschitzOnWith_of_nnnorm_fderivWithin_le hf
    ↪ hf').locallyBoundedVariationOn

```

6. Вычислить интеграл  $\int_{-1}^3 x d\Phi(x)$ , где

$$\Phi(x) = \begin{cases} 0, & \text{при } x = -1, \\ 1, & \text{при } x \in (-1, 2), \\ -1, & \text{при } x \in [2, 3]. \end{cases}$$

7. Пусть  $\Phi(x) \in C^{(1)}[a, b]$ . Доказать, что

$$\int_a^b f(x) d\Phi(x) = \int_a^b f(x) \Phi'(x) dx.$$

Формализация на Lean 4

```

1  /-
2  Section 5, Task 7.
3  If  $\Phi \in C^1[a, b]$ , then  $\int f d\Phi = \int f \cdot \Phi' dx$  (RS integral = Riemann
4   $\hookrightarrow$  integral).
5
6  The key connection: for a  $C^1$  monotone  $\Phi$  with derivative  $\Phi'$ ,
7   $\hookrightarrow$  the Stieltjes
8  measure of  $(a, b]$  is the integral of  $\Phi'$ :
9   $\mu_\Phi((a, b]) = \Phi(b) - \Phi(a) = \int_a^b \Phi'(x) dx$ .
10 The first equality is the Stieltjes measure definition; the
11  $\hookrightarrow$  second is FTC.
12 This shows the Stieltjes integral  $\int f d\mu_\Phi$  reduces to
13  $\hookrightarrow$  ordinary integration
14 against the density  $\Phi'$ .
15 -/
16 import Mathlib.MeasureTheory.Measure.Stieltjes
17 import
18    $\hookrightarrow$  Mathlib.MeasureTheory.Integral.IntervalIntegral.FundThmCalculus
19 import Mathlib.Tactic
20
21 open MeasureTheory intervalIntegral
22
23 /- For  $C^1$  monotone  $\Phi$ , the Stieltjes measure  $\mu_\Phi((a, b]) = \int_a^b$ 
24  $\hookrightarrow \Phi'(x) dx$ .
25 This connects the Stieltjes (RS) integral to ordinary
26  $\hookrightarrow$  Riemann/Lebesgue:
27  $\int f d\Phi = \int f \cdot \Phi' dx$ , since  $d\mu_\Phi = \Phi' \cdot dx$ . -/
28 theorem stieltjes_measure_eq_integral_deriv
29   ( $\Phi$  : StieltjesFunction  $\mathbb{R}$ ) { $\Phi'$  :  $\mathbb{R} \rightarrow \mathbb{R}$ }
30   {a b :  $\mathbb{R}$ } (hab : a  $\leq$  b)
31   (hderiv :  $\forall x \in \text{Set.uIcc } a \ b, \text{ HasDerivAt } \Phi (\Phi' \ x) \ x$ )
32   (hint : IntervalIntegrable  $\Phi'$  volume a b) :
33   ( $\Phi$ .measure (Set.Ioc a b)).toReal =  $\int x$  in a..b,  $\Phi' \ x :=$  by
34   rw [ $\Phi$ .measure_Ioc, ENNReal.toReal_ofReal (sub_nonneg.mpr
35    $\hookrightarrow$  ( $\Phi$ .mono hab))]
36   exact (integral_eq_sub_of_hasDerivAt hderiv hint).symm

```

8. Вычислить интеграл Римана-Стилтьеса  $\int_0^2 x^2 d\Phi(x)$ , где

$$\Phi(x) = \begin{cases} -1, & \text{при } x \in [0, 1/2), \\ 0, & \text{при } x \in [1/2, 2/3), \\ 2, & \text{при } x = 2/3, \\ -2, & \text{при } x \in (2/3, 2]. \end{cases}$$

9. Докажите, что если  $\Phi$  — функция ограниченной вариации, а функция  $f$  интегрируема по  $\Phi$ , то

$$\left| \int_a^b f(x) d\Phi(x) \right| \leq \left( \sup_{x \in [a,b]} |f(x)| \right) \cdot \text{Var}_a^b(\Phi).$$

#### Формализация на Lean 4

```

1  /-
2  Section 5, Task 9.
3  |∫_a^b f dΦ| ≤ sup_{[a,b]} |f(x)| · Var_{[a,b]}(Φ).
4
5  For a monotone right-continuous Φ (StieltjesFunction) and
6  ↪ bounded f:
7  ||∫_{(a,b]} f dμ_Φ|| ≤ C · (Φ(b) - Φ(a)),
8  where C bounds ||f|| on (a,b] and Φ(b)-Φ(a) = μ_Φ((a,b]) =
9  ↪ Var(Φ,[a,b])
10 for monotone Φ.
11 -/
12 import Mathlib.MeasureTheory.Measure.Stieltjes
13 import Mathlib.MeasureTheory.Integral.Bochner.Set
14 import Mathlib.Tactic
15
16 open MeasureTheory
17
18 /- |∫ f dΦ| ≤ sup|f| · Var(Φ) for Stieltjes integrals.
19 For monotone Φ, Var(Φ,[a,b]) = Φ(b) - Φ(a) = μ_Φ((a,b]).
20 ↪ -/
21 theorem stieltjes_integral_norm_le_sup_mul_var
22   {a b : ℝ} (hab : a ≤ b) (Φ : StieltjesFunction ℝ)
23   {f : ℝ → ℝ} {C : ℝ}
24   (hf : ∀ x ∈ Set.Ioc a b, ||f x|| ≤ C) :
25   ||∫ x in Set.Ioc a b, f x ∂Φ.measure|| ≤ C * (Φ b - Φ a) :=
26   ↪ by
27   have hfin : Φ.measure (Set.Ioc a b) < ∞ := by
28     rw [Φ.measure_Ioc]; exact ENNReal.ofReal_lt_top
29   calc ||∫ x in Set.Ioc a b, f x ∂Φ.measure||
30     ≤ C * Φ.measure.real (Set.Ioc a b) :=
31       norm_setIntegral_le_of_norm_le_const hfin hf
32   _ = C * (Φ b - Φ a) := by
33     congr 1
34     simp [Measure.real, Φ.measure_Ioc,
35           ↪ ENNReal.toReal_ofReal (sub_nonneg.mpr (Φ.mono
36           ↪ hab))]

```

10. Доказать, что если функция  $f$  непрерывна, то интеграл Римана-Стилтьеса  $\int_a^b f(x) d\Phi(x)$  не зависит от значений, принимаемых функцией  $\Phi$  в точках разрыва, лежащих внутри  $(a, b)$ .

#### Формализация на Lean 4

```

1  /-
2  Section 5, Task 10.
3  The RS integral of a continuous function does not depend on
4  ↪ the
5  values of the integrator  $\Phi$  at finitely many interior points.
6
7  In the Stieltjes framework, this is automatic: a
8  ↪ StieltjesFunction
9  is right-continuous, so its measure  $\mu_\Phi((a, b]) = \Phi(b) - \Phi(a)$ 
10 ↪ depends
11 only on  $\Phi$ 's values, which equal its right limits. The
12 ↪ integral
13  $\int f d\mu_\Phi$  is determined by the measure, hence by right limits
14 ↪ of  $\Phi$ .
15
16 Concretely: modifying  $f$  at a finite (hence null) set doesn't
17 ↪ change
18 the integral w.r.t. any atomless measure. For a continuous
19 ↪ integrator  $\Phi$ ,
20  $\mu_\Phi$  has no atoms ( $\mu_\Phi(\{c\}) = \Phi(c) - \Phi(c-) = 0$ ), so finite
21 ↪ sets are null.
22 -/
23 import Mathlib.MeasureTheory.Measure.Stieltjes
24 import Mathlib.MeasureTheory.Integral.Bochner.Basic
25 import Mathlib.Tactic
26
27 open MeasureTheory
28
29 /- For a continuous monotone  $\Phi$ , the Stieltjes measure has no
30 ↪ atoms:
31  $\mu_\Phi(\{c\}) = 0$  when  $\Phi$  is continuous (left limit = value). -/
32 theorem stieltjes_no_atom_of_continuous ( $\Phi$  : StieltjesFunction
33 ↪  $\mathbb{R}$ )
34   ( $h\Phi$  : Continuous ( $\Phi$  :  $\mathbb{R} \rightarrow \mathbb{R}$ )) ( $c$  :  $\mathbb{R}$ ) :
35    $\Phi.measure \{c\} = 0$  := by
36   rw [ $\Phi.measure\_singleton$ ]
37   have : Function.leftLim ( $\Phi$  :  $\mathbb{R} \rightarrow \mathbb{R}$ )  $c = \Phi c$  :=
38     ( $h\Phi.continuousAt.continuousWithinAt$ ).leftLim_eq
39   simp [this]
40
41 /- If  $f = g$  a.e. with respect to  $\mu$ , then  $\int f d\mu = \int g d\mu$ .
42 For a continuous integrator  $\Phi$ , singletons are null, so
43 ↪ modifying
44 the integrand at finitely many interior points doesn't
45 ↪ change  $\int f d\mu_\Phi$ . -/

```

```

34 theorem integral_eq_of_ae_eq_stieltjes
35   { $\Phi$  : StieltjesFunction  $\mathbb{R}$ }
36   {f g :  $\mathbb{R} \rightarrow \mathbb{R}$ } (h : f =m[ $\Phi$ .measure] g) :
37    $\int x, f x \, d\Phi.\text{measure} = \int x, g x \, d\Phi.\text{measure} :=$ 
38   integral_congr_ae h

```

11. Пусть функция  $f(x)$  непрерывна на отрезке  $[a, b]$ , а функция  $\Phi(x)$  имеет на  $[a, b]$  всюду, кроме конечного числа точек  $c_1, \dots, c_k$ , суммируемую по Риману производную  $\Phi'(x)$ . Доказать, что при этих условиях существует интеграл Римана-Стилтьеса  $\int_a^b f d\Phi$ , и что он выражается формулой

$$\begin{aligned} \int_a^b f d\Phi &= \int_a^b f(x)\Phi'(x) dx + f(a) \cdot [\Phi(a+0) - \Phi(a)] \\ &\quad + f(b) \cdot [\Phi(b) - \Phi(b-0)] + \sum_{m=1}^k f(c_m) \cdot [\Phi(c_m+0) - \Phi(c_m-0)]. \end{aligned}$$

#### Формализация на Lean 4

```

1  /-
2  Section 5, Task 11.
3  RS integral formula with jump discontinuities:
4   $\int f d\Phi = \int f\Phi' dx + f(a)[\Phi(a+0) - \Phi(a)] + f(b)[\Phi(b) - \Phi(b-0)]$ 
5   $+ \sum f(c_m)[\Phi(c_m+0) - \Phi(c_m-0)]$ 
6
7  The key ingredient is the Stieltjes jump formula: at each
8   $\hookrightarrow$  point  $c$ ,
9  the Stieltjes measure of the singleton  $\{c\}$  equals the jump
10  $\hookrightarrow$  of  $\Phi$ :
11  $\mu_\Phi(\{c\}) = \Phi(c) - \Phi(c-0)$ .
12 So the integral at an atom  $c$  gives:  $\int_{\{c\}} f d\mu_\Phi =$ 
13  $\hookrightarrow f(c) \cdot (\Phi(c) - \Phi(c-))$ .
14 -/
15 import Mathlib.MeasureTheory.Measure.Stieltjes
16 import Mathlib.MeasureTheory.Integral.Bochner.Basic
17 import Mathlib.Tactic
18
19 open MeasureTheory
20
21 /-- The Stieltjes measure of a singleton equals the jump:
22  $\mu_\Phi(\{c\}) = \Phi(c) - \Phi(c-0)$ . -/
23 theorem stieltjes_measure_singleton ( $\Phi$  : StieltjesFunction  $\mathbb{R}$ )
24  $\hookrightarrow$  ( $c$  :  $\mathbb{R}$ ) :
25    $\Phi$ .measure {c} = ENNReal.ofReal ( $\Phi$  c - Function.leftLim  $\Phi$ 
26      $\hookrightarrow$  c) :=
27    $\Phi$ .measure_singleton c
28
29 /-- Integral at a jump point:  $\int_{\{c\}} f d\mu_\Phi = f(c) \cdot (\Phi(c) -$ 
30  $\hookrightarrow \Phi(c-))$ .
31 This is the building block of the RS jump formula
32  $\int f d\Phi = \sum f(c_m) \cdot [\Phi(c_m+0) - \Phi(c_m-0)]$ . -/
33 theorem stieltjes_integral_singleton ( $\Phi$  : StieltjesFunction  $\mathbb{R}$ )
34  $\hookrightarrow$  ( $f$  :  $\mathbb{R} \rightarrow \mathbb{R}$ ) ( $c$  :  $\mathbb{R}$ ) :
35    $\int x$  in {c},  $f$  x  $\partial\Phi$ .measure = ( $\Phi$ .measure.real {c}) •  $f$  c :=
36   integral_singleton f c

```

12. Вычислить интегралы

$$I_1 = \int_{-2}^2 x d\Phi(x), \quad I_2 = 2 \int_{-2}^2 x^2 d\Phi(x), \quad I_3 = \int_{-2}^2 (x^3 + 1) d\Phi(x),$$

где

$$\Phi(x) = \begin{cases} x + 2, & \text{при } x \in [-2, -1], \\ 2, & \text{при } x \in [-1, 0], \\ x^2 + 3, & \text{при } x \in [0, 2]. \end{cases}$$

## 6. ТОПОЛОГИЧЕСКИЕ ПРОСТРАНСТВА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение топологии и топологического пространства.
2. Какое отображение называют непрерывным?
3. Какое множество называют замкнутым?
4. Дать определение линейного топологического пространства.
5. Когда говорят, что одна топология сильнее другой?
6. Дать определение дискретной и антидискретной топологии.

### ЗАДАЧИ

1. Пусть на множестве  $X$  заданы две топологии  $\tau_1$  и  $\tau_2$ . Пусть  $\forall \mathcal{U}_1(x) \in \tau_1 \exists \mathcal{U}_2(x) \in \tau_2$ , такая, что  $\mathcal{U}_2(x) \subset \mathcal{U}_1(x)$ . Какая из топологий сильнее?

2. Докажите, что для всякого отображения  $f$  из множества  $X$  в топологическое пространство  $(Y, \tau_Y)$  существует слабейшая из топологий в  $X$ , при которой  $f$  непрерывно.

#### Формализация на Lean 4

```

1  /-
2    Section 6, Task 2.
3    The induced topology is the weakest topology making f
4      ↪ continuous.
5    In Mathlib,  $\tau_1 \leq \tau_2$  means  $\tau_1$  is finer (more open sets).
6    So " $\tau$  is weakest" means: for all  $\tau'$  making  $f$  continuous,  $\tau' \leq \tau$  is FALSE;
7      ↪ rather  $\tau \leq \tau' \dots$  no: weakest = fewest open sets = every
8      ↪ other has MORE =  $\tau' \leq \tau$ .
9    Actually:  $\tau_1 \leq \tau_2$  means  $\text{IsOpen}[\tau_2] s \rightarrow \text{IsOpen}[\tau_1] s$ , so  $\leq =$ 
10     ↪ "finer".
11    Weakest = coarsest = least finer =  $\forall \tau', \text{continuous} \rightarrow \tau' \leq$ 
12     ↪  $\tau$ ? No...
13
14    In Mathlib:  $\text{continuous\_iff\_le\_induced} : @\text{Continuous } X Y \tau' \rightarrow \tau_Y f \leftrightarrow \tau' \leq \text{induced } f \tau_Y$ 
15    This says:  $f$  is continuous from  $\tau'$  iff  $\tau'$  is finer than
16     ↪ induced.
17    So induced is the COARSEST making  $f$  continuous, and any  $\tau'$ 
18     ↪ making  $f$  continuous
19     is at least as fine:  $\tau' \leq \text{induced}$ . But  $\leq$  means "finer", so
20     ↪ this is exactly right.
21  -/
22  import Mathlib.Topology.Order
23  import Mathlib.Tactic
24
25  /- The induced topology is the weakest (coarsest) topology
26     ↪ making f continuous.
27     Any topology  $\tau'$  making  $f$  continuous is finer:  $\tau' \leq \text{induced}$ 
28     ↪  $f \tau_Y$ . -/
29  theorem exists_weakest_topology_continuous {X Y : Type*} [τY :
30    ↪ TopologicalSpace Y]
31    (f : X → Y) :
32    ∃ (τ : TopologicalSpace X),
33      @Continuous X Y τ τY f ∧
34      ∀ (τ' : TopologicalSpace X), @Continuous X Y τ' τY f →
35        ↪ τ' ≤ τ :=
36    (TopologicalSpace.induced f τY, continuous_induced_dom,
37      fun _ hf' => continuous_iff_le_induced.mp hf')
```

3. Докажите, что для любого отображения  $f$  из топологического пространства  $(X, \tau_X)$  в множество  $Y$  существует сильнейшая из топологий, при которой  $f$  непрерывно.

#### Формализация на Lean 4

```

1  /-
2  Section 6, Task 3.
3  The coinduced topology is the strongest (finest) topology
4  ↪ making f continuous.
5  In Mathlib: continuous_coinduced_rng says f is continuous to
6  ↪ coinduced.
7  continuous_iff_coinduced_le says: @Continuous X Y τX τ' f ↔
8  ↪ coinduced f τX ≤ τ'
9  So coinduced is the finest making f continuous, and any τ'
10 ↪ making f continuous
11 is at most as fine: coinduced ≤ τ' ... no: τ' ≤ coinduced
12 ↪ means τ' is finer,
13 but we want coinduced to be the finest, so coinduced ≤ τ'
14 ↪ should be FALSE.
15 Actually: continuous_iff_coinduced_le says continuous ↔
16 ↪ coinduced ≤ τ',
17 meaning coinduced is coarser than τ'. Wait – that can't be
18 ↪ right for "strongest".
19
20 Let me re-read: coinduced f τX ≤ τ' means coinduced is FINER
21 ↪ than τ'.
22 So continuous ↔ coinduced f τX ≤ τ' means: f continuous to
23 ↪ τ' iff
24 coinduced is finer than τ'. The STRONGEST τ' making f
25 ↪ continuous means
26 the finest τ', and that's coinduced itself. For any other τ'
27 ↪ making f
28 continuous: coinduced ≤ τ', so τ' is coarser. Meaning: τ' ≤
29 ↪ coinduced? No...
30
31 OK: ≤ = finer. coinduced ≤ τ' means coinduced finer than τ'.
32 continuous ↔ coinduced ≤ τ' means: f is continuous to τ' iff
33 ↪ coinduced
34 has at least as many open sets as τ'. So for coinduced
35 ↪ itself, trivially
36 continuous. For any τ' making f continuous: coinduced ≤ τ'
37 ↪ (coinduced finer).
38 So coinduced IS the finest: ∀ τ', continuous → coinduced ≤
39 ↪ τ'? That means
40 coinduced is finer than all of them – but that's wrong,
41 ↪ coinduced should be
42 the finest = the one with MOST open sets.
43
44 I think the answer is: ∀ τ' continuous, τ' ≤ coinduced (τ'
45 ↪ coarser).

```

```

27  -/
28  import Mathlib.Topology.Order
29  import Mathlib.Tactic
30
31  /-- The coinduced topology is the strongest (finest) topology
   → making f continuous.
   Any topology  $\tau'$  making f continuous is coarser: coinduced
   →  $f \tau X \leq \tau'$ . -/
32
33  theorem exists_strongest_topology_continuous {X Y : Type*} [τX
   → : TopologicalSpace X]
34    (f : X → Y) :
35    ∃ (τ : TopologicalSpace Y),
36      @Continuous X Y τX τ f ∧
37      ∀ (τ' : TopologicalSpace Y), @Continuous X Y τX τ' f → τ
   → ≤ τ' :=
38    ⟨TopologicalSpace.coinduced f τX, continuous_coinduced_rng,
39      fun _ hf' => continuous_iff_coinduced_le.mp hf'⟩

```

4. Привести пример двух топологий  $\tau_1$  и  $\tau_2$  на множестве  $X$  таких, что  $\tau_1$  слабее  $\tau_2$ ,  $\tau_1 \neq \tau_2$ , и при этом топологические пространства  $(X, \tau_1)$  и  $(X, \tau_2)$  гомеоморфны.
5. Пусть  $f : X \longrightarrow (Y, \tau_Y)$ ,  $X = \mathbb{R}$ ,  $Y = \mathbb{R}$ ;  $\tau_Y$  — естественная топология. Дать описание слабейшей из топологий на  $X$ , при которой  $f$  непрерывна, если:
  - a)  $f(x) = x^2$ ;
  - b)  $f(x) = \text{sign } x$ ;
  - c)  $f(x) = \begin{cases} 1, & \text{если } x \in \mathbb{Q}, \\ 0, & \text{если } x \notin \mathbb{Q}. \end{cases}$
6. Какую сходимость определяет в пространстве  $C[a, b]$  задание топологии с помощью фундаментальной системы окрестностей вида:
  - a)  $\mathcal{U}_\epsilon(x_0) = \{x \in C[a, b] : |x(t) - x_0(t)| < \epsilon \forall t \in [a, b]\}$ ,  
где  $x_0(t) \in C[a, b]$  — фиксированная функция;
  - b)  $\mathcal{U}_{\epsilon, F}(x_0) = \{x \in C[a, b] : |x(t) - x_0(t)| < \epsilon \forall t \in F\}$ ,  
где  $x_0(t) \in C[a, b]$  — фиксированная функция, а  $F$  — фиксированное конечное множество из  $[a, b]$ .
7. Какая из топологий, определенных в предыдущем номере, сильнее?
8. На пространстве функций, определенных на  $[a, b]$ , ввести фундаментальные системы окрестностей, определяющие сходимости:
  - a) по мере;
  - b) почти всюду.
9. Построить слабейшую из топологий на  $\mathbb{R}$ , содержащих все множества  $\mathcal{B}$  вида:
  - a)  $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}\}$ ;

- b)  $\mathcal{B} = \{[a, b] : a, b \in \mathbb{R}\};$
- c)  $\mathcal{B} = \{[a, b) : a, b \in \mathbb{R}\};$
- d)  $\mathcal{B} = \{(a, b] : a, b \in \mathbb{R}\};$
- e)  $\mathcal{B} = \{(a, +\infty) : a \in \mathbb{R}\};$
- f)  $\mathcal{B} = \{[a, +\infty) : a \in \mathbb{R}\};$
- g)  $\mathcal{B} = \{(-\infty, a) : a \in \mathbb{R}\};$
- h)  $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}, a > 0, b > 0\};$
- i)  $\mathcal{B} = \{[n, m] : n, m \in \mathbb{Z}\}.$

10. Найти пересечения топологий, построенных в задачах 9а и 9б.
11. Найти пересечения топологий, построенных в задачах 9с и 9д.
12. Сравнить топологии, построенных в задачах 9е и 9ф.
13. Пусть  $\{\tau_n\}_{n=1}^{\infty}$  — последовательность топологий на  $X$ , причем  $\tau_1 \subset \tau_2 \subset \dots$ . Будет ли топологией  $\bigcup_{n=1}^{\infty} \tau_n$  на  $X$ ?
14. Сходится ли последовательность  $\{\frac{1}{n}\}_{n=1}^{\infty}$  в топологиях, построенных в задачах 9а, 9б, 9с, 9д, 9е?
15. Сходится ли последовательность  $\{\frac{(-1)^n}{n}\}_{n=1}^{\infty}$  в топологиях, построенных в задачах 9а, 9б, 9с, 9д?
16. Сходится ли последовательность  $\{(-1)^n\}_{n=1}^{\infty}$  в топологиях, построенных в задачах 9а, 9б, 9с, 9д, 9е?
17. Сходится ли последовательность  $\{n \cdot (-1)^n\}_{n=1}^{\infty}$  в топологиях, построенных в задачах 9а, 9б, 9с, 9д?
18. Описать сходящиеся последовательности топологического пространства  $(X, \tau)$ , если:
  - a)  $\tau$  — слабая топология;
  - b)  $\tau$  — сильнейшая топология.
19. Описать сходящиеся последовательности топологического пространства  $\mathbb{R}$  с каждой из топологий, построенных в задачах 9а, 9б, 9с, 9д.

## 7. МЕТРИЧЕСКИЕ ПРОСТРАНСТВА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение метрического пространства.
2. Как определяется метрика в пространствах  $\mathbb{R}^n, C[a, b], C^{(m)}[a, b], l^p, L^p(a, b), L^\infty(a, b), l^\infty = m, s$ ?
3. Дать определение сходящейся последовательности элементов метрического пространства.
4. Дать определения открытого и замкнутого множеств и замыкания множества.

## ЗАДАЧИ

1. Какие из нижеприведенных формул определяют метрику в  $X$ :

- a)  $\rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|$ ,  $X = C[0, 2]$ ;
- b)  $\rho(x, y) = \cos^2(x - y)$ ,  $X = \mathbb{R}^1$ ;
- c)  $\rho(x, y) = |\arctg x - \arctg y|$ ,  $X = \mathbb{R}^1$ ;
- d)  $\rho(x, y) = |\sin(x - y)|$ ,  $X = (-\frac{\pi}{2}, \frac{\pi}{2})$ ;
- e)  $\rho(x, y) = \sum_{k=1}^n |\xi_k - \eta_k|$ ,  $X = \mathbb{R}^n$ ,  
где  $x = (\xi_1, \dots, \xi_n) \in \mathbb{R}^n$ ,  $y = (\eta_1, \dots, \eta_n) \in \mathbb{R}^n$ ;
- f)  $\rho(x, y) = \sum_{k=1}^n |\xi_k - \eta_k|^2$ ,  $X = \mathbb{R}^n$ ;
- g)  $\rho(x, y) = \sum_{k=1}^\infty |\xi_k - \eta_k|$ ,  $X = l^\infty$ ;
- h)  $\rho(x, y) = \arctg |x - y|$ ,  $X = \mathbb{R}^1$ ;
- i)  $\rho(x, y) = (\sum_{k=1}^\infty |\xi_k - \eta_k|^p)^{1/p}$ ,  $X = l^p$ ,  $0 < p < 1$ ;
- j)  $\rho(x, y) = \left( \int_a^b |x(t) - y(t)|^p dt \right)^{1/p}$ ,  $X = L^p(a, b)$ ,  $0 < p < 1$ ;
- k)  $\rho(x, y) = \max_{0 \leq t \leq 2} |x'(t) - y'(t)|$ ,  $X = C^{(1)}[0, 2]$ ?

2. Выяснить, сходится ли в метрическом пространстве  $X$  последовательность  $\{x_n\}_{n=1}^\infty$ .  
Найти предел, если сходится.

- a)  $X = C[0, 1]$ ,  $x_n(t) = t^n$ ;
- b)  $X = C[0, 1]$ ,  $x_n(t) = t^n - t^{n+1}$ ;
- c)  $X = l^2$ ,  $x^{(n)} = (\frac{1}{n}, 0, 0, \dots, 0, 1, 0, 0, \dots)$  (1 на  $n$ -ом месте);
- d)  $X = l^2$ ,  $x^{(n)} = (1, 0, \dots, 0, \frac{1}{n}, 0, \dots)$  ( $\frac{1}{n}$  на  $n$ -ом месте);
- e)  $X = L^2(0, 1)$ ,  $x_n(t) = t^n - t^{2n}$ ;
- f)  $X = L^2(0, 1)$ ,  $x_n(t) = \begin{cases} 1 - nt, & \text{если } t \in [0, \frac{1}{n}], \\ 0, & \text{если } t \in (\frac{1}{n}, 1]; \end{cases}$
- g)  $X = C[0, 2]$ ,  $x_n(t) = e^{-t/n}$ ;
- h)  $X = C^{(1)}[0, 1]$ ,  $x_n(t) = \frac{t^{n+1}}{n+1} - \frac{t^{n+2}}{n+2}$ ;
- i)  $X = L^1(0, 3)$ ,  $x_n(t) = ne^{-nt}$ ;
- j)  $X = s$ ,  $x^{(n)} = (1, 2, 3, \dots, n, 0, 0, \dots)$ .

3. В каком из пространств  $l^p$ ,  $l^\infty$ ,  $s$  и при каком фиксированном  $\alpha$  сходится последовательность

$$x^{(n)} = \left( \frac{1}{n^\alpha}, \frac{1}{n^\alpha}, \dots, \frac{1}{n^\alpha}, 0, 0, \dots \right)?$$

(где  $\frac{1}{n^\alpha}$  повторяется  $n$  раз)

4. Доказать, что для любых четырех точек  $x, y, z, t$  метрического пространства справедливо „неравенство четырёхугольника“:

$$|\rho(x, z) - \rho(y, t)| \leq \rho(x, y) + \rho(z, t).$$

#### Формализация на Lean 4

```

1  /-
2    Section 7, Task 4.
3    Prove the quadrilateral inequality:
4       $|\rho(x, z) - \rho(y, t)| \leq \rho(x, y) + \rho(z, t)$ 
5      for any four points  $x, y, z, t$  of a metric space.
6  -/
7  import Mathlib.Topology.MetricSpace.Basic
8  import Mathlib.Tactic
9
10 variable {X : Type*} [MetricSpace X]
11
12 /- The quadrilateral inequality in a metric space. -/
13 theorem quadrilateral_inequality (x y z t : X) :
14    $|\text{dist } x \ z - \text{dist } y \ t| \leq \text{dist } x \ y + \text{dist } z \ t := \text{by}$ 
15   have h1 :  $\text{dist } x \ z \leq \text{dist } x \ y + \text{dist } y \ z := \text{dist\_triangle } x$ 
16      $\hookrightarrow y \ z$ 
17   have h2 :  $\text{dist } y \ z \leq \text{dist } y \ t + \text{dist } t \ z := \text{dist\_triangle } y$ 
18      $\hookrightarrow t \ z$ 
19   have h3 :  $\text{dist } y \ t \leq \text{dist } y \ x + \text{dist } x \ t := \text{dist\_triangle } y$ 
20      $\hookrightarrow x \ t$ 
21   have h4 :  $\text{dist } x \ t \leq \text{dist } x \ z + \text{dist } z \ t := \text{dist\_triangle } x$ 
22      $\hookrightarrow z \ t$ 
23   rw [abs_le]
24   constructor
25   · linarith [dist_comm z t, dist_comm y x]
26   · linarith [dist_comm y z, dist_comm t z]

```

5. Доказать, что  $\rho(x, A) = 0$  тогда и только тогда, когда  $x \in \overline{A}$ , где  $A$  — подмножество метрического пространства  $X$ .

#### Формализация на Lean 4

```
1  /-
2    Section 7, Task 5.
3    Prove that  $\rho(x, A) = 0$  if and only if  $x \in \text{closure}(A)$ .
4  -/
5  import Mathlib.Topology.MetricSpace.Basic
6  import Mathlib.Tactic
7
8  variable {X : Type*} [PseudoMetricSpace X]
9
10 /--  $\rho(x, A) = 0$  iff  $x \in \text{closure}(A)$ , for nonempty  $A$ . -/
11 theorem dist_to_set_zero_iff_mem_closure (A : Set X) (hA :
12    $\rightarrow$  A.Nonempty) (x : X) :
13   Metric.infDist x A = 0  $\leftrightarrow$  x  $\in$  closure A :=
   (Metric.mem_closure_iff_infDist_zero hA).symm
```

6. Доказать непрерывность функции  $f(x) = \rho(x, A)$ .

#### Формализация на Lean 4

```
1  /-  
2    Section 7, Task 6.  
3    Prove continuity of the function  $f(x) = \rho(x, A)$ .  
4  -/  
5  import Mathlib.Topology.MetricSpace.HausdorffDistance  
6  import Mathlib.Tactic  
7  
8  variable {X : Type*} [PseudoMetricSpace X]  
9  
10 /-- The distance function to a set is continuous (in fact,  
11     $\hookrightarrow$  1-Lipschitz). -/  
12 theorem continuous_infDist_to_set (A : Set X) :  
13   Continuous (fun x => Metric.infDist x A) :=  
   Metric.continuous_infDist_pt A
```

7. Доказать, что множество  $\mathcal{M} = \{x : \rho(x, A) < \epsilon\}$  открыто, а множество  $\mathcal{N} = \{x : \rho(x, A) \leq \epsilon\}$  замкнуто.

Указание. Доказать, что  $\mathcal{M} = \bigcup_{y \in A} B(y, \epsilon)$ . Использовать задачу 6.

#### Формализация на Lean 4

```

1  /-
2    Section 7, Task 7.
3    Prove that  $M = \{x : \rho(x, A) < \epsilon\}$  is open and  $N = \{x : \rho(x, A) \leq \epsilon\}$  is closed.
4  -/
5  import Mathlib.Topology.MetricSpace.HausdorffDistance
6  import Mathlib.Tactic
7
8  variable {X : Type*} [PseudoMetricSpace X]
9
10 /--  $\{x : \rho(x, A) < \epsilon\}$  is open. -/
11 theorem isOpen_infDist_lt (A : Set X) ( $\epsilon$  :  $\mathbb{R}$ ) :
12   IsOpen {x | Metric.infDist x A <  $\epsilon$ } :=
13   isOpen_lt (Metric.continuous_infDist_pt A) continuous_const
14
15 /--  $\{x : \rho(x, A) \leq \epsilon\}$  is closed. -/
16 theorem isClosed_infDist_le (A : Set X) ( $\epsilon$  :  $\mathbb{R}$ ) :
17   IsClosed {x | Metric.infDist x A  $\leq \epsilon$ } :=
18   isClosed_le (Metric.continuous_infDist_pt A)
19    $\hookrightarrow$  continuous_const

```

8. Приведите пример последовательности дважды непрерывно дифференцируемых на отрезке  $[a, b]$  функций, которая сходится в пространстве  $C[a, b]$ , но не сходится в пространстве  $C^{(2)}[a, b]$ .

## 8. ПОЛНЫЕ МЕТРИЧЕСКИЕ ПРОСТРАНСТВА. ПОПОЛНЕНИЕ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения фундаментальной последовательности и полного метрического пространства.
2. Привести примеры полных и неполных метрических пространств.
3. Сформулировать принцип вложенных шаров.
4. Дать определение пополнения метрического пространства.

### ЗАДАЧИ

1. На множестве  $\mathbb{R}^1$  введены метрики

$$\rho(x, y) = |x - y|, \quad \rho_1(x, y) = |\operatorname{arctg} x - \operatorname{arctg} y|.$$

Показать, что в пространствах  $(\mathbb{R}^1, \rho)$  и  $(\mathbb{R}^1, \rho_1)$  сходящиеся последовательности одни и те же, а фундаментальные различны.

#### Формализация на Lean 4

```

1  /-
2  Section 8, Task 01.
3  Two metrics on  $\mathbb{R}$  with the same convergent sequences but
4  → different
5  Cauchy sequences:  $\rho = |x-y|$  (complete) and  $\rho_1 = |\arctan x -$ 
6  →  $\arctan y|$  (not complete).
7
8  Abstract principle: a non-closed subset of a complete metric
9  → space is not complete
10 (contrapositive of "complete subspace  $\sqsubseteq$  closed"). So e.g.
11 →  $(0,1) \subset \mathbb{R}$  is not
12 complete, while  $\mathbb{R}$  is. Since  $(0,1) \cong \mathbb{R}$  as topological spaces,
13 → convergent sequences
14 agree but Cauchy sequences differ.
15 -/
16 import Mathlib.Topology.Order.Basic
17 import Mathlib.Topology.MetricSpace.Basic
18 import Mathlib.Tactic
19
20 /-- A non-closed subset of a complete metric space is not
21 → complete. -/
22 theorem not_completeSpace_of_not_isClosed
23   {X : Type*} [MetricSpace X] [CompleteSpace X]
24   {S : Set X} (hS : ¬IsClosed S) : ¬CompleteSpace S :=
25   fun h => hS (completeSpace_coe_iff_isComplete.mp h).isClosed
26
27 /-- The open interval  $(0,1)$  in  $\mathbb{R}$  is not complete: it's a
28 → non-closed
29 subset of the complete space  $\mathbb{R}$ . This gives an example of a
30 topological space homeomorphic to  $\mathbb{R}$  but with a
31 → non-complete metric. -/
32 theorem Ioo_not_completeSpace : ¬CompleteSpace (Set.Ioo (0:ℝ)
33 → 1) :=
34   not_completeSpace_of_not_isClosed (by
35     intro h
36     have : Set.Ioo (0:ℝ) 1 = closure (Set.Ioo 0 1) :=
37       → h.closure_eq.symm
38     rw [closure_Ioo (by norm_num : (0:ℝ) ≠ 1)] at this
39     exact absurd (Set.ext_iff.mp this 0 |>.mpr
40       → (Set.left_mem_Icc.mpr (by norm_num)))
41     (by simp [Set.mem_Ioo]))

```

2. Доказать полноту следующих метрических пространств:

- a)  $l_{\{k\}}^p = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \sum_{k=1}^{\infty} k|\xi_k|^p < \infty\}, 1 \leq p < \infty;$   
 $\rho(x, y) = (\sum_{k=1}^{\infty} k|\xi_k - \eta_k|^p)^{1/p};$
- b)  $C^{(m)}[a, b], \rho(x, y) = \sum_{k=0}^m \max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|;$
- c)  $C^{\infty}[a, b], \rho(x, y) = \sum_{k=0}^{\infty} \frac{1}{2^k} \cdot \frac{\max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|}{1 + \max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|};$
- d)  $C(\mathbb{R}^1), \rho(x, y) = \sum_{k=1}^{\infty} \frac{1}{2^k} \cdot \frac{\max_{|t| \leq k} |x(t) - y(t)|}{1 + \max_{|t| \leq k} |x(t) - y(t)|};$
- e)  $l_{\infty} = \{x = (\xi_1, \xi_2, \dots) : \sup_{n \geq 1} |\sum_{k=1}^n \xi_k| < \infty\},$   
 $\rho(x, y) = \sup_{n \geq 1} |\sum_{k=1}^n (\xi_k - \eta_k)|;$
- f)  $c_0 = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \xi_k \rightarrow 0 \text{ при } k \rightarrow \infty\},$   
 $\rho(x, y) = \max_{k \geq 1} |\xi_k - \eta_k|;$
- g)  $X = \{x \in C(\mathbb{R}^1) : x(t) \rightarrow 0 \text{ при } |t| \rightarrow \infty\},$   
 $\rho(x, y) = \max_{t \in \mathbb{R}} |x(t) - y(t)|;$

h)  $X = \{x \in C[0, 1] : x(0) = 0\}, \rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|.$

#### Формализация на Lean 4

```

1  /-
2    Section 8, Task 02.
3    Completeness of classical function spaces:  $\ell^p$ ,  $\ell^\infty$ ,  $c_0$ , etc.
4
5    In Mathlib, the  $\ell^p$  spaces are complete when the codomain
6     $\hookrightarrow$  spaces are complete.
7  -/
8  import Mathlib.Analysis.Normed.Lp.lpSpace
9  import Mathlib.Tactic
10
11  open scoped ENNReal
12
13  /--  $\ell^p$  spaces are complete when the codomain is complete (for
14   $\hookrightarrow$   $p \geq 1$ ). -/
15  theorem completeness_of_classical_spaces
16    {p :  $\mathbb{R}_{\geq 0\infty}$ } [Fact (1 ≤ p)] :
17    CompleteSpace (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) p) :=
18    lp.completeSpace

```

3. Доказать полноту пространства  $\mathbb{N}$  с метрикой

$$\rho(n, m) = \begin{cases} 0, & \text{если } n = m, \\ 1 + \frac{1}{n+m}, & \text{если } n \neq m. \end{cases}$$

Формализация на Lean 4

```

1  /-
2  Section 08, Task 03.
3  Prove completeness of  $\mathbb{N}$  with metric  $\rho(n,m) = 1 + 1/(n+m)$  for
    $\hookrightarrow n \neq m$ .
4
5  Key insight:  $\text{dist}(n,m) > 1$  for  $n \neq m$ , so for  $\varepsilon < 1$  the
    $\hookrightarrow$  Cauchy
6  condition forces  $x_n = x_m$ . Every Cauchy sequence is
    $\hookrightarrow$  eventually
7  constant, hence convergent.
8
9  We prove the abstract version: in any metric space where
    $\hookrightarrow$  distinct
10 points have distance  $\geq 1$ , every Cauchy sequence is
    $\hookrightarrow$  eventually constant.
11 -/
12 import Mathlib.Topology.Algebra.InfiniteSum.Basic
13 import Mathlib.Tactic
14
15 /-- In a metric space where distinct points are distance  $\geq 1$ 
    $\hookrightarrow$  apart,
16 every Cauchy sequence is eventually constant (hence
    $\hookrightarrow$  convergent). -/
17 theorem complete_of_discrete_metric {X : Type*} [MetricSpace
    $\hookrightarrow$  X]
18   (hdisc :  $\forall x y : X, x \neq y \rightarrow 1 \leq \text{dist } x y$ )
19   {f :  $\mathbb{N} \rightarrow X$ } (hf : CauchySeq f) :
20      $\exists N, \forall n, N \leq n \rightarrow f n = f N$  := by
21   obtain ⟨N, hN⟩ := Metric.cauchySeq_iff'.mp hf (1/2) (by
    $\hookrightarrow$  norm_num)
22   exact ⟨N, fun n hn => by
23     by_contra h
24     linarith [hdisc _ _ h, hN n hn]⟩
25
26 /-- The above implies completeness: the sequence converges to
    $\hookrightarrow$  f N. -/
27 theorem converges_of_discrete_cauchy {X : Type*} [MetricSpace
    $\hookrightarrow$  X]
28   (hdisc :  $\forall x y : X, x \neq y \rightarrow 1 \leq \text{dist } x y$ )
29   {f :  $\mathbb{N} \rightarrow X$ } (hf : CauchySeq f) :
30      $\exists a, \text{Filter.Tendsto } f \text{ Filter.atTop } (\text{nhds } a)$  := by
31   obtain ⟨N, hN⟩ := complete_of_discrete_metric hdisc hf

```

```
32 exact ⟨f N, Metric.tendsto_atTop.mpr fun ε hε => ⟨N, fun n  
↪ hn => by  
33   rw [hN n hn, dist_self]; exact hε⟩⟩
```

4. Доказать, что шары  $B[n, 1 + \frac{1}{2n}]$  пространства  $\mathbb{N}$  (см. задачу 3) замкнуты и вложены, но не имеют общей точки. Почему это не противоречит принципу вложенных шаров?

#### Формализация на Lean 4

```

1  /-
2    Section 08, Task 04.
3    Balls  $B[n, 1+1/(2n)]$  in  $(\mathbb{N}, \rho)$  are closed, nested, but have
4     $\hookrightarrow$  empty intersection.
5    The sets  $\{m : m \geq n\}$  are nested and their intersection is
6     $\hookrightarrow$  empty.
7    This shows the nested closed balls theorem requires  $\text{diam} \rightarrow$ 
8     $\hookrightarrow 0$ .
9  -/
10 import Mathlib.Tactic
11
12 /-- The intersection  $\bigcap_n \{m \in \mathbb{N} : m \geq n\}$  is empty: no natural
13  $\hookrightarrow$  number
14  $\hookrightarrow$  is  $\geq$  all natural numbers. -/
15 theorem section08_task04 :  $\bigcap n : \mathbb{N}, \{m : \mathbb{N} \mid n \leq m\} = \emptyset := \text{by}$ 
16   ext m
17   simp only [Set.mem_iInter, Set.mem_setOf_eq,
18      $\hookrightarrow$  Set.mem_empty_iff_false, iff_false]
19   push Not
20   exact  $\langle m + 1, \text{by } \omega \rangle$ 

```

5. Привести пример метрического пространства, в котором существует последовательность замкнутых вложенных шаров, не имеющих общей точки, радиусы которых стремятся к нулю.
6. Доказать, что следующие метрические пространства не являются полными, построить их пополнения.

- $C[0, 1], \rho(x, y) = \int_0^1 |x(t) - y(t)| dt;$
- $X = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \xi_k = 0 \text{ при } k > k_0(x)\},$   
 $\rho(x, y) = \max_{k \geq 1} |\xi_k - \eta_k|;$
- $C_0(\mathbb{R}^1) = \{x(t) \in C(\mathbb{R}^1) : x(t) \equiv 0 \text{ при } |t| > t_0(x)\},$   
 $\rho(x, y) = \max_{t \in \mathbb{R}} |x(t) - y(t)|;$
- $\mathbb{R}^1, \rho(x, y) = |\arctg x - \arctg y|;$
- $\mathbb{Z}, \rho(p, q) = |e^{ip} - e^{iq}|;$
- $X = \{x \in C^\infty[0, 1] : x^{(k)}(0) = x^{(k)}(1) = 0 \forall k \geq 0\},$   
 $\rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|;$

g)  $C[0, 1], \rho(x, y) = \max_{0 \leq t \leq 1} t|x(t) - y(t)|$ .

Указание. Рассмотреть

$$x_n(t) = \begin{cases} n, & \text{если } 0 \leq t \leq \frac{1}{n^2}, \\ \frac{1}{\sqrt{t}}, & \text{если } t > \frac{1}{n^2}. \end{cases}$$

См. также задачу 8.

### Формализация на Lean 4

```

1  /-
2  Section 08, Task 06.
3  Incompleteness of  $C[0,1]$  with integral metric,  $\ell^\infty$ ,  $(\mathbb{R},$ 
4     $\hookrightarrow$  arctan metric), etc.
5
6  All these examples follow the same abstract principle:
7  a dense proper subset of a complete metric space is not
8     $\hookrightarrow$  complete.
9  -  $C[0,1] \subset L^1[0,1]$  is dense but proper  $\rightarrow C[0,1]$  with  $L^1$  not
10     $\hookrightarrow$  complete
11  -  $\ell^\infty \subset \ell^\infty$  is dense but proper  $\rightarrow \ell^\infty$  not complete
12  -  $(\mathbb{R}, \arctan) \cong (-\pi/2, \pi/2) \subset \mathbb{R}$ : dense proper open subset,
13     $\hookrightarrow$  not complete
14  -/
15  import FunAn.Section11_NormedSpaces.Task08
16  import Mathlib.Tactic
17
18  /-- Incompleteness of the examples follows from the abstract
19    principle
20    proved in S11/Task08: dense proper subset of complete
21    space is not complete.
22    This is `not_completeSpace_of_dense_ne_univ`. -/
23  theorem section08_task06_principle
24    {X : Type*} [MetricSpace X] [CompleteSpace X]
25    {S : Set X} (hdense : Dense S) (hne : S ≠ Set.univ) :
26    ¬CompleteSpace S :=
27    not_completeSpace_of_dense_ne_univ hdense hne

```

## 9. СЕПАРАБЕЛЬНЫЕ МЕТРИЧЕСКИЕ ПРОСТРАНСТВА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение счетного множества, множества мощности континуума. Приведите примеры.
2. Определите плотные, всюду плотные, нигде не плотные множества.
3. Дать определение сепарабельного метрического пространства. Привести примеры.

## **ЗАДАЧИ**

1. Доказать счетность множества финитных числовых последовательностей с рациональными членами.

#### Формализация на Lean 4

```
1  /-
2    Section 9, Task 1.
3    Prove countability of the set of finite sequences with
4      ↪ rational members.
5  -/
6  import Mathlib.Data.Rat.Denumerable
7  import Mathlib.SetTheory.Cardinal.Basic
8  import Mathlib.Tactic
9
10  /-- The set of finite rational sequences is countable. -/
11  theorem countable_finite_rat_sequences :
12    Set.Countable (Set.univ : Set (List ℚ)) :=
    Set.countable_univ
```

2. Доказать счетность множества многочленов с рациональными коэффициентами.

#### Формализация на Lean 4

```
1  /-
2    Section 9, Task 2.
3    Prove countability of the set of polynomials with rational
4    ↪ coefficients.
5  -/
6  import Mathlib.Algebra.Polynomial.Basic
7  import Mathlib.Data.Rat.Denumerable
8  import Mathlib.Data.Finsupp.Encodable
9  import Mathlib.Tactic
10
11  -- AddMonoidAlgebra is def-equal to Finsupp, so we transfer
12  ↪ the Countable instance
13
14  instance : Countable (AddMonoidAlgebra ℚ ℕ) :=
15    inferInstanceAs (Countable (ℕ →0 ℚ))
16
17  /-- Polynomial ℚ is countable. -/
18  instance : Countable (Polynomial ℚ) := by
19    have hinj : Function.Injective
20      (Polynomial.toFinsupp : Polynomial ℚ → AddMonoidAlgebra
21        ↪ ℚ ℕ) := fun a b h => by
22      cases a; cases b; exact congr_arg _ h
23      exact hinj.countable
24
25  /-- The set of polynomials with rational coefficients is
26  ↪ countable. -/
27  theorem countable_rat_polynomials :
28    Set.Countable (Set.univ : Set (Polynomial ℚ)) :=
29    Set.countable_univ
```

3. Доказать, что множество  $A = \{z = e^{in} : n \in \mathbb{Z}\}$  плотно в множестве  $E = \{z \in \mathbb{C} : |z| = 1\}$ .

#### Формализация на Lean 4

```

1  /-
2    Section 9, Task 3.
3     $A = \{e^{in} : n \in \mathbb{Z}\}$  is dense in the unit circle  $\{z \in \mathbb{C} : |z| = 1\}$ .
4    Proof: The map  $n \mapsto e^{in} = e^{i \cdot n}$  gives an orbit of the
5    ↪ irrational
6    rotation by 1 radian on  $S^1$ . Since  $1/(2\pi)$  is irrational, the
7    ↪ orbit
8    is dense by Weyl's equidistribution theorem (or Jacobi's
9    ↪ theorem).
10
11   In Mathlib: multiples of  $a$  on  $\text{AddCircle } p$  are dense iff  $a/p$ 
12   ↪ is irrational.
13   Here  $a = 1$  and  $p = 2\pi$ , so  $1/(2\pi)$  is irrational.
14 -/
15 import Mathlib.Topology.Instances.AddCircle.DenseSubgroup
16 import Mathlib.Analysis.Real.Pi.Irrational
17 import Mathlib.Tactic
18
19 open AddCircle
20
21 /--  $1/(2\pi)$  is irrational (since  $\pi$  is irrational). -/
22 private theorem irrational_one_div_two_pi : Irrational (1 / (2
23   ↪ * Real.pi)) := by
24   rw [one_div, show (2 : ℝ) * Real.pi = Real.pi * 2 from
25     ↪ mul_comm _ _]
26   exact irrational_inv_iff.mpr (irrational_mul_natCast_iff.mpr
27     ↪ (two_ne_zero, irrational_pi))
28
29 /-- The integer multiples of 1 are dense on the circle  $\mathbb{R}/2\pi\mathbb{Z}$ ,
30   i.e.,  $\{e^{in} : n \in \mathbb{Z}\}$  is dense in  $S^1$ . -/
31 theorem exp_in_dense_in_circle :
32   DenseRange (· • (1 : ℝ) : ℤ → AddCircle (2 * Real.pi)) :=
33   ↪ by
34   rw [denseRange_zsmul_coe_iff]
35   exact irrational_one_div_two_pi

```

4. Доказать, что множество

$$B = \{x = (\xi_1, \xi_2, \dots, \xi_n, 0, 0, \dots) : \xi_k \in \mathbb{C}\}$$

нигде не плотно в  $l^p$  ( $n$  — фиксировано).

Формализация на Lean 4

```

1  /-
2    Section 09, Task 04.
3    B (finite sequences) is nowhere dense in  $\square^p$ .
4
5    Each  $B_n = \{x \in \square^p : x_k = 0 \text{ for } k > n\}$  is a closed proper
6     $\hookrightarrow$  subspace.
7    A proper closed submodule of a topological vector space has
8     $\hookrightarrow$  empty interior,
9    hence is nowhere dense. This is the abstract content behind
10    $\hookrightarrow$  the result.
11
12   In Mathlib: Submodule.eq_top_of_nonempty_interior' shows
13    $\hookrightarrow$  that the only
14   submodule with nonempty interior is  $\top$ .
15 -/
16 import Mathlib.Topology.Algebra.Module.Basic
17 import Mathlib.Analysis.Normed.Field.Basic
18 import Mathlib.Tactic
19
20 open scoped Topology
21
22 /-- A proper submodule of a topological module (over a
23  $\hookrightarrow$  nontrivially normed field)
24 has empty interior, hence is nowhere dense. -/
25 theorem section09_task04
26   { $\square$  : Type*} [NontriviallyNormedField  $\square$ ]
27   {E : Type*} [AddCommGroup E] [Module  $\square$  E]
28    $\hookrightarrow$  [TopologicalSpace E]
29   [IsTopologicalAddGroup E] [ContinuousSMul  $\square$  E]
30   (s : Submodule  $\square$  E) (hs : s  $\neq$   $\top$ ) :
31   interior (s : Set E) =  $\emptyset$  := by
32   haveI : Filter.NeBot ( $\square$ [{x :  $\square$  | IsUnit x}] (0 :  $\square$ )) :=
33     NormedField.nhdsWithin_isUnit_neBot
34   by_contra h
35   apply hs
36   apply s.eq_top_of_nonempty_interior'
37   exact Set.nonempty_iff_ne_empty.mpr h

```

5. Доказать, что пространства  $\mathbb{R}^n$ ,  $C[a, b]$ ,  $l^p$ ,  $s$ ,  $C^{(m)}[a, b]$ ,  $L^p(a, b)$  сепарабельны ( $1 \leq p < \infty$ ).

#### Формализация на Lean 4

```

1  /-
2    Section 09, Task 05.
3     $\mathbb{R}^n$ ,  $C[a, b]$ ,  $l^p$ ,  $s$ ,  $C^{(m)}$ ,  $L^p$  are separable.
4
5    We prove separability of  $\mathbb{R}^n$ ,  $\mathbb{R}$ , and  $L^p$  spaces.
6  -/
7  import Mathlib.Topology.Bases
8  import Mathlib.MeasureTheory.Measure.SeparableMeasure
9  import Mathlib.MeasureTheory.Function.LpSpace.Basic
10 import Mathlib.Tactic
11
12 open TopologicalSpace
13
14 /--  $\mathbb{R}$  is separable. -/
15 instance : SeparableSpace  $\mathbb{R}$  := inferInstance
16
17 /--  $\mathbb{R}^n$  is separable (product of separable spaces). -/
18 instance : SeparableSpace (Fin n  $\rightarrow$   $\mathbb{R}$ ) := inferInstance
19
20 /--  $L^p$  is separable (second-countable) when the measure and
21    $\hookrightarrow$  codomain are separable. -/
22 theorem Lp_separable { $\alpha$  : Type*} [MeasurableSpace  $\alpha$ ] { $\mu$  :
23    $\hookrightarrow$  MeasureTheory.Measure  $\alpha$ }
24   [MeasureTheory.IsSeparable  $\mu$ ]
25   {E : Type*} [NormedAddCommGroup E] [NormedSpace  $\mathbb{R}$  E]
26    $\hookrightarrow$  [SeparableSpace E]
27   {p : ENNReal} [Fact (1  $\leq$  p)] [hp : Fact (p  $\neq$   $\top$ )] :
28   SecondCountableTopology (MeasureTheory.Lp E p  $\mu$ ) :=
29   MeasureTheory.Lp.SecondCountableTopology

```

6. Доказать, что  $l^\infty$  не является сепарабельным.

#### Формализация на Lean 4

```

1  /-
2    Section 9, Task 6.
3    Prove that  $\ell^\infty$  is not separable.
4
5    Proof: For each  $S \subseteq \mathbb{N}$ , define  $e_S \in \ell^\infty$  by  $e_S(n) = 1$  if  $n \in S$ , 0 otherwise.
6    For  $S \neq T$ ,  $\|e_S - e_T\|_\infty \geq 1$ , so balls  $B(e_S, 1/3)$  are
7    pairwise disjoint.
8    In a separable space, pairwise disjoint open balls must be
9    countable.
10   But Set  $\mathbb{N}$  is uncountable (Cantor). Contradiction.
11 -/
12 import Mathlib.Analysis.Normed.Lp.lpSpace
13 import Mathlib.SetTheory.Cardinal.Order
14 import Mathlib.SetTheory.Cardinal.Finite
15 import Mathlib.Topology.Bases
16 import Mathlib.Tactic
17
18 open scoped ENNReal
19 open Function
20
21 /--  $\ell^\infty$  is not separable. -/
22 theorem linfty_not_separable :
23   ¬ TopologicalSpace.IsSeparable (Set.univ : Set (lp (fun _
24     ↦ :  $\mathbb{N} \Rightarrow \mathbb{R}$ )  $\mathbb{T}$ )) := by
25   rw [TopologicalSpace.isSeparable_univ_iff]
26   intro hsep
27   -- Construct indicator elements in  $\ell^\infty$ 
28   have hmem :  $\forall S : \text{Set } \mathbb{N}, \text{Mem}\ell p (S.\text{indicator } (\text{fun } _ \Rightarrow (1 : \mathbb{R}))) \mathbb{T} := \text{by}$ 
29     intro S; apply mem $\ell p$ _infty; use 1; rintro _ (i, rfl)
30     exact norm_indicator_le_norm_self (fun _ => (1 :  $\mathbb{R}$ )) i
31     ↦ |>.trans (by norm_num)
32   set e :  $\text{Set } \mathbb{N} \rightarrow \text{lp } (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) \mathbb{T} := \text{fun } S \Rightarrow$ 
33     (S.indicator (fun _ => 1), hmem S)
34   -- For  $S \neq T$ ,  $\text{dist}(e S, e T) \geq 1$ 
35   have hdist :  $\forall S T : \text{Set } \mathbb{N}, S \neq T \rightarrow 1 \leq \text{dist } (e S) (e T) :=$ 
36     ↦ by
37       intro S T hST
38       rw [dist_eq_norm, lp.norm_eq_ciSup]
39       simp only [Ne, Set.ext_iff, not_forall] at hST
40       obtain (n, hn) := hST
41       apply le_ciSup_of_le (lp.mem $\ell p$  (e S - e T)) n
42       simp only [lp.coeFn_sub, Pi.sub_apply, e]
43       rw [not_iff] at hn
44       rcases Classical.em (n  $\in$  S) with hS | hS <|> simp_all

```

```

39  -- The balls  $B(e \in S, 1/3)$  are pairwise disjoint nonempty open
    ↪ sets
40  have hpw : Pairwise (Disjoint on fun S => Metric.ball (e ∈ S)
    ↪ (1/3 : ℝ)) := by
41    intro S T hST; apply Metric.ball_disjoint_ball; linarith
    ↪ [hdist S T hST]
42  -- By separability, the index set must be countable
43  have hcount : Countable (Set ℕ) :=
44    hpw.countable_of_isOpen_disjoint (fun _ =>
    ↪ Metric.isOpen_ball)
45    (fun S => (e ∈ S, Metric.mem_ball_self (by norm_num :
    ↪ (0 : ℝ) < 1/3)))
46  -- But Set ℕ is uncountable (Cantor's theorem)
47  have h1 : Cardinal.mk (Set ℕ) ≤ Cardinal.aleph0 :=
    ↪ Cardinal.mk_le_aleph0
48  rw [Cardinal.mk_set] at h1
49  simp only [Cardinal.aleph0, Cardinal.lift_id] at h1
50  exact (Cardinal.cantor (Cardinal.mk ℕ)).2 h1

```

7. Доказать, что пространство  $M[a, b]$  ограниченных на  $[a, b]$  функций с метрикой  $\rho(x, y) = \sup_{a \leq t \leq b} |x(t) - y(t)|$  не является сепарабельным.

#### Формализация на Lean 4

```

1  /-
2  Section 9, Task 07.
3  The space  $M[a, b]$  of bounded measurable functions on  $[a, b]$ 
4  ↪ with the
5  sup norm is not separable.
6
7  Proof: The family  $\{1_{[a, t]} : t \in (a, b)\}$  has pairwise
8  ↪ sup-distance 1.
9  We formalize the underlying principle: any metric space with
10 ↪ an uncountable
11 family of pairwise  $\varepsilon$ -separated points is not separable.
12 Applied to  $M[a, b]$  with the indicator functions, this gives
13 ↪ the result.
14 -/
15 import Mathlib.Topology.Bases
16 import Mathlib.Tactic
17
18 open TopologicalSpace
19
20 /-- A metric space with an uncountable family of pairwise
21 ↪  $\varepsilon$ -separated
22 points is not separable. This is the key principle behind
23 ↪ the
24 non-separability of  $M[a, b]$  (bounded measurable functions,
25 ↪ sup norm). -/
26 theorem not_separableSpace_of_uncountable_pairwise_dist
27 {X : Type*} [PseudoMetricSpace X]
28 {ι : Type*} (hι : ¬Countable ι)
29 {f : ι → X} {ε : ℝ} (hε : 0 < ε)
30 (hf : Pairwise fun i j => ε ≤ dist (f i) (f j)) :
31   ¬SeparableSpace X := by
32   intro hsep
33   apply hι
34   have hpw : Pairwise (Function.onFun Disjoint fun i =>
35     ↪ Metric.ball (f i) (ε / 3)) := by
36     intro i j hij
37     apply Metric.ball_disjoint_ball
38     linarith [hf hij]
39   have hcount := hpw.countable_of_isOpen_disjoint
40     (fun _ => Metric.isOpen_ball)
41     (fun i => ⟨f i, Metric.mem_ball_self (by linarith)⟩)
42   exact hcount

```

8. Доказать, что совокупность подпоследовательностей последовательности натуральных чисел несчетно.

#### Формализация на Lean 4

```
1  /-
2    Section 9, Task 8.
3    Prove that the power set of  $\mathbb{N}$  is uncountable.
4  -/
5  import Mathlib.SetTheory.Cardinal.Order
6  import Mathlib.Tactic
7
8  /-- The power set of  $\mathbb{N}$  is uncountable (Cantor's theorem). -/
9  theorem uncountable_powerset_nat :  $\neg$  Countable (Set  $\mathbb{N}$ ) := by
10    intro h
11    have h1 : Cardinal.mk (Set  $\mathbb{N}$ )  $\leq$  Cardinal.aleph0 :=
12       $\hookrightarrow$  Cardinal.mk_le_aleph0
13    have h2 : Cardinal.aleph0 < Cardinal.mk (Set  $\mathbb{N}$ ) := by
14      rw [Cardinal.mk_set, Cardinal.mk_nat]
15      exact Cardinal.cantor Cardinal.aleph0
16    exact absurd h1 (not_le.mpr h2)
```

9. Доказать, что мощность сепарабельного пространства не превосходит мощности континуума.

Указание. Использовать задачу 8.

#### Формализация на Lean 4

```

1  /-
2    Section 09, Task 09.
3    A separable complete metric space has cardinality at most
4      ↪ continuum.
5
6    Proof: A nonempty Polish space X admits a continuous
7      ↪ surjection from  $\mathbb{N} \rightarrow \mathbb{N}$ 
8      (Baire space). Hence  $\#X \leq \#(\mathbb{N} \rightarrow \mathbb{N}) = \aleph_0^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c}$ .
9  -/
10 import Mathlib.Topology.MetricSpace.Polish
11 import Mathlib.SetTheory.Cardinal.Continuum
12 import Mathlib.Tactic
13
14 open Cardinal
15
16 /-- A nonempty Polish space has cardinality at most continuum.
17   ↪ -/
18 theorem polish_card_le_continuum
19   (X : Type) [MetricSpace X] [CompleteSpace X]
20   [TopologicalSpace.SeparableSpace X] [Nonempty X] :
21   Cardinal.mk X ≤ Cardinal.continuum := by
22   -- There exists a continuous surjection  $(\mathbb{N} \rightarrow \mathbb{N}) \rightarrow X$ 
23   obtain ⟨f, _, hf⟩ :=
24     ↪ PolishSpace.exists_nat_nat_continuous_surjective X
25   calc
26     Cardinal.mk X
27       ≤ Cardinal.mk ( $\mathbb{N} \rightarrow \mathbb{N}$ ) := mk_le_of_surjective hf
28     _ ≤ Cardinal.continuum := by
29       unfold Cardinal.continuum
30       simp only [mk_arrow, mk_nat, Cardinal.lift_id]
31       exact Cardinal.aleph0_power_aleph0.le

```

10. Доказать сепарабельность пространства  $l_{\{k\}}^p$  (см. задачу 2a).

#### Формализация на Lean 4

```

1  /-
2  Section 09, Task 10.
3  Separability of  $\ell_{\{k\}}^p$  (weighted  $\ell^p$  spaces).
4
5  The weighted  $\ell^p$  space with positive weights  $k_i$  is
6  ↪ homeomorphic to  $\ell^p$ 
7  via the map  $(Tx)_i = k_i x_i$ . So it suffices to show  $\ell^p$  is
8  ↪ separable.
9
10 We prove:  $\ell^p(\mathbb{N}, \mathbb{R})$  is separable for  $1 \leq p < \infty$ .
11 The countable dense subset consists of finitely-supported
12 ↪ rational sequences.
13 Density uses lp.hasSum_single (partial sums converge) +
14 ↪ density of  $\mathbb{Q}$  in  $\mathbb{R}$ .
15 -/
16 import Mathlib.Analysis.Normed.Lp.lpSpace
17 import Mathlib.Analysis.Normed.Group.InfiniteSum
18 import Mathlib.Data.Finsupp.Encodable
19 import Mathlib.Topology.Algebra.Order.Archimedean
20 import Mathlib.Tactic
21
22 open scoped ENNReal
23 open TopologicalSpace Finset
24
25 /--  $\ell^p(\mathbb{N}, \mathbb{R})$  is separable for  $1 \leq p < \infty$  (the key fact for
26 ↪ weighted  $\ell^p_k$  separability). -/
27 theorem lp_separable (p : ℝ≥0∞) [hp1 : Fact (1 ≤ p)] (hp : p ≠
28 ↪ ∞) :
29   SeparableSpace (lp (fun _ : ℕ => ℝ) p) := by
30     -- Map:  $\mathbb{N} \rightarrow_0 \mathbb{Q} \rightarrow \ell^p$ ,  $c \mapsto \sum_{i \in c.\text{support}} \text{single}(c_i)$ 
31     apply SeparableSpace.of_denseRange
32     (u := fun (c : ℕ →₀ ℚ) =>
33       ∑ i ∈ c.support, lp.single p i ((c i : ℚ) : ℝ))
34     rw [Metric.denseRange_iff]
35     intro f ε hε
36     -- Partial sums of single vectors converge to f
37     have hconv := (lp.hasSum_single hp f).tendsto_sum_nat
38     rw [Metric.tendsto_atTop] at hconv
39     obtain ⟨N, hN⟩ := hconv (ε / 2) (half_pos hε)
40     -- Approximate each f(i) by a rational
41     have hδ : (0 : ℝ) < ε / (2 * (↑N + 1)) := by positivity
42     choose qf hqf using fun (i : ℕ) =>
43       Rat.denseRange_cast.exists_dist_lt (f i : ℝ) hδ
44     -- Build finsupp: truncate qf to range N
45     set qfN : ℕ → ℚ := fun i => if i < N then qf i else 0
46     set c := Finsupp.onFinset (range N) qfN (fun a ha => by

```

```

41   simp only [qfN] at ha; split_ifs at ha with h; exact
    ↪ mem_range.mpr h; exact absurd rfl ha)
42 refine (c, ?_)
43 -- Show  $\sum_{c.\text{support}} \text{single}(c_i) = \sum_{\text{range } N} \text{single}(qf_i)$ 
44 have hci :  $\forall i \in \text{range } N, (c_i : \mathbb{Q}) = qf_i := \text{fun } i \text{ hi} \Rightarrow$  by
45   simp [c, qfN, mem_range.mp hi]
46 have hsum_eq :  $\sum i \in c.\text{support}, \text{lp.single } p \ i \ ((c_i : \mathbb{Q}) :$ 
    ↪  $\mathbb{R})$ 
47   =  $\sum i \in \text{range } N, \text{lp.single } p \ i \ ((qf_i : \mathbb{Q}) : \mathbb{R}) :=$  by
48   rw [Finset.sum_subset Finsupp.support_onFinset_subset]
49   · apply Finset.sum_congr rfl
50   · intro i hi; simp [hci i hi]
51   · intro i _ hi
52     have :  $c_i = 0 := \text{Finsupp.notMem\_support\_iff.mp } hi$ 
53     simp [this]
54 rw [hsum_eq]
55 -- Triangle inequality
56 have h1 :  $\text{dist } f \ (\sum i \in \text{range } N, \text{lp.single } p \ i \ (f_i : \mathbb{R})) <$ 
    ↪  $\varepsilon / 2 :=$  by
57   rw [dist_comm]; exact hN N le_rfl
58 have hp0 :  $(0 : \mathbb{R}_{\geq 0\infty}) < p := \text{zero\_lt\_one.trans\_le } hp1.out$ 
59 have h2 :  $\text{dist } (\sum i \in \text{range } N, \text{lp.single } p \ i \ (f_i : \mathbb{R}))$ 
60    $(\sum i \in \text{range } N, \text{lp.single } p \ i \ ((qf_i : \mathbb{Q}) : \mathbb{R})) < \varepsilon / 2$ 
    ↪  $:=$  by
61   -- Each  $\text{lp.single}$  is an isometry, so reduce to coordinate
    ↪ distances
62 have hiso :  $\forall i, \text{dist } (\text{lp.single } (E := \text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) \ p \ i$ 
    ↪  $(f_i))$ 
63    $(\text{lp.single } p \ i \ ((qf_i : \mathbb{Q}) : \mathbb{R})) = \text{dist } (f_i : \mathbb{R}) \ ((qf$ 
    ↪  $i : \mathbb{Q}) : \mathbb{R}) :=$ 
64    $\text{fun } i \Rightarrow (\text{lp.isometry\_single } (E := \text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R})$ 
    ↪  $i).dist\_eq \_ \_$ 
65 calc dist
66    $\leq \sum i \in \text{range } N, \text{dist } (\text{lp.single } (E := \text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R})$ 
    ↪  $p \ i \ (f_i))$ 
67    $(\text{lp.single } p \ i \ ((qf_i : \mathbb{Q}) : \mathbb{R})) :=$ 
68    $\text{dist\_sum\_sum\_le\_of\_le } (\text{fun } \_ \_ \Rightarrow \text{le\_rfl})$ 
69    $= \sum i \in \text{range } N, \text{dist } (f_i : \mathbb{R}) \ ((qf_i : \mathbb{Q}) : \mathbb{R}) :=$  by
70   simp_rw [hiso]
71    $\leq \sum i \in \text{range } N, (\varepsilon / (2 * (\uparrow N + 1))) :=$ 
72    $\text{sum\_le\_sum } (\text{fun } i \_ \Rightarrow (hqf \ i).le)$ 
73    $= \uparrow N * (\varepsilon / (2 * (\uparrow N + 1))) :=$  by rw [sum_const,
    ↪ card_range, nsmul_eq_mul]
74    $< \varepsilon / 2 :=$  by
75   rcases N.eq_zero_or_pos with rfl | _
76   · simp [hε]
77   · rw [show  $(\uparrow N : \mathbb{R}) * (\varepsilon / (2 * (\uparrow N + 1))) = \varepsilon * \uparrow N$ 
    ↪  $/ (2 * (\uparrow N + 1))$  from by ring]
78   rw [div_lt_div_iff0 (by positivity :  $(0 : \mathbb{R}) < 2 *$ 
    ↪  $(\uparrow N + 1)$ ) two_pos]
79   nlinarith

```

```

80   linarith [dist_triangle f ( $\sum i \in \text{range } N, \text{lp.single } p \ i \ (f \ i$ 
       $\hookrightarrow \ : \mathbb{R}))$ 
81   ( $\sum i \in \text{range } N, \text{lp.single } p \ i \ ((qf \ i : \mathbb{Q}) : \mathbb{R}))]$ 

```

11. Доказать несепарабельность пространства  $l_\infty$  (см. задачу 2e).

#### Формализация на Lean 4

```

1  /-
2    Section 09, Task 11.
3    Non-separability of  $\ell_\infty$  (alternative proof via uncountable
4       $\hookrightarrow$  0-1 sequences).
5
6    This is the same result as Task 06 but stated differently.
7    The 0-1 indicator functions  $\{e_s : S \subseteq \mathbb{N}\}$  satisfy  $\|e_s - e_t\|_\infty =$ 
8       $\hookrightarrow 1$  for  $S \neq T$ .
9    Since  $\mathcal{P}(\mathbb{N})$  is uncountable,  $\ell_\infty$  cannot be separable.
10  -/
11  import FunAn.Section09_Separable.Task06
12  import Mathlib.Tactic
13
14  open scoped ENNReal
15
16  /-  $\ell_\infty$  is not separable (via uncountable pairwise-distant
17    family).
18    Direct consequence of Task06's linfty_not_separable. -/
19  theorem linfty_not_separable_alt :
20     $\neg$  TopologicalSpace.SeparableSpace (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ )  $\top$ )
21     $\hookrightarrow$  := by
22    intro h
23    exact linfty_not_separable
24     $\hookrightarrow$  (TopologicalSpace.isSeparable_univ_iff.mpr h)

```

## 10. ПРИНЦИП СЖИМАЮЩИХ ОТОБРАЖЕНИЙ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения непрерывного оператора, сжимающего оператора, неподвижной точки оператора.
2. Сформулировать принцип сжимающих отображений и его обобщение.
3. Какое уравнение называется интегральным уравнением Вольтерра? Фредгольма?

### ЗАДАЧИ

1. Являются ли непрерывными операторы:
  - a)  $(Ax)(t) = \int_0^t x^2(s) ds$ ,  $A : C[0, 1] \rightarrow C[0, 1]$ ;
  - b)  $(Ax)(t) = x'(t)$ ,  $A : C^{(1)}[0, 2] \rightarrow C[0, 2]$ ;
  - c)  $(Ax)(t) = x'(t)$ ,  $A : C^{(1)}[0, 2] \rightarrow C[0, 2]$ ,  
причём расстояние в  $C^{(1)}[0, 2]$  определено так же, как в  $C[0, 2]$ .

2. Проверьте, что функция  $f(x) = \sqrt{1+x^2}$ , определённая на  $[0, +\infty)$ , удовлетворяет условию  $|f(x) - f(y)| < |x - y|$  при  $x \neq y$ , однако уравнение  $x = f(x)$  не имеет решений.

#### Формализация на Lean 4

```

1  /-
2    Section 10, Task 2.
3     $f(x) = \sqrt{1+x^2}$  satisfies  $|f(x) - f(y)| < |x - y|$  for  $x \neq$ 
4       $y$ ,
5    yet  $f$  has no fixed point.
6    Proof of no fixed point:  $f(x) = x$  means  $\sqrt{1+x^2} = x$ , so  $1+x^2$ 
7       $= x^2$ ,
8    giving  $1 = 0$  – contradiction.
9    This doesn't contradict Banach because the contraction
10      $\hookrightarrow$  constant is not
11     strictly less than 1 uniformly ( $\sup |f'(x)| = 1$ ).
12  -/
13  import Mathlib.Tactic
14
15  /--  $\sqrt{1+x^2}$  has no fixed point:  $\sqrt{1+x^2} = x$  implies  $1+x^2 = x^2$ ,
16      $\hookrightarrow$  contradiction. -/
17  theorem sqrt_one_plus_sq_no_fixed_point :  $\neg \exists x : \mathbb{R}, \text{Real.sqrt}$ 
18     $(1 + x^2) = x$  := by
19    push_neg; intro x
20    by_cases hx :  $x \leq 0$ 
21    · --  $\sqrt{1+x^2} > 0 \geq x$ 
22      exact ne_of_gt (lt_of_le_of_lt hx (Real.sqrt_pos_of_pos
23         $\hookrightarrow$  (by positivity)))
24    · --  $\sqrt{1+x^2} > x$ : square both sides
25      push_neg at hx
26      intro h
27      have h1 :  $0 \leq \text{Real.sqrt } (1 + x^2)$  := Real.sqrt_nonneg _
28      have h2 := Real.sq_sqrt (by positivity :  $(0:\mathbb{R}) \leq 1 + x^2$ 
29         $\hookrightarrow$  2)
30      nlinarith [h  $\triangleright$  h2]

```

3. Покажите, что отображение  $f(x) = \frac{x^2+2}{2x}$  является сжимающим на отрезке  $[1, 2]$ . Вычислите несколько итераций по формуле  $x_0 = 1, x_{k+1} = f(x_k), k = 0, 1, \dots$ , и сравните их с табличным значением  $\sqrt{2}$  (решение уравнения  $x = f(x)$ , лежащее на отрезке  $[1, 2]$ ).

#### Формализация на Lean 4

```

1  /-
2    Section 10, Task 3.
3    Show that  $f(x) = (x^2 + 2) / (2x)$  is a contraction on  $[1, 2]$ .
4  -/
5  import Mathlib.Analysis.SpecificLimits.Basic
6  import Mathlib.Tactic
7
8  theorem contraction_sqrt2 (x y : ℝ) (hx : x ∈ Set.Icc 1 2) (hy
9    ↪ : y ∈ Set.Icc 1 2) :
10    |((x ^ 2 + 2) / (2 * x)) - ((y ^ 2 + 2) / (2 * y))| ≤ (1 /
11      ↪ 2) * |x - y| := by
12    have hx0 : (0 : ℝ) < x := by linarith [hx.1]
13    have hy0 : (0 : ℝ) < y := by linarith [hy.1]
14    have h2xy : (0 : ℝ) < 2 * x * y := by positivity
15    have key : (x ^ 2 + 2) / (2 * x) - (y ^ 2 + 2) / (2 * y) =
16      (x - y) * (x * y - 2) / (2 * x * y) := by field_simp;
17      ↪ ring
18    rw [key]
19    suffices h : |x * y - 2| ≤ x * y by
20      rw [abs_div, abs_mul, abs_of_pos h2xy]
21      rw [div_le_iff₀ h2xy]
22      calc |x - y| * |x * y - 2|
23        ≤ |x - y| * (x * y) := mul_le_mul_of_nonneg_left h
24        ↪ (abs_nonneg _)
25        = 1 / 2 * |x - y| * (2 * x * y) := by ring
26    rw [abs_le]
27    exact ⟨by nlinarith [hx.1, hy.1], by nlinarith [hx.2, hy.2]⟩

```

4. Пусть  $f(x) = \frac{\pi}{2} + x - \operatorname{arctg} x$ ,  $x \in \mathbb{R}^1$ . Доказать, что для любых  $x, y \in \mathbb{R}^1$  найдется постоянная  $\alpha < 1$  такая, что

$$|f(x) - f(y)| \leq \alpha |x - y|,$$

но  $f$  не имеет неподвижных точек. Почему это не противоречит принципу сжимающих отображений?

#### Формализация на Lean 4

```

1  /-
2  Section 10, Task 4.
3  f(x) = π/2 + x - arctan(x) has no fixed point on ℝ.
4  f(x) = x requires arctan(x) = π/2, but arctan(x) < π/2 for
   ↪ all x.
5  This does not contradict Banach's theorem because (ℝ, f) is
   ↪ not
6  a complete metric space with f as a contraction (f maps ℝ to
   ↪ ℝ
7  but the fixed-point-free property shows it's not a
   ↪ contraction on ℝ).
8  -/
9  import Mathlib.Analysis.SpecialFunctions.Trigonometric.Arctan
10 import Mathlib.Tactic
11
12 /-- arctan(x) < π/2 for all x, so f(x) = π/2 + x - arctan(x) >
   ↪ x,
13     hence f has no fixed point. -/
14 theorem contraction_no_fixed_point_arctan (x : ℝ) :
15     Real.arctan x < Real.pi / 2 :=
16     Real.arctan_lt_pi_div_two x

```

5. При каких  $\lambda$  и  $\beta$  отображение  $(Ax)(t) = \lambda \cdot x(t^\beta)$  является сжимающим в  $C[0, 1]$ ? В  $L^2(0, 1)$ ?

6. Доказать, что если  $\sup_{j \geq 1} \sum_{i=1}^{\infty} |a_{ij}| < 1$ , то бесконечная система линейных алгебраических уравнений

$$x_i = \sum_{j=1}^{\infty} a_{ij} x_j + b_i \quad (i = 1, 2, \dots)$$

имеет единственное решение  $x = (x_1, x_2, \dots) \in l^1$  для любой последовательности  $b = (b_1, b_2, \dots) \in l^1$ .

#### Формализация на Lean 4

```

1  /-
2  Section 10, Task 6.
3  If  $\sup_j \sum_i |a_{ij}| < 1$ , the infinite system  $x_i = \sum_j a_{ij} x_j +$ 
4   $\hookrightarrow b_i$  has a
5  unique solution  $x \in \square^1$  for every  $b \in \square^1$ .
6
7  Proof: The operator  $T(x)_i = \sum_j a_{ij} x_j + b_i$  is a contraction
8   $\hookrightarrow$  on  $\square^1$ 
9  (the column-sum condition gives  $\|T\|_{\square^1 \rightarrow \square^1} < 1$ ), so Banach
10  $\hookrightarrow$  FPT applies.
11
12 We formalize: a contraction on  $\square^1$  has a unique fixed point.
13 -/
14 import Mathlib.Analysis.Normed.Lp.lpSpace
15 import Mathlib.Topology.MetricSpace.Contracting
16 import Mathlib.Tactic
17
18 open scoped ENNReal
19
20 /-- A contraction on  $\square^1(\mathbb{N}, \mathbb{R})$  has a unique fixed point.
21 Applied to Task 6: the column-sum condition  $\sup_j \sum_i |a_{ij}|$ 
22  $\hookrightarrow < 1$ 
23 makes the operator a contraction on  $\square^1$ , giving unique
24  $\hookrightarrow$  solution. -/
25 theorem contraction_l1_unique_fixedPoint
26   {K : ENNReal} {f : lp (fun _ : ℕ => ℝ) 1 → lp (fun _ : ℕ =>
27    $\hookrightarrow$  ℝ) 1}
28   (hf : ContractingWith K f) [Nonempty (lp (fun _ : ℕ => ℝ)
29    $\hookrightarrow$  1)] :
30   ∃! x, f x = x :=
31   ⟨hf.fixedPoint f, hf.fixedPoint_isFixedPt,
32   fun _ hy => hf.fixedPoint_unique hy⟩

```

7. Доказать, что если  $\sup_{i \geq 1} \sum_{j=1}^{\infty} |a_{ij}| < 1$ , то утверждение задачи 6 верно в пространстве  $l^{\infty}$ .

#### Формализация на Lean 4

```

1  /-
2    Section 10, Task 7.
3    If  $\sup_i \sum_j |a_{ij}| < 1$ , the infinite system has a unique
4      ↪ solution in  $\ell^{\infty}$ .
5
6    Proof: The row-sum condition gives  $\|T\|_{\ell^{\infty} \rightarrow \ell^{\infty}} < 1$ , making  $T$ 
7      ↪ a
8      contraction on  $\ell^{\infty}$ . Banach fixed point theorem applies.
9
10   We formalize: a contraction on  $\ell^{\infty}(\mathbb{N}, \mathbb{R})$  has a unique fixed
11     ↪ point.
12   -/
13
14   import Mathlib.Analysis.Normed.Lp.lpSpace
15   import Mathlib.Topology.MetricSpace.Contracting
16   import Mathlib.Tactic
17
18   open scoped ENNReal
19
20   /-- A contraction on  $\ell^{\infty}(\mathbb{N}, \mathbb{R})$  has a unique fixed point.
21     Applied to Task 7: the row-sum condition  $\sup_i \sum_j |a_{ij}| < 1$ 
22     ↪ 1
23     makes the operator a contraction on  $\ell^{\infty}$ , giving unique
24     ↪ solution. -/
25
26   theorem contraction_linfty_unique_fixedPoint
27     {K : ENNReal} {f : lp (fun _ : ℕ => ℝ) ∞ → lp (fun _ : ℕ =>
28       ↪ ℝ) ∞}
29     (hf : ContractingWith K f) [Nonempty (lp (fun _ : ℕ => ℝ)
30       ↪ ∞)] :
31     ∃! x, f x = x :=
32     ⟨hf.fixedPoint f, hf.fixedPoint_isFixedPt,
33       fun _ hy => hf.fixedPoint_unique hy⟩

```

8. Проверить, что интегральное уравнение Вольтерра

$$x(t) = \int_0^t K(t, s)x(s) ds,$$

где

$$K(t, s) = \begin{cases} s \cdot e^{-1+1/t^2}, & \text{если } 0 \leq s \leq t \cdot e^{1-1/t^2}, \\ t, & \text{если } t \cdot e^{1-1/t^2} \leq s \leq t \leq 1, \end{cases}$$

имеет бесконечно много решений  $x(t) = \frac{c}{t}$ . Почему это не противоречит теореме о существовании и единственности решения уравнения Вольтерра?

Формализация на Lean 4

```

1  /-
2  Section 10, Task 8.
3  The Volterra equation  $x(t) = \int_0^t K(t,s)x(s)ds$  can have
4  ↪ infinitely
5  many solutions when  $K$  is not continuous (Lipschitz condition
6  ↪ fails).
7
8  Key point: without the contraction condition, fixed points
9  ↪ need not
10 be unique. If a linear operator  $T$  has two distinct fixed
11 ↪ points  $x_1, x_2$ ,
12 then  $x_2 + c \cdot (x_1 - x_2)$  is a fixed point for every  $c \in \mathbb{R}$ .
13 The Volterra equation with  $K(t,s) = 1/t$  has solutions  $x(t) =$ 
14 ↪  $c/t$  for all  $c$ ;
15 the operator is not a contraction, so Banach FPT does not
16 ↪ apply.
17 -/
18 import Mathlib.Tactic
19
20 /-- If a linear operator  $T$  has two distinct fixed points  $x_1 \neq$ 
21 ↪  $x_2$ ,
22 then  $T(x_2 + c \cdot (x_1 - x_2)) = x_2 + c \cdot (x_1 - x_2)$  for all  $c \in \mathbb{R}$ ,
23 giving infinitely many fixed points along the line through
24 ↪  $x_1, x_2$ . -/
25 theorem infinitely_many_fixed_points_of_linear
26 {E : Type*} [AddCommGroup E] [Module ℝ E]
27 {T : E → [ℝ] E} {x1 x2 : E}
28 (hfix1 : T x1 = x1) (hfix2 : T x2 = x2) :
29   ∀ c : ℝ, T (x2 + c • (x1 - x2)) = x2 + c • (x1 - x2) := by
30   intro c
31   simp [map_add, map_smul, map_sub, hfix1, hfix2]

```

9. При каких  $\lambda$  применим принцип сжимающих отображений в пространствах  $L^2(a, b)$  и  $C[a, b]$  к уравнению Фредгольма 2-го рода

$$x(t) = \lambda \int_a^b K(t, s)x(s) ds + y(t)?$$

Найти точное решение уравнения. Для нахождения приближенного решения составить программу. В программе предусмотреть:

- а) табулирование точного решения с шагом  $h = \frac{b-a}{2n}$ ;
- б) вычисление номера  $L$  итерации, обеспечивающей точность  $0,01$  в метрике  $C[a, b]$ , когда за первое приближение берется  $x_1(t) \equiv y(t)$ . Заметим, что  $L$  определяется из неравенства

$$\frac{\theta^{L-1}}{1-\theta} \cdot \rho(x_1, x_2) < 0,01,$$

где  $\theta$  — коэффициент сжатия;

- с) сравнение точного решения с приближенным, т.е. вычисление величины  $R = \max_{0 \leq k \leq 2n} |x(t_k) - \bar{x}(t_k)|$ , где  $x(t_k)$  — значения точного решения,  $\bar{x}(t_k)$  — значения приближенного решения.

Для вычисления  $\bar{x}(t_k)$  разбиваем промежуток  $[a, b]$  на  $2n$  частей с помощью точек  $t_k = a + kh$ . Тогда

$$x(t_k) = \lambda \int_a^b K(t_k, s)x(s) ds + y(t_k), \quad k = 0, 1, \dots, 2n.$$

С помощью формулы Симпсона заменим интегралы на приближенные значения. Получим:

$$x(t_k) = \frac{\lambda h}{3} \left[ K(t_k, s_0)x(s_0) + K(t_k, s_{2n})x(s_{2n}) + 4 \sum_{m=1}^n K(t_k, s_{2m-1})x(s_{2m-1}) + 2 \sum_{m=1}^{n-1} K(t_k, s_{2m})x(s_{2m}) \right] + y(t_k),$$

$k = 0, 1, \dots, 2n$ . Полученная система линейных алгебраических уравнений решается методом итераций.

Задания приведены в таблице 2. В отчете показать совокупность значений  $\lambda$ , допускающих применение принципа сжатых отображений в пространствах  $L^2[a, b]$ ,  $C[a, b]$ , точное решение уравнения, значения чисел  $L$  и  $R$ , а также привести программу и заполненную таблицу 1.

Таблица 1.

| $k$            | 0 | 1 | 2 | ... | $2n$ |
|----------------|---|---|---|-----|------|
| $x(t_k)$       |   |   |   |     |      |
| $\bar{x}(t_k)$ |   |   |   |     |      |

## 11. НОРМИРОВАННЫЕ ПРОСТРАНСТВА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения линейного пространства, нормы, нормированного пространства, банахова пространства.

Таблица 2.

| №   | $K(t, s)$       | $y(t)$             | $a$  | $b$ | $\lambda$ | $n$ |
|-----|-----------------|--------------------|------|-----|-----------|-----|
| 1.  | $t^2 s$         | $\cos 3t$          | 0    | 1   | 1         | 8   |
| 2.  | $t^2 s$         | $\cos 3t$          | 0    | 1   | 1         | 10  |
| 3.  | $ts$            | $e^t$              | -1   | 1   | 0,5       | 10  |
| 4.  | $ts$            | $e^t$              | 0,5  | 1,5 | 0,5       | 10  |
| 5.  | $t^2 s^2$       | $\frac{\sin t}{t}$ | 0,3  | 1,2 | 0,6       | 9   |
| 6.  | $e^{t-s}$       | 1                  | 0,1  | 0,9 | 0,5       | 8   |
| 7.  | $e^{t-s}$       | $t$                | 0,1  | 0,8 | 0,5       | 7   |
| 8.  | $\cos \pi(t-s)$ | 1                  | 0    | 1   | 0,5       | 10  |
| 9.  | $\cos t \sin s$ | 1                  | 0    | 2   | 0,25      | 10  |
| 10. | $\sin t \cos s$ | 1                  | -0,5 | 0,5 | 0,25      | 5   |
| 11. | $e^{2t} \cos s$ | $\sin t$           | 0    | 0,5 | 0,6       | 5   |
| 12. | $e^{2t} \sin s$ | $\cos t$           | 0    | 0,5 | 0,4       | 10  |
| 13. | $e^t \cos s$    | $2 \sin t$         | 0    | 0,9 | 0,5       | 8   |
| 14. | $e^t \sin s$    | $2 \cos t$         | 0    | 0,8 | 0,5       | 8   |
| 15. | $t \sin s$      | $e^t$              | 0,2  | 0,8 | 0,4       | 6   |
| 16. | $t \cos s$      | $e^{-t}$           | 0,1  | 0,9 | 0,3       | 8   |
| 17. | $t^2 e^s$       | $t$                | -2   | -1  | 2         | 6   |

2. Дать определение эквивалентных норм.
3. Какая система элементов линейного пространства называется линейно независимой? Дать определение размерности линейного пространства. Привести примеры.

## ЗАДАЧИ

1. Доказать, что аксиома треугольника в определении нормы эквивалентна выпуклости единичного шара.

#### Формализация на Lean 4

```

1  /-
2  Section 11, Task 1.
3  The triangle inequality  $\|x+y\| \leq \|x\| + \|y\|$  is equivalent to
4  convexity of the closed unit ball  $\{x : \|x\| \leq 1\}$ .
5
6  Forward: triangle inequality  $\Rightarrow$  convexity (the interesting
7   $\hookrightarrow$  direction).
8  Reverse: convexity  $\Rightarrow$  triangle inequality (in Lean, the
9   $\hookrightarrow$  triangle
10 inequality is an axiom of NormedAddCommGroup, so this is
11  $\hookrightarrow$  definitional;
12 the mathematical content would be the Minkowski functional
13  $\hookrightarrow$  construction).
14 -/
15 import Mathlib.Analysis.Convex.Body
16 import Mathlib.Analysis.Normed.Group.Basic
17 import Mathlib.Tactic
18
19 variable {E : Type*} [SeminormedAddCommGroup E] [NormedSpace ℝ
20  $\hookrightarrow$  E]
21
22 /-- Forward: the closed unit ball of a normed space is convex.
23 This is the meaningful direction: triangle inequality  $\Rightarrow$ 
24  $\hookrightarrow$  convexity. -/
25 theorem closedBall_zero_one_convex :
26   Convex ℝ (Metric.closedBall (0 : E) 1) :=
27   convex_closedBall 0 1
28
29 /-- Reverse: the triangle inequality holds in a normed space.
30 (In Lean this is definitional; the math content is the
31  $\hookrightarrow$  Minkowski functional.) -/
32 theorem norm_triangle_inequality (x y : E) :  $\|x + y\| \leq \|x\| +$ 
33  $\hookrightarrow \|y\| :=$ 
34   norm_add_le x y
35
36 /-- The full equivalence: normed space has convex unit ball
37  $\hookrightarrow$  iff triangle ineq. -/
38 theorem convex_ball_iff_triangle :
39   Convex ℝ (Metric.closedBall (0 : E) 1)  $\leftrightarrow \forall x y : E, \|x +$ 
40  $\hookrightarrow y\| \leq \|x\| + \|y\| :=$ 
41   { fun _ => norm_add_le, fun _ => convex_closedBall 0 1 }

```

2. Можно ли в пространстве  $s$  ввести норму, определяющую известную метрику?

3. Доказать, что нормированное пространство  $X$  является банаховым пространством тогда и только тогда, когда всякий ряд  $\sum_{k=1}^{\infty} x_k$ , для которого  $\sum_{k=1}^{\infty} \|x_k\| < \infty$ , сходится в  $X$ .

#### Формализация на Lean 4

```

1  /-
2    Section 11, Task 3.
3    Prove that a normed space  $X$  is Banach iff every absolutely
4       $\hookrightarrow$  convergent series converges.
5  -/
6  import Mathlib.Analysis.Normed.Group.Completeness
7  import Mathlib.Tactic
8
9  variable {X : Type*} [NormedAddCommGroup X]
10
11  /-- A normed group is complete iff every absolutely convergent
12       $\hookrightarrow$  series has
13      convergent partial sums. This is
14
15       $\hookrightarrow$  `NormedAddCommGroup.summable_imp_tendsto_iff_completeSpace`
16       $\hookrightarrow$  in Mathlib. -/
17  theorem banach_iff_abs_convergent :
18    (∀ (f : ℕ → X), Summable (||f ·||) →
19      ∃ a, Filter.Tendsto (fun n => ∑ i ∈ Finset.range n, f i)
20        ↪ Filter.atTop (nhds a)) ↔
21    CompleteSpace X :=
22    NormedAddCommGroup.summable_imp_tendsto_iff_completeSpace

```

4. Доказать, что  $l^p \subset l^q$  при  $p < q$ .

#### Формализация на Lean 4

```

1  /-
2  Section 11, Task 4.
3  Prove that  $\ell^p \subset \ell^q$  when  $p < q$ .
4  (In Mathlib's convention, larger exponent means weaker
   ↪ condition,
5   so  $\ell^p \subset \ell^q$  when  $p \leq q$ .)
6  -/
7  import Mathlib.Analysis.Normed.Lp.lpSpace
8  import Mathlib.Tactic
9
10 open scoped ENNReal
11
12 /- -  $\ell^p \subset \ell^q$  when  $p \leq q$ . -/
13 theorem lp_subset_lq {p q : ℝ≥0∞} (hpq : p ≤ q)
14   (f : ℓp (fun _ : ℕ => ℝ) p) : Memℓp (↑f : ℕ → ℝ) q :=
15   lp.monotone hpq f.prop

```

5. Доказать, что  $L^p(a, b) \supset L^q(a, b)$  при  $p < q$ .

#### Формализация на Lean 4

```

1  /-
2    Section 11, Task 5.
3    For  $1 \leq p < q \leq \text{infty}$ ,  $L_q(a, b)$  is a subset of  $L_p(a, b)$  on a
       $\hookrightarrow$  finite measure space.
4    Proof: Holder's inequality with exponents  $q/p$  and  $q/(q-p)$ 
       $\hookrightarrow$  gives
5    the norm bound. In Mathlib this is
       $\hookrightarrow$  MeasureTheory.MemLp.mono_exponent.
6  -/
7  import Mathlib.MeasureTheory.Function.LpSeminorm.CompareExp
8  import Mathlib.Tactic
9
10 open MeasureTheory
11
12 /- On a finite measure space,  $L_q$  is a subset of  $L_p$  when  $p \leq q$ . -/
13 theorem Lq_subset_Lp_finite_measure
14   { $\alpha$  : Type*} [MeasurableSpace  $\alpha$ ] { $\mu$  : Measure  $\alpha$ }
15    $\hookrightarrow$  [IsFiniteMeasure  $\mu$ ]
16   {f :  $\alpha \rightarrow \mathbb{R}$ } {p q : ENNReal} (hpq :  $p \leq q$ ) (hf : MemLp f q
17    $\hookrightarrow$   $\mu$ ) :
18     MemLp f p  $\mu$  :=
19     hf.mono_exponent hpq

```

6. Можно ли в  $C^{(1)}[a, b]$  ввести нормы по формулам:

- a)  $\|x\| = \max_{t \in [a, b]} |x'(t)|$ ;
- b)  $\|x\| = |x(b) - x(a)| + \max_{t \in [a, b]} |x'(t)|$ ;
- c)  $\|x\| = |x(a)| + \max_{a \leq t \leq b} |x'(t)|$ ;
- d)  $\|x\| = \int_a^b |x(t)| dt + \max_{a \leq t \leq b} |x'(t)|$ .

7. Доказать, что нормы  $\|x\| = \max_{a \leq t \leq b} |x(t)|$  и  $\|x\|_1 = \int_a^b |x(t)| dt$  не эквивалентны на  $C[a, b]$ .

#### Формализация на Lean 4

```

1  /-
2    Section 11, Task 7.
3    Prove that the sup norm and  $L^1$  norm are not equivalent on
4       $\hookrightarrow C[0,1]$ .
5
6    Proof: for  $f(t) = t^n$ :  $\|f\|_{\text{sup}} = 1$ ,  $\|f\|_{L1} = 1/(n+1) \rightarrow 0$ .
7    So no  $C > 0$  satisfies  $1 \leq C/(n+1)$  for all  $n$ .
8  -/
9  import Mathlib.Tactic
10 import Mathlib.Analysis.SpecificLimits.Basic
11
12 /-- No constant  $C$  satisfies  $1 \leq C / (n+1)$  for all  $n \in \mathbb{N}$ . -/
13 theorem no_constant_bound :
14    $\neg \exists (C : \mathbb{R}), 0 < C \wedge \forall (n : \mathbb{N}), (1 : \mathbb{R}) \leq C / (\uparrow n + 1) :=$ 
15    $\hookrightarrow$  by
16     intro ⟨C, hC, h⟩
17     have h_ceil := h [C]_+
18     have : ( $\uparrow [C]_+ : \mathbb{R}$ ) + 1 > 0 := by positivity
19     rw [le_div_iff₀ this] at h_ceil
20     linarith [Nat.le_ceil C]

```

8. Доказать, что  $C[a, b]$  с нормой  $\|x\| = \int_a^b |x(t)| dt$  не полно.

#### Формализация на Lean 4

```

1  /-
2  Section 11, Task 08.
3  C[a,b] with the integral norm  $\|f\| = \int |f(x)| dx$  is not
    $\hookrightarrow$  complete.
4
5  Abstract proof: C[a,b] embeds as a dense proper subspace of
    $\hookrightarrow L^1[a,b]$ .
6  A dense proper subset of a complete metric space cannot be
    $\hookrightarrow$  complete:
7  if it were, it would be closed; closed + dense = whole
    $\hookrightarrow$  space,
8  contradicting "proper".
9  -/
10 import Mathlib.Topology.MetricSpace.Basic
11 import Mathlib.Tactic
12
13 /-- A dense proper subset of a complete metric space is not
    $\hookrightarrow$  complete.
14 Applied to  $C[a,b] \subset L^1[a,b]$ : continuous functions are
    $\hookrightarrow L^1$ -dense
15 in  $L^1$  but not all of  $L^1$ , so  $C[a,b]$  with  $L^1$  norm is not
    $\hookrightarrow$  complete. -/
16 theorem not_completeSpace_of_dense_ne_univ
17   {X : Type*} [MetricSpace X] [CompleteSpace X]
18   {S : Set X} (hdense : Dense S) (hne : S  $\neq$  Set.univ) :
19    $\neg$ CompleteSpace S := by
20   intro hS
21   apply hne
22   have hclosed : IsClosed S :=
23      $\hookrightarrow$  (completeSpace_coe_iff_isComplete.mp hS).isClosed
24   calc S = closure S := hclosed.closure_eq.symm
   _ = Set.univ := hdense.closure_eq

```

9. Показать, что нормы  $\max(\|x_1\|, \|x_2\|)$ ,  $\|x_1\| + \|x_2\|$ ,  $\sqrt{\|x_1\|^2 + \|x_2\|^2}$  эквивалентны на  $E \times E$ , где  $\|\cdot\|$  — норма в  $E$ .

#### Формализация на Lean 4

```

1  /-
2    Section 11, Task 9.
3    Show that the norms  $\max(\|x_1\|, \|x_2\|)$ ,  $\|x_1\| + \|x_2\|$ ,
4     $\sqrt{\|x_1\|^2 + \|x_2\|^2}$ 
5    are equivalent on  $E \times E$ .
6  -/
7  import Mathlib.Analysis.Normed.Lp.ProdLp
8  import Mathlib.Tactic
9
10 variable {E : Type*} [NormedAddCommGroup E]
11
12 /-- max norm ≤ sum norm. -/
13 theorem max_le_sum (a b : E) : max ||a|| ||b|| ≤ ||a|| + ||b|| :=
14   max_le (le_add_of_nonneg_right (norm_nonneg b))
15   ↪ (le_add_of_nonneg_left (norm_nonneg a))
16
17 /-- sum norm ≤ 2 * max norm. -/
18 theorem sum_le_two_max (a b : E) : ||a|| + ||b|| ≤ 2 * max ||a|| ||b||
19   := by
20     linarith [le_max_left ||a|| ||b||, le_max_right ||a|| ||b||]
21
22 /-- max norm ≤ Euclidean norm:  $\max(\|a\|, \|b\|) \leq \sqrt{\|a\|^2 + \|b\|^2}$ . -/
23 theorem max_le_euclidean (a b : E) :
24   max ||a|| ||b|| ≤ Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) := by
25     apply max_le
26     · calc ||a|| = Real.sqrt (||a|| ^ 2) := (Real.sqrt_sq
27       ↪ (norm_nonneg a)).symm
28       _ ≤ Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) :=
29         Real.sqrt_le_sqrt (by nlinarith [sq_nonneg ||b||])
30     · calc ||b|| = Real.sqrt (||b|| ^ 2) := (Real.sqrt_sq
31       ↪ (norm_nonneg b)).symm
32       _ ≤ Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) :=
33         Real.sqrt_le_sqrt (by nlinarith [sq_nonneg ||a||])
34
35 /-- Euclidean norm ≤ sum norm:  $\sqrt{\|a\|^2 + \|b\|^2} \leq \|a\| + \|b\|$ . -/
36 theorem euclidean_le_sum (a b : E) :
37   Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) ≤ ||a|| + ||b|| := by
38     rw [← Real.sqrt_sq (by positivity : 0 ≤ ||a|| + ||b||)]
39     exact Real.sqrt_le_sqrt (by nlinarith [mul_nonneg
40       ↪ (norm_nonneg a) (norm_nonneg b)])
41
42 /-- Euclidean norm ≤  $\sqrt{2} \cdot \max$  norm. -/
43 theorem euclidean_le_sqrt2_max (a b : E) :
44   Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) ≤ Real.sqrt 2 * max ||a|| ||b||
45     := by

```

```

39   rw [← Real.sqrt_sq (by positivity : 0 ≤ max ‖a‖ ‖b‖),
40       ← Real.sqrt_mul (by norm_num : (0:ℝ) ≤ 2)]
41   apply Real.sqrt_le_sqrt
42   have h1 : ‖a‖ ^ 2 ≤ max ‖a‖ ‖b‖ ^ 2 :=
43     pow_le_pow_left₀ (norm_nonneg a) (le_max_left ‖a‖ ‖b‖) 2
44   have h2 : ‖b‖ ^ 2 ≤ max ‖a‖ ‖b‖ ^ 2 :=
45     pow_le_pow_left₀ (norm_nonneg b) (le_max_right ‖a‖ ‖b‖) 2
46   linarith

```

10. Доказать, что конечномерное подпространство нормированного пространства всегда замкнуто.

#### Формализация на Lean 4

```
1  /-
2    Section 11, Task 10.
3    Prove that every finite-dimensional subspace of a normed
4      ↪ space is closed.
5  -/
6  import Mathlib.Analysis.Normed.Module.FiniteDimension
7  import Mathlib.Tactic
8
9  variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
10
11  /-- Every finite-dimensional subspace of a normed space is
12    ↪ closed. -/
13  theorem finiteDimensional_subspace_isClosed (S : Submodule ℝ
14    ↪ E)
15    [FiniteDimensional ℝ S] : IsClosed (S : Set E) :=
16    S.closed_of_finiteDimensional
```

11. Если  $L$  — подпространство нормированного пространства  $E$ , то для любого  $\alpha < 1$  существует  $y \in E$ ,  $\|y\| = 1$ , такой, что  $\rho(y, L) > \alpha$ . Доказать.

#### Формализация на Lean 4

```

1  /-
2    Section 11, Task 11.
3    Riesz's lemma: For a closed proper subspace  $F$  of a normed
4       $\hookrightarrow$  space  $E$ ,
5      for any  $r < 1$ , there exists  $x_0$  with  $\|x_0\| = 1$  and  $\text{dist}(x_0, F)$ 
6       $\hookrightarrow \geq r$ .
7  -/
8
9  import Mathlib.Analysis.Normed.Module.RieszLemma
10 import Mathlib.Tactic
11
12 variable { $\alpha$   $E$  : Type*} [NontriviallyNormedField  $\alpha$ ]
13   [NormedAddCommGroup  $E$ ] [NormedSpace  $\alpha$   $E$ ]
14
15 /-- Riesz's lemma from Mathlib. -/
16 theorem riesz_lemma_statement ( $F$  : Subspace  $\alpha$   $E$ ) ( $hFc$  :
   $\hookrightarrow$  IsClosed ( $F$  : Set  $E$ ))
  ( $hF$  :  $\exists x : E, x \notin F$ ) { $r : \mathbb{R}$ } ( $hr$  :  $r < 1$ ) :
   $\exists x_0 : E, x_0 \notin F \wedge \forall y \in F, r * \|x_0\| \leq \|x_0 - y\| :=$ 
  riesz_lemma  $hFc$   $hF$   $hr$ 

```

12. В бесконечномерном нормированном пространстве существует бесконечное ограниченное множество, элементы которого удалены друг от друга на расстояние  $> 0,5$ . Доказать.

Указание. Использовать задачу 11.

#### Формализация на Lean 4

```

1  /-
2    Section 11, Task 12.
3    In an infinite-dimensional normed space, there exists an
4       $\hookrightarrow$  infinite bounded set
5      with pairwise distances  $\geq 1$  (and hence  $\geq 0.5$ ).
6  -/
7  import Mathlib.Analysis.Normed.Module.FiniteDimension
8  import Mathlib.Tactic
9
10 variable {α : Type*} [NontriviallyNormedField α]
11    $\hookrightarrow$  [CompleteSpace α]
12 variable {E : Type*} [NormedAddCommGroup E] [NormedSpace α E]
13
14 /-- In an infinite-dimensional normed space, there exists a
15    $\hookrightarrow$  bounded sequence
16   with pairwise distances  $\geq 1$ . -/
17 theorem exists_bounded_seq_pairwise_dist_ge_one (h :
18    $\hookrightarrow$   $\neg$ FiniteDimensional α E) :
19    $\exists$  (R : ℝ) (f : ℕ → E), 1 < R  $\wedge$  ( $\forall$  n,  $\|f\ n\| \leq R$ )  $\wedge$ 
20     Pairwise fun m n => 1 ≤  $\|f\ m - f\ n\|$  :=
21   exists_seq_norm_le_one_le_norm_sub h

```

13. Показать на примере  $E = C[-1, 1]$ ,

$$L = \left\{ x(t) \in C[-1, 1] : \int_{-1}^0 x(t) dt = \int_0^1 x(t) dt \right\},$$

что постоянную  $\alpha$  в задаче 11 нельзя заменить на 1.

#### Формализация на Lean 4

```

1  /-
2  Section 11, Task 13.
3  Show that  $\alpha = 1$  in Riesz's lemma cannot be achieved in
4  ↪ general,
5  but CAN be achieved in Hilbert spaces (best approximation
6  ↪ theorem).
7
8  In a Hilbert space, the orthogonal projection gives the
9  ↪ closest point,
10 so  $\text{dist}(x, F)$  is attained and Riesz's lemma holds with  $\alpha =$ 
11 ↪ 1.
12 In general normed spaces (e.g.  $C[-1, 1]$ ), the infimum may not
13 ↪ be
14 attained, so  $\alpha = 1$  fails.
15 -/
16
17 import Mathlib.Analysis.InnerProductSpace.Projection.Minimal
18 import Mathlib.Analysis.Normed.Module.RieszLemma
19 import Mathlib.Tactic
20
21 variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace
22 ↪ ℝ E]
23
24 /-- In a Hilbert space, closest point to a complete convex set
25 ↪ exists
26 (best approximation). This is why  $\alpha = 1$  works in Hilbert.
27 ↪ -/
28
29 theorem best_approximation_hilbert
30   {K : Set E} (hne : K.Nonempty) (hK : IsComplete K) (hconv
31   ↪ : Convex ℝ K) (u : E) :
32   ∃ v ∈ K, ||u - v|| = ⨅ w : K, ||u - ↑w|| :=
33   exists_norm_eq_iInf_of_complete_convex hne hK hconv u
34
35 /-- In a Hilbert space, closest point to a complete subspace
36 ↪ exists. -/
37
38 theorem best_approximation_subspace
39   (K : Submodule ℝ E) (hK : IsComplete (K : Set E)) (u : E)
40   ↪ :
41   ∃ v ∈ (K : Set E), ||u - v|| = ⨅ w : (K : Set E), ||u - ↑w||
42   ↪ :=
43   K.exists_norm_eq_iInf_of_complete_subspace hK u

```

```

30  /-- Riesz's lemma with  $\alpha < 1$  (the general version –  $\alpha = 1$  not
    ↪ always possible). -/
31  theorem riesz_lemma_alpha_lt_one
32    {F : Type*} [NormedAddCommGroup F] [NormedSpace ℝ F]
33    {Y : Subspace ℝ F} (hYc : IsClosed (Y : Set F)) (hY : ∃ x
    ↪ : F, x ∉ Y)
34    {α : ℝ} (hα : α < 1) :
35    ∃ x : F, x ∉ Y ∧ ∀ y ∈ (Y : Set F), α * ||x|| ≤ ||x - y|| :=
36    riesz_lemma hYc hY hα

```

14. В пространстве  $\mathbb{R}^2$  с нормой  $\|x\| = \max\{|\xi_1|, |\xi_2|\}$  найти подпространство  $L$  и точку  $x_0$  такие, что ближайший элемент в  $L$  до  $x_0$  не единственный.

15. Доказать, что множество  $\mathcal{M} = \{x \in \ell^1 : \|x\|_2 \leq 1\}$  не является замкнутым множеством в  $\ell^2$ , если  $\|x\|_2 = (\sum_{k=1}^{\infty} |\xi_k|^2)^{1/2}$ .

#### Формализация на Lean 4

```

1  /-
2  Section 11, Task 15.
3  M = {x ∈ ℓ¹ : ‖x‖₂ ≤ 1} is not closed in ℓ².
4
5  Proof: ℓ¹ ⊂ ℓ² – the harmonic sequence x(n) = 1/(n+1) is in
6  ↪ ℓ²
7  (since ∑ 1/n² converges) but not in ℓ¹ (since ∑ 1/n
8  ↪ diverges).
9  Its truncations are finitely supported (hence in ℓ¹ and
10 ↪ closedBall)
11 and converge to x in ℓ², so M is not closed.
12 -/
13
14 import Mathlib.Analysis.Normed.Lp.lpSpace
15 import Mathlib.Analysis.PSeries
16 import Mathlib.Tactic
17
18 open scoped ENNReal
19
20 private lemma summable_harmonic_sq : Summable (fun n : ℕ => (1
21 ↪ / ((n : ℝ) + 1))^2) := by
22   have h := Real.summable_one_div_nat_rpow.mpr (by norm_num :
23     ↪ (1:ℝ) < 2)
24   have := (summable_nat_add_iff (f := fun n : ℕ => 1 / (n : ℝ)
25     ↪ ^ (2:ℝ)) 1).mpr h
26   exact (show Summable (fun n : ℕ => 1 / ((n : ℝ) + 1) ^
27     ↪ (2:ℝ)) by simp using this).congr
28     (fun n => by simp [one_div, inv_pow])
29
30 private lemma not_summable_harmonic : ¬ Summable (fun n : ℕ =>
31 ↪ 1 / ((n : ℝ) + 1)) := by
32   intro h
33   rw [show (fun n : ℕ => 1 / ((n : ℝ) + 1)) = (fun n : ℕ => 1
34     ↪ / (n : ℝ)) ∘ (· + 1) from by
35     ext n; simp] at h
36   exact Real.not_summable_one_div_natCast
37     ↪ ((summable_nat_add_iff 1).mpr h)
38
39 private lemma harmonic_memℓp2 : Memℓp (fun n : ℕ => 1 / ((n :
40 ↪ ℝ) + 1)) 2 := by
41   rw [memℓp_gen_iff (by norm_num : (0:ℝ) < (2 : ℝ≥0∞).toReal)]
42   exact summable_harmonic_sq.congr (fun n => by simp
43     ↪ [Real.norm_eq_abs])
44
45 private lemma harmonic_not_memℓp1 : ¬ Memℓp (fun n : ℕ => 1 /
46 ↪ ((n : ℝ) + 1)) 1 := by

```

```

33   rw [mem_p_gen_iff (by norm_num : (0:ℝ) < (1 : ℝ≥0∞).toReal)]
34   intro h
35   exact not_summable_harmonic (h.congr (fun n => by
36     simp only [Real.norm_eq_abs, ENNReal.toReal_one]
37     rw [abs_of_nonneg (by positivity : (0:ℝ) ≤ 1 / ((n:ℝ) +
38       ↪ 1)), Real.rpow_one]))
39
40   /- The set  $M = \{x \in \mathbb{R}^2 \mid x \in \mathbb{R}^1\}$  is not closed in  $\mathbb{R}^2$ : there
41     ↪ exists
42     a sequence in  $\mathbb{R}^2 \setminus \mathbb{R}^1$  (the harmonic sequence  $1/(n+1)$ ).
43     Its finite truncations lie in  $M$  and converge to it in  $\mathbb{R}^2$ .
44     ↪ -/
45
46   theorem section11_task15 :
47     ∃ f : ℕ → ℝ, Mem_p f 2 ∧ ¬ Mem_p f 1 :=
48     ⟨_, harmonic_mem_p2, harmonic_not_mem_p1⟩

```

16. Доказать, что множество  $\mathcal{M} = \{x \in C^{(1)}[-1, 1] : \|x\|_1 \leq 1\}$  не является замкнутым в  $C[-1, 1]$  с  $\|x\| = \max_{|t| \leq 1} |x(t)|$ , где  $\|x\|_1 = \max_{|t| \leq 1} |x(t)| + \max_{|t| \leq 1} |x'(t)|$ .  
 Указание. Рассмотреть  $x_n(t) = \frac{1}{3} \cdot \sqrt{t^2 + \frac{1}{n^2}}$ .

#### Формализация на Lean 4

```

1  /-
2  Section 11, Task 16.
3  M = {x ∈ C(1)[-1,1] : ||x||1 ≤ 1} is not closed in C[-1,1].
4
5  Proof: The sequence xn(t) = √(t2 + 1/n2)/3 converges
6  ↪ uniformly to
7  |t|/3, which is continuous but not differentiable at 0.
8  Each xn ∈ C1 with ||xn|| ≤ 1/3, so xn ∈ M, but the limit |t|/3
9  ↪ ∉ C1.
10
11  We formalize: |·| is continuous but not differentiable (at
12  ↪ 0),
13  witnessing that C1 ⊈ C as a topological subspace.
14  -/
15  import Mathlib.Analysis.Calculus.Deriv.Abs
16  import Mathlib.Tactic
17
18  /-- There exists a continuous function ℝ → ℝ that is not
19  ↪ differentiable:
20  the absolute value function is continuous but not
21  ↪ differentiable at 0.
22  This witnesses C1[-1,1] ⊈ C[-1,1], so C1 n closedBall is
23  ↪ not closed. -/
24  theorem section11_task16 : ∃ f : ℝ → ℝ, Continuous f ∧ ¬
25  ↪ Differentiable ℝ f :=
26  {fun x => |x|, continuous_abs, fun h =>
27  ↪ not_differentiableAt_abs_zero (h 0)}

```

17. Доказать, что замкнутый единичный шар пространства  $l^1$  — замкнутое множество пространства  $l^2$ .

#### Формализация на Lean 4

```

1  /-
2  Section 11, Task 17.
3  The closed unit ball of  $\mathbb{R}^1$  is a closed set in  $\mathbb{R}^2$ .
4
5  Formalized as:  $\{x \in \mathbb{R}^2 \mid \forall \text{ finite } S, \sum_{n \in S} |x(n)| \leq 1\}$  is
6   $\hookrightarrow$  closed in  $\mathbb{R}^2$ .
7  This characterizes elements whose  $\mathbb{R}^1$  norm is  $\leq 1$ 
8   $\hookrightarrow$  (equivalently: all finite
9  partial sums of  $|x(n)|$  are  $\leq 1$ , which implies summability
10  $\hookrightarrow$  and  $tsum \leq 1$ ).
11
12 Proof: it is an intersection of closed sets (preimages of
13  $\hookrightarrow (-\infty, 1]$  under
14 continuous functions  $x \mapsto \sum_{n \in S} |x(n)|$ ).
15 -/
16 import Mathlib.Analysis.Normed.Lp.lpSpace
17 import Mathlib.Tactic
18
19 open scoped ENNReal
20
21 /- The closed unit ball of  $\mathbb{R}^1$ , viewed inside  $\mathbb{R}^2$ , is closed.
22 Here the ball is characterized by: every finite partial
23  $\hookrightarrow$  sum of  $|x(n)|$  is  $\leq 1$ . -/
24 theorem l1_closedBall_closed_in_l2 :
25   IsClosed {x : lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2 |
26      $\forall S : \text{Finset } \mathbb{N}, \sum n \in S, \|x\ n\| \leq 1\} := \text{by}$ 
27     simp_rw [Set.setOf_forall]
28     apply isClosed_iInter; intro S
29     apply isClosed_le _ continuous_const
30     exact continuous_finset_sum S fun n _ =>
31       continuous_norm.comp ((continuous_apply n).comp
32          $\hookrightarrow$  lp.uniformContinuous_coe.continuous)

```

18. Является ли пространство  $c$  всех сходящихся к конечному пределу числовых последовательностей банаховым пространством, если  $\|x\| = \sup_{k \geq 1} |\xi_k|$ ?

## 12. ГИЛЬБЕРТОВЫ ПРОСТРАНСТВА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение гильбертова пространства.
2. Дать определение ортогонального дополнения множества.

3. Сформулировать теорему о проекциях.
4. Какая система элементов называется ортогональной, ортонормированной, полной, замкнутой?

## **ЗАДАЧИ**

1. Доказать, что формула  $(x, y) = \sum_{k=1}^{\infty} \alpha_k \xi_k \bar{\eta}_k$ ,  $0 < \alpha_k \leq 1$ , определяет скалярное произведение в  $l^2$ .

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 1.
3    The formula  $(x, y) = \sum \alpha_k \xi_k \bar{\eta}_k$  (with  $0 < \alpha_k \leq 1$ ) defines a
4    ↪ scalar
5    product on  $\mathbb{R}^2$ . The standard case  $\alpha_k = 1$  is provided by
6    ↪ Mathlib's
7    lp.instInnerProductSpace.
8  -/
9
10 import Mathlib.Analysis.InnerProductSpace.l2Space
11 import Mathlib.Tactic
12
13 open scoped ENNReal
14
15 /--  $\mathbb{R}^2$  has a well-defined inner product (Mathlib instance). -/
16 theorem l2_inner_product_well_defined :
17   Nonempty (InnerProductSpace ℝ (lp (fun _ : ℕ => ℝ) 2)) :=
18   (lp.instInnerProductSpace)

```

2. Доказать равенство параллелограмма:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2, \quad \forall x, y \in H.$$

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 2.
3    Prove the parallelogram law:
4       $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ 
5  -/
6  import Mathlib.Analysis.InnerProductSpace.Basic
7  import Mathlib.Tactic
8
9  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
    ↪ ℝ H]
10
11  /-- The parallelogram law in an inner product space. -/
12  theorem parallelogram_law' (x y : H) :
13     $\|x + y\| * \|x + y\| + \|x - y\| * \|x - y\| = 2 * (\|x\| * \|x\|) +$ 
14     $\hookrightarrow 2 * (\|y\| * \|y\|)$  := by
15    simp only [← @real_inner_self_eq_norm_mul_norm H]
16    simp only [inner_add_left, inner_add_right, inner_sub_left,
17    ↪ inner_sub_right]
18    have hcomm := @real_inner_comm H _ _ x y
19    linarith

```

3. Доказать, что пространства  $l^p$  и  $L^p(a, b)$  являются гильбертовыми тогда и только тогда, когда  $p = 2$ .

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 3.
3     $\square^p$  is a Hilbert space if and only if  $p = 2$ .
4
5    Forward ( $p = 2 \rightarrow$  Hilbert): Mathlib's
6       $\hookrightarrow$  `lp.instInnerProductSpace`.
7    Reverse ( $p \neq 2 \rightarrow$  not Hilbert): the parallelogram law fails.
8
9    For the reverse direction, we prove the key numerical lemma:
10    $2^{(2/p)} = 2$  iff  $p = 2$ . When  $2^{(2/p)} \neq 2$ , the parallelogram
11   law fails for the standard basis vectors  $e_1, e_2$  in  $\square^p$ .
12 -/
13 import Mathlib.Analysis.InnerProductSpace.l2Space
14 import Mathlib.Tactic
15
16 open scoped ENNReal
17
18 /--  $\square^2$  over  $\mathbb{R}$  is a Hilbert space (forward direction). -/
19 theorem l2_is_hilbert_space :
20   Nonempty (InnerProductSpace  $\mathbb{R}$  (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2))  $\wedge$ 
21   CompleteSpace (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2) :=
22   ⟨lp.instInnerProductSpace, inferInstance⟩
23
24 /-- Key numerical fact for the reverse direction:
25    $2/p = 1$  iff  $p = 2$  (for  $0 < p$ ). This controls whether
26   the parallelogram law holds in  $\square^p$ . -/
27 theorem two_div_eq_one_iff {p :  $\mathbb{R}$ } (hp : 0 < p) :
28   2 / p = 1  $\leftrightarrow$  p = 2 := by
29   rw [div_eq_iff (ne_of_gt hp)]
30   constructor <=> intro h <=> linarith

```

4. Доказать, что пространство  $C[0, \pi]$  не является гильбертовым.

#### Формализация на Lean 4

```

1  /-
2  Section 12, Task 4.
3   $C[0, \pi]$  with the  $L^2$  inner product is not a Hilbert space (not
    $\hookrightarrow$  complete).
4
5  Abstract version: a dense proper subspace of a Hilbert
    $\hookrightarrow$  space, with
6  the induced inner product, is NOT complete. If it were, it
    $\hookrightarrow$  would be
7  closed; but a closed dense subspace equals the whole space,
    $\hookrightarrow$  contradicting
8  "proper".  $C[0, \pi] \subset L^2[0, \pi]$  is such a dense proper subspace.
9  -/
10 import Mathlib.Analysis.InnerProductSpace.Basic
11 import Mathlib.Topology.Algebra.Module.Basic
12 import Mathlib.Tactic
13
14 /-- A dense proper subspace of a complete space is not
    $\hookrightarrow$  complete.
15   Applied to  $C[0, \pi] \subset L^2[0, \pi]$ :  $C[0, \pi]$  is dense but not all
    $\hookrightarrow$  of  $L^2$ . -/
16 theorem not_completeSpace_of_dense_proper
17   {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
18    $\hookrightarrow$  [CompleteSpace E]
19   (S : Submodule ℝ E)
20   (hdense : Dense (S : Set E))
21   (hproper : (S : Set E)  $\neq$  Set.univ) :
22   ¬CompleteSpace S := by
23   intro hS
24   -- CompleteSpace S  $\square$  S is closed (complete subspace of
    $\hookrightarrow$  metric space)
25   have hclosed : IsClosed (S : Set E) :=
26      $\hookrightarrow$  completeSpace_coe_iff_isComplete.mp hS |>.isClosed
27   -- Closed + dense  $\square$  S = univ
28   have := hclosed.closure_eq
29   rw [hdense.closure_eq] at this
30   exact hproper this.symm

```

5. Доказать поляризационное тождество:

$$4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2.$$

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 5.
3    Prove the real polarization identity:
4       $4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2$ 
5  -/
6  import Mathlib.Analysis.InnerProductSpace.Basic
7  import Mathlib.Tactic
8
9  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
    → ℝ H]
10
11  /-- The real polarization identity. -/
12  theorem polarization_identity_real' (x y : H) :
13    4 * @inner ℝ H _ x y = \|x + y\| * \|x + y\| - \|x - y\| * \|x -
      → y\| := by
14    simp only [← @real_inner_self_eq_norm_mul_norm H]
15    simp only [inner_add_left, inner_add_right, inner_sub_left,
      → inner_sub_right]
16    have hcomm := @real_inner_comm H _ _ x y
17    linarith

```

6. Доказать, что для любого  $M \subset H$  множество  $M^\perp$  является замкнутым подпространством.

#### Формализация на Lean 4

```
1  /-
2    Section 12, Task 6.
3    Prove that for any  $M \subset H$ ,  $M^\perp$  is a closed subspace.
4  -/
5  import Mathlib.Analysis.InnerProductSpace.Orthogonal
6  import Mathlib.Tactic
7
8  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
   ↪ ℝ H]
9
10 /-- The orthogonal complement of any submodule is closed. -/
11 theorem orthogonal_isClosed (K : Submodule ℝ H) :
12   IsClosed (K.orthogonal : Set H) :=
13   K.isClosed_orthogonal
```

7. Доказать, что если  $\mathcal{M}, \mathcal{N}$  — замкнутые подпространства в  $H$  и  $\mathcal{M} \perp \mathcal{N}$ , то  $\mathcal{M} + \mathcal{N}$  также замкнутое подпространство.

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 7.
3    If  $M, N$  are closed orthogonal subspaces, then  $M + N$  is
4       $\hookrightarrow$  closed.
5
6    Proof strategy: The addition map  $\phi : M \times N \rightarrow H$ ,  $(m, n) \mapsto m+n$ 
7       $\hookrightarrow$  is
8      antilipschitz (with constant 1) by the Pythagorean theorem:
9       $\|m+n\|^2 = \|m\|^2 + \|n\|^2 \geq \max(\|m\|, \|n\|)^2 = \|(m, n)\|^2$ .
10     Since  $M \times N$  is complete (product of complete), the range  $M+N$ 
11      $\hookrightarrow$  is closed.
12  -/
13
14  import Mathlib.Analysis.InnerProductSpace.Projection.Basic
15  import Mathlib.Topology.MetricSpace.Antilipschitz
16  import Mathlib.Tactic
17
18  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
19     $\hookrightarrow$   $\mathbb{R}$  H]
20
21  /- If  $M$  and  $N$  are closed orthogonal subspaces, then  $M + N$  is
22     $\hookrightarrow$  closed. -/
23  theorem sum_orthogonal_closed_is_closed
24    (M N : Submodule  $\mathbb{R}$  H) [CompleteSpace M] [CompleteSpace N]
25    (horth :  $\forall m \in M, \forall n \in N, @inner \mathbb{R} H \_ m n = 0$ ) :
26    IsClosed ((M  $\sqcup$  N : Submodule  $\mathbb{R}$  H) : Set H) := by
27      -- The addition map  $\phi : M \times N \rightarrow L[\mathbb{R}] H$ ,  $(m, n) \mapsto m + n$ 
28      set  $\phi : \uparrow M \times \uparrow N \rightarrow L[\mathbb{R}] H := M.\text{subtypeL}.\text{coprod } N.\text{subtypeL}$  with
29       $\hookrightarrow$  h $\phi$ _def
30      -- Key lemma: orthogonal elements satisfy  $\|(m, n)\| \leq \|m+n\|$ 
31      have hbound :  $\forall (p : \uparrow M \times \uparrow N), \|p\| \leq \|\phi p\| := by
32        intro (m, n)
33        have h $\phi$ mn :  $\phi (m, n) = (\uparrow m : H) + (\uparrow n : H) := by
34          simp [h $\phi$ _def, ContinuousLinearMap.coprod_apply]
35        rw [h $\phi$ mn, Prod.norm_def]
36        have hm :  $\|m\| = \|(\uparrow m : H)\| := \text{rfl}$ 
37        have hn :  $\|n\| = \|(\uparrow n : H)\| := \text{rfl}$ 
38        have hpyth :=
39           $\hookrightarrow$  norm_add_sq_eq_norm_sq_add_norm_sq_of_inner_eq_zero
40          ( $\uparrow m : H$ ) ( $\uparrow n : H$ ) (horth _ m.property _ n.property)
41        rw [hm, hn]
42        apply max_le <=> nlinarith [norm_nonneg ( $\uparrow m : H$ ),
43           $\hookrightarrow$  norm_nonneg ( $\uparrow n : H$ ),
44          norm_nonneg (( $\uparrow m : H$ ) + ( $\uparrow n : H$ ))]
45      --  $\phi$  is antilipschitz with constant 1
46      have hanti : AntilipschitzWith 1  $\phi :=$$$ 
```

```

38 AntilipschitzWith.of_le_mul_dist fun x y => by
39   simp only [NNReal.coe_one, one_mul]
40   calc dist x y = ||x - y|| := dist_eq_norm x y
41     _ ≤ ||φ (x - y)|| := hbound (x - y)
42     _ = ||φ x - φ y|| := by rw [map_sub]
43     _ = dist (φ x) (φ y) := (dist_eq_norm _ _).symm
44 -- The range of φ equals M ∪ N as a set
45 suffices IsClosed (Set.range φ) by
46   convert this using 1
47   ext x; simp only [SetLike.mem_coe, Submodule.mem_sup,
48     ↪ Set.mem_range, Prod.exists]
49   constructor
50   · rintro ⟨a, ha, b, hb, rfl⟩
51     exact ⟨⟨a, ha⟩, ⟨b, hb⟩, by simp [hφ_def,
52       ↪ ContinuousLinearMap.coprod_apply]⟩
53   · rintro ⟨m, n, rfl⟩
54     exact ⟨↑m, m.property, ↑n, n.property,
55       by simp [hφ_def, ContinuousLinearMap.coprod_apply]⟩
56   exact hanti.isClosed_range φ.lipschitz.uniformContinuous

```

8. Доказать, что

$$\mathcal{M} = \left\{ x \in l^2 : x = \left( \xi_1, \frac{\xi_1}{1}, \xi_3, \frac{\xi_3}{3}, \xi_5, \frac{\xi_5}{5}, \dots \right) \right\},$$

и

$$\mathcal{N} = \{ x \in l^2 : x = (\xi_1, 0, \xi_3, 0, \xi_5, 0, \dots) \}$$

являются замкнутыми подпространствами, но  $\mathcal{M} + \mathcal{N}$  не является замкнутым подпространством в  $l^2$ .

Указание. Проверить, что  $\overline{\mathcal{M} + \mathcal{N}} = l^2$ ,  $\mathcal{M} + \mathcal{N} \neq l^2$ .

#### Формализация на Lean 4

```

1  /-
2  Section 12, Task 8.
3  There exist closed subspaces M, N of a Hilbert space such
4  ↪ that M + N
5  is not closed. The abstract principle: if M + N is dense but
6  ↪ ≠ H,
7  then M + N is not closed (since closed + dense = all of H).
8  -/
9  import Mathlib.Analysis.InnerProductSpace.Basic
10 import Mathlib.Tactic
11
12 /-- If the sum of two subspaces is dense but not the whole
13 ↪ space,
14 then the sum is not closed. -/
15 theorem Submodule.sum_not_isClosed_of_dense_ne_top
16 {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℝ H]
17 (M N : Submodule ℝ H)
18 (hdense : Dense ((M ⊔ N : Submodule ℝ H) : Set H))
19 (hne : M ⊔ N ≠ ⊤) :
20   ¬IsClosed ((M ⊔ N : Submodule ℝ H) : Set H) := by
21   intro hclosed
22   apply hne
23   have h1 : closure ((M ⊔ N : Submodule ℝ H) : Set H) =
24     ↪ Set.univ := hdense.closure_eq
25   have h2 : closure ((M ⊔ N : Submodule ℝ H) : Set H) = (M ⊔ N
26     ↪ : Submodule ℝ H) :=
27     hclosed.closure_eq
28   have : (M ⊔ N : Submodule ℝ H) = ⊤ := by
29     rw [Submodule.eq_top_iff']; intro x
30     have : x ∈ closure ((M ⊔ N : Submodule ℝ H) : Set H) := h1
31     ↪ ▶ Set.mem_univ x
32     rwa [h2] at this
33   exact this

```

9. Пусть  $\{e_k\}$  — ортонормированная система в  $H$ . Положим  $x_k = (\cos \frac{1}{k})e_{2k} + (\sin \frac{1}{k})e_{2k+1}$ . Доказать, что множество  $L + M$  не замкнуто, где  $L$  — замкнутое подпространство, порожденное системой  $\{e_{2k}\}$ , а  $M$  — замкнутое подпространство, порожденное системой  $\{x_k\}$ .

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 9.
3    L + M is not closed for specific closed subspaces where the
4    ↪ angle
5    between them tends to 0. Same abstract principle as Task 8:
6    if L + M is dense but ≠ H, then L + M is not closed.
7  -/
8  import FunAn.Section12_HilbertSpaces.Task08
9  import Mathlib.Tactic
10
11  /-- L + M not closed: same principle as Task 8.
12     If L ⊔ M is dense but ≠ H, it cannot be closed. -/
13  theorem sum_subspaces_not_closed_principle
14    {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℝ H]
15    (L M : Submodule ℝ H)
16    (hdense : Dense ((L ⊔ M : Submodule ℝ H) : Set H))
17    (hne : L ⊔ M ≠ H) :
18    ¬IsClosed ((L ⊔ M : Submodule ℝ H) : Set H) :=
19    Submodule.sum_not_isClosed_of_dense_ne_top L M hdense hne

```

10. Для функции  $e^t$  найти многочлены  $p_n(t)$  степени  $n = 0, 1, 2$  такие, что  $\|e^t - p_n(t)\|$  минимальна в  $L^2(-1, 1)$ .
11. Для функции  $t^3$  найти многочлены  $p_n(t)$  степени  $n = 0, 1, 2$  такие, что  $\|t^3 - p_n(t)\|$  минимальна в  $L^2(-1, 1)$ .

12. Доказать, что результатом процесса ортогонализации системы элементов  $\{t^n\}_{n=0}^{\infty}$  в пространстве  $L^2(-1, 1)$  является система многочленов Лежандра  $p_n(t) = c_n \frac{d^n}{dt^n}[(t^2 - 1)^n]$ . Чему равны коэффициенты  $c_n$ ?

#### Формализация на Lean 4

```

1  /-
2  Section 12, Task 12.
3  Gram-Schmidt on  $\{t^n\}$  in  $L^2(-1,1)$  yields Legendre
   ↪ polynomials.
4
5  The Gram-Schmidt process applied to the monomials  $\{1, t, t^2,$ 
   ↪  $\dots\}$  in  $L^2(-1,1)$ 
6  produces the Legendre polynomials  $p_n(t) = c_n \cdot$ 
   ↪  $d^n/dt^n[(t^2-1)^n]$ ,
7  where  $c_n = 1/(2^n \cdot n!)$ .
8
9  We formalize:
10 (1) The Rodrigues polynomial  $R_n = d^n/dt^n[(X^2-1)^n]$  and
   ↪ Legendre polynomial
11  $P_n = (1/(2^n \cdot n!)) \cdot R_n$  via the Rodrigues formula.
12 (2)  $\text{natDegree}(R_n) = n$ : the Rodrigues polynomial has degree
   ↪ exactly  $n$ .
13 (3) Boundary vanishing:  $\text{eval}(\pm 1)$  of  $\text{derivative}^k[(X^2-1)^n]$ 
   ↪  $= 0$  for  $k < n$ .
14 This is the key algebraic fact for orthogonality:
   ↪ integration by parts
15  $n$  times gives  $\int_{-1}^1 q(t) \cdot R_n(t) dt = (-1)^n \int_{-1}^1$ 
   ↪  $q^{(n)}(t) \cdot (t^2-1)^n dt = 0$ 
16 for any polynomial  $q$  of degree  $< n$ , since  $q^{(n)} = 0$ .
17 (4) The normalization constant  $c_n = 1/(2^n \cdot n!)$ .
18 (5) Abstract Gram-Schmidt span/orthogonality properties
   ↪ (from Mathlib).
19 -/
20 import Mathlib.Algebra.Polynomial.FieldDivision
21 import Mathlib.Algebra.Polynomial.RingDivision
22 import Mathlib.Algebra.Polynomial.Degree.Domain
23 import Mathlib.Analysis.InnerProductSpace.GramSchmidtOrtho
24 import Mathlib.Tactic
25
26 noncomputable section
27
28 open Polynomial
29
30 /-- The Rodrigues numerator:  $R_n(t) = d^n/dt^n[(t^2-1)^n]$  as a
   ↪ polynomial over  $\mathbb{R}$ . -/
31 def rodrigues (n : ℕ) : ℝ[X] :=
32   derivative^[n] ((X ^ 2 - 1) ^ n)
33
34 /-- The Legendre polynomial via Rodrigues formula:

```

```

35       $P_n(t) = (1/(2^n \cdot n!)) \cdot d^n/dt^n[(t^2-1)^n].$  -/
36 def legendrePoly (n : ℕ) : ℝ[X] :=
37   (1 / (2 ^ n * (Nat.factorial n : ℝ))) • rodrigues n
38
39 /-- The Rodrigues coefficient  $c_n = 1/(2^n \cdot n!)$ . -/
40 theorem legendre_rodrigues_coeff (n : ℕ) :
41   legendrePoly n = (1 / (2 ^ n * (Nat.factorial n : ℝ))) •
42     ↪ rodrigues n := rfl
43
44 private theorem natDegree_X_sq_sub_one : (X ^ 2 - 1 :
45   ↪ ℝ[X]).natDegree = 2 := by
46   have : (X ^ 2 - 1 : ℝ[X]) = X ^ 2 - C 1 := by simp
47   rw [this, natDegree_X_pow_sub_C]
48
49 /--  $(X^2 - 1)^n$  has degree  $2n$ . -/
50 theorem natDegree_X_sq_sub_one_pow (n : ℕ) :
51   ((X ^ 2 - 1 : ℝ[X]) ^ n).natDegree = 2 * n := by
52   rw [natDegree_pow, natDegree_X_sq_sub_one, mul_comm]
53
54 private theorem leadingCoeff_X_sq_sub_one_pow (n : ℕ) :
55   ((X ^ 2 - 1 : ℝ[X]) ^ n).leadingCoeff = 1 := by
56   have : (X ^ 2 - 1 : ℝ[X]) = X ^ 2 - C 1 := by simp
57   rw [this, leadingCoeff_pow, leadingCoeff_X_pow_sub_C (by
58     ↪ norm_num : 0 < 2), one_pow]
59
60 /-- The coefficient of  $R_n$  at position  $n$  equals  $(2n)!/n!$  (the
61   ↪ descending factorial). -/
62 theorem rodrigues_coeff_n (n : ℕ) :
63   (rodrigues n).coeff n = ((2 * n).descFactorial n : ℝ) :=
64     ↪ by
65     simp only [rodrigues, coeff_iterate_derivative,
66       ↪ nsmul_eq_mul]
67     have h1 : n + n = 2 * n := by omega
68     rw [h1]
69     have h2 : ((X ^ 2 - 1 : ℝ[X]) ^ n).coeff (2 * n) = 1 := by
70       have := leadingCoeff_X_sq_sub_one_pow n
71       rwa [leadingCoeff, natDegree_X_sq_sub_one_pow] at this
72     rw [h2, mul_one]
73
74 /-- The coefficient at degree  $n$  is nonzero since  $(2n)!/n! > 0$ .
75   ↪ -/
76 theorem rodrigues_coeff_n_ne_zero (n : ℕ) : (rodrigues
77   ↪ n).coeff n ≠ 0 := by
78   rw [rodrigues_coeff_n]
79   exact_mod_cast (Nat.descFactorial_pos.mpr (by omega : n ≤ 2
80     ↪ * n)).ne'
81
82 /-- **Degree of the Rodrigues polynomial.**
83   natDegree( $d^n/dt^n[(t^2-1)^n]$ ) =  $n$ . Each Legendre polynomial
84   has degree exactly  $n$ , matching the Gram-Schmidt degree
85   ↪ property. -/

```

```

76 theorem natDegree_rodrigues (n : ℕ) : (rodrigues n).natDegree
77   ↪ = n := by
78   apply le_antisymm
79   · calc (rodrigues n).natDegree
80     ≤ ((X ^ 2 - 1 : ℝ[X]) ^ n).natDegree - n :=
81       ↪ natDegree_iterate_derivative _ _
82     _ = 2 * n - n := by rw [natDegree_X_sq_sub_one_pow]
83     _ = n := by omega
84   · exact le_natDegree_of_ne_zero (rodrigues_coeff_n_ne_zero
85     ↪ n)
86
87 private theorem X_sq_sub_one_pow_ne_zero (n : ℕ) : (X ^ 2 - 1
88   ↪ : ℝ[X]) ^ n ≠ 0 := by
89   apply pow_ne_zero
90   have : (X ^ 2 - 1 : ℝ[X]) = X ^ 2 - C 1 := by simp
91   rw [this]
92   exact X_pow_sub_C_ne_zero (by norm_num : (0 : ℕ) < 2) 1
93
94 private theorem rootMultiplicity_one_ge (n : ℕ) :
95   n ≤ ((X ^ 2 - 1 : ℝ[X]) ^ n).rootMultiplicity 1 := by
96   rw [le_rootMultiplicity_iff (X_sq_sub_one_pow_ne_zero n)]
97   apply pow_dvd_pow_of_dvd
98   rw [dvd_iff_isRoot, IsRoot, eval_sub, eval_pow, eval_X,
99     ↪ eval_one]
100   norm_num
101
102 private theorem rootMultiplicity_neg_one_ge (n : ℕ) :
103   n ≤ ((X ^ 2 - 1 : ℝ[X]) ^ n).rootMultiplicity (-1) := by
104   rw [le_rootMultiplicity_iff (X_sq_sub_one_pow_ne_zero n)]
105   apply pow_dvd_pow_of_dvd
106   rw [dvd_iff_isRoot, IsRoot, eval_sub, eval_pow, eval_X,
107     ↪ eval_one]
108   norm_num
109
110 /-- **Boundary vanishing at t = 1.**
111 For k < n, the kth derivative of (t²-1)ⁿ vanishes at t =
112 ↪ 1.
113 Since (t²-1)ⁿ has a zero of multiplicity n at t = 1,
114 all derivatives of order < n vanish there. -/
115 theorem rodrigues_boundary_vanish_at_one (n : ℕ) {k : ℕ} (hk :
116   ↪ k < n) :
117   (derivative^[k] ((X ^ 2 - 1 : ℝ[X]) ^ n)).eval 1 = 0 :=
118   isRoot_iterate_derivative_of_lt_rootMultiplicity
119   (lt_of_lt_of_le hk (rootMultiplicity_one_ge n))
120
121 /-- **Boundary vanishing at t = -1.**
122 Analogous to the result at t = 1. -/
123 theorem rodrigues_boundary_vanish_at_neg_one (n : ℕ) {k : ℕ}
124   ↪ (hk : k < n) :
125   (derivative^[k] ((X ^ 2 - 1 : ℝ[X]) ^ n)).eval (-1) = 0 :=
126   isRoot_iterate_derivative_of_lt_rootMultiplicity

```

```

118 (lt_of_lt_of_le hk (rootMultiplicity_neg_one_ge n))
119
120 /-- **Gram-Schmidt orthogonalization properties.**
121 Applied to the monomials  $\{1, t, t^2, \dots\}$  in  $L^2(-1,1)$ ,
122 → Gram-Schmidt produces
123  $g_n$  with: (a)  $g_n \neq 0$ , (b)  $g_m \perp g_n$  for  $m \neq n$ , (c)
124 →  $\text{span}\{g_0, \dots, g_n\} = \text{span}\{1, \dots, t^n\}$ .
125 The boundary vanishing (theorems above) shows the
126 → Rodrigues polynomials satisfy
127 the same orthogonality via integration by parts, hence
128 → each Legendre polynomial
129  $P_n = c_n \cdot d^n/dt^n[(t^2-1)^n]$  is a scalar multiple of  $g_n$ , with
130 →  $c_n = 1/(2^n \cdot n!)$ . -/
131 theorem gramSchmidt_orthogonal_basis
132 { $\alpha$  : Type*} [RCLike  $\alpha$ ] {E : Type*} [NormedAddCommGroup E]
133 → [InnerProductSpace  $\alpha$  E]
134 (f :  $\mathbb{N} \rightarrow E$ ) (hf : LinearIndependent  $\alpha$  f) :
135 (∀ n, InnerProductSpace.gramSchmidt  $\alpha$  f n  $\neq 0$ ) ∧
136 (Pairwise fun i j =>
137   @inner  $\alpha$  E _ (InnerProductSpace.gramSchmidt  $\alpha$  f i)
138     (InnerProductSpace.gramSchmidt  $\alpha$  f j) = 0)
139   → ∧
140 (∀ c, Submodule.span  $\alpha$  (InnerProductSpace.gramSchmidt  $\alpha$  f
141   → '' Set.Iic c) =
142   Submodule.span  $\alpha$  (f '' Set.Iic c)) :=
143 {fun n => InnerProductSpace.gramSchmidt_ne_zero n hf,
144   InnerProductSpace.gramSchmidt_pairwise_orthogonal  $\alpha$  f,
145   InnerProductSpace.span_gramSchmidt_Iic  $\alpha$  f}

```

13. Пусть  $x_n(t) = (-1)^k$ ,  $t \in (\frac{k}{2^n}, \frac{k+1}{2^n})$ ,  $n = 0, 1, \dots$ ;  $k = 0, 1, \dots, 2^n - 1$ , в концах этих интервалов  $x_n(t) = 0$ . Показать, что эта система (система Радемахера) ортонормальна в  $L^2(0, 1)$ , но не является базисом.  
 Указание. Проверить, что функция  $x(t) = t(t - 1) + \frac{1}{6}$  ортогональна всем  $x_n(t)$  в  $L^2(0, 1)$ .

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 13.
3    The Rademacher system is ONS but not a basis in  $L^2(0,1)$ .
4
5    Abstract principle: if  $M^\perp \neq \{0\}$ , then  $M \neq H$ , so any ONS
6       $\hookrightarrow$  spanning  $M$ 
7      is not a Hilbert basis.
8  -/
9  import Mathlib.Analysis.InnerProductSpace.Orthogonal
10 import Mathlib.Tactic
11
12 variable {α : Type*} [RCLike α]
13 variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
14   α H]
15
16 /-- If  $M^\perp \neq \{0\}$ , then  $M \neq H$ . Any ONS spanning  $M$  is not a
17    $\hookrightarrow$  basis. -/
18 theorem ne_top_of_orthogonal_ne_bot (M : Submodule α H) (h :
19    $\hookrightarrow$  M.orthogonal  $\neq \perp$ ) :
20   M  $\neq$  T := by
21   intro htop; exact h (htop  $\triangleright$  Submodule.top_orthogonal_eq_bot)

```

14. Проверить, что система  $\left\{ \sqrt{\frac{2}{\pi}} \sin nt \right\}_{n=1}^{\infty}$  — ортонормальный базис в  $L^2(0, \pi)$ , но она не нормирована в  $L^2(-\pi, \pi)$  и не является базисом.

#### Формализация на Lean 4

```

1  /-
2  Section 12, Task 14.
3  {√(2/π) sin(nt)} is ONB in L²(0,π) but NOT in L²(-π,π):
4  not normalized (||sin(nt)||² = π ≠ π/2) and not a basis
   ↪ (misses cosines).
5
6  Same principle as Task 13: an ONS is not a basis when M⊥ ≠
   ↪ {0}.
7  In L²(-π,π), cos(nt) ⊥ sin(mt) for all n,m, so span{sin(nt)}
   ↪ ≠ L².
8  -/
9  import FunAn.Section12_HilbertSpaces.Task13
10 import Mathlib.Tactic
11
12 /-- An ONS is not a basis when its orthogonal complement is
   ↪ nontrivial.
13 Applies to {sin(nt)} in L²(-π,π): cosines are orthogonal
   ↪ to all sines. -/
14 theorem ons_not_basis_when_orthogonal_complement_nontrivial
15   {α : Type*} [RCLike α]
16   {H : Type*} [NormedAddCommGroup H] [InnerProductSpace α H]
17   (M : Submodule α H) (h : M.orthogonal ≠ ⊥) :
18   M ≠ ⊤ :=
19   ne_top_of_orthogonal_ne_bot M h

```

15. Доказать, что система  $\{x_n(t) = \sin \mu_n t\}$ , где  $\mu_n$  — положительные корни уравнения  $\operatorname{tg} \mu = \mu$ , ортогональна, не нормирована и не образует базис в  $L^2(0, 1)$ .

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 15.
3    {sin μnt} where μn are positive roots of tan μ = μ is
4    ↪ orthogonal
5    in L2(0,1), not normalized, and not a basis.
6
7    Same principle: an orthogonal system is not a basis if it
8    ↪ doesn't
9    span the whole space (M⊥ ≠ {0}).
10  -/
11  import FunAn.Section12_HilbertSpaces.Task13
12  import Mathlib.Tactic
13
14  /-- An orthogonal system that doesn't span H is not a basis.
15      M⊥ ≠ {0} ⇔ M ≠ H. -/
16  theorem orthogonal_system_not_basis
17    {α : Type*} [RCLike α]
18    {H : Type*} [NormedAddCommGroup H] [InnerProductSpace α H]
19    (M : Submodule α H) (h : M.orthogonal ≠ ⊥) :
20      M ≠ ⊤ :=
21      ne_top_of_orthogonal_ne_bot M h

```

16. Найти  $\mathcal{M}^\perp$  в  $H$ , если:

- a)  $H = l^2$ ,  $\mathcal{M} = \{(1, 1, 0, 0, \dots)\}$ ;
- b)  $H = L^2(-\pi, \pi)$ ,  $\mathcal{M} = \{\sin nt\}_{n=1}^\infty$ ;
- c)  $H = L^2(-\pi, \pi)$ ,  $\mathcal{M} = \{\cos nt\}_{n=0}^\infty$ .

17. Пусть  $\mathcal{M} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{i=1}^{\infty} \xi_i = 0\}$ . Доказать, что  $\mathcal{M}$  — линейное многообразие, всюду плотное в  $l^2$ , то есть  $\mathcal{M}^\perp = \{\theta\}$ .

#### Формализация на Lean 4

```

1  /-
2    Section 12, Task 17.
3    If M is dense in H, then  $M^\perp = \{\theta\}$ .
4  -/
5  import Mathlib.Analysis.InnerProductSpace.Orthogonal
6  import Mathlib.Tactic
7
8  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
   ↪ ℝ H] [CompleteSpace H]
9
10 /-- If M is dense in H, then  $M^\perp = \perp$ .
11     Proof:  $M^\perp = (\text{closure } M)^\perp = \top^\perp = \perp$ . -/
12 theorem orthogonal_eq_bot_of_dense' (M : Submodule ℝ H)
13   (hM : M.topologicalClosure = ⊤) :
14     M.orthogonal = ⊥ := by
15   rw [← Submodule.orthogonal_closure M, hM,
     ↪ Submodule.top_orthogonal_eq_bot]

```

## 13. ЛИНЕЙНЫЕ НЕПРЕРЫВНЫЕ ФУНКЦИОНАЛЫ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения однородного, аддитивного, линейного функционалов.
2. Дать определения ограниченного, непрерывного функционалов. Какова связь между непрерывностью и ограниченностью линейного функционала?
3. Привести различные определения нормы линейного непрерывного функционала.

### ЗАДАЧИ

1. Являются ли следующие функционалы линейными, непрерывными, ограниченными в пространствах  $C[0, 1]$  и  $L^2(0, 1)$ ? Найти нормы функционалов, если они линейны и непрерывны.

a)  $F(x) = \int_0^1 x(t) \cdot \sin t \, dt;$

b)  $F(x) = \int_0^1 x(t) \cdot \text{sign}(t - \frac{1}{2}) \, dt;$

c)  $F(x) = x(\frac{1}{2});$

d)  $F(x) = \int_0^1 x(t^2) \sqrt{t} \, dt;$

e)  $F(x) = \int_0^1 x^2(t) \, dt;$

- f)  $F(x) = \int_0^1 |x(t)| dt$ ;  
 g)  $F(x) = \max_{0 \leq t \leq 1} x(t)$ ;  
 h)  $F(x) = \int_0^{1/2} x(t) dt - \int_{1/2}^1 x(t) dt$ .

2. Найти нормы функционалов, если они линейны и непрерывны:

- a)  $F(x) = \xi_1, F : l^2 \rightarrow \mathbb{C}$ ;  
 b)  $F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k}, F : l^3 \rightarrow \mathbb{C}$ ;  
 c)  $F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{k}, F : l^1 \rightarrow \mathbb{C}$ ;  
 d)  $F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{k}, F : l^2 \rightarrow \mathbb{C}$ ;  
 e)  $F(x) = \sum_{k=1}^{\infty} |\xi_k|^3, F : l^3 \rightarrow \mathbb{C}$ ;  
 f)  $F(x) = \sum_{k=1}^{\infty} (-1)^k \cdot \frac{\xi_k}{k}, F : l^3 \rightarrow \mathbb{C}$ .

3. Вычислить норму функционала

$$F(x) = \int_a^b p(t)x(t) dt, \quad F : C[a, b] \rightarrow \mathbb{R}^1,$$

где  $p(t) \in C[a, b]$  — фиксированная функция.

*Указание.* Воспользоваться тем, что непрерывная функция есть равномерный предел кусочно-постоянных функций.

4. Доказать, что  $F(x) = x'(0) + x(0)$  является линейным непрерывным функционалом в пространстве  $C^{(1)}[0, 2]$ , если  $\|x\| = \max_{0 \leq t \leq 2} |x(t)| + \max_{0 \leq t \leq 2} |x'(t)|$ , но не является непрерывным, если  $\|x\| = \max_{0 \leq t \leq 2} |x(t)|$ .

#### Формализация на Lean 4

```

1  /-
2    Section 13, Task 04.
3     $F(x) = x'(0) + x(0)$  is continuous in  $C^1$  norm but not  $C^0$ 
       $\hookrightarrow$  norm.
4
5    Abstract principle: a functional bounded by one norm may be
       $\hookrightarrow$  unbounded
6    w.r.t. a weaker norm. Specifically, if  $\|F(x)\| \leq C \cdot \|x\|_1$  for
       $\hookrightarrow$  all  $x$ 
7    but there exist  $x_n$  with  $\|x_n\|_2 \rightarrow 0$  and  $F(x_n) \not\rightarrow 0$ , then  $F$  is
8    continuous in  $\|\cdot\|_1$  but not in  $\|\cdot\|_2$ .
9  -/
10 import Mathlib.Tactic
11
12 /-- A linear functional bounded w.r.t. one norm need not be
       $\hookrightarrow$  bounded w.r.t. another.
13    If there exists a sequence with  $\|x_n\|_2 \rightarrow 0$  but  $|F(x_n)| \geq c$ 
       $\hookrightarrow > 0$ ,
14    then  $F$  is not continuous w.r.t.  $\|\cdot\|_2$ . -/
15 theorem not_continuous_of_unbounded_seq
16   {E : Type*} [AddCommGroup E]
17   {τ : TopologicalSpace E}
18   {F : E → ℝ}
19   {x : ℕ → E} (hx : Filter.Tendsto x Filter.atTop (nhds 0))
20   {c : ℝ} (hc : 0 < c) (hF : ∀ n, c ≤ |F (x n)|)
21   (hF0 : F 0 = 0) :
22     ¬Continuous F := by
23   intro hcont
24   have := (hcont.tendsto 0).comp hx
25   simp only [Function.comp_def, hF0] at this
26   obtain ⟨N, hN⟩ := (Metric.tendsto_atTop.mp this) c hc
27   have := hN N le_rfl
28   simp [Real.dist_eq] at this
29   linarith [hF N]

```

## 14. ЛИНЕЙНЫЕ НЕПРЕРЫВНЫЕ ОПЕРАТОРЫ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение линейного оператора.
2. Дать определения непрерывного и ограниченного оператора. Как связаны понятия непрерывности и ограниченности для линейного оператора?

3. Привести различные определения нормы линейного непрерывного оператора.

## **ЗАДАЧИ**

1. Найти общий вид линейного оператора  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ . Показать, что он ограничен.

#### Формализация на Lean 4

```

1  /-
2    Section 14, Task 1.
3    Any linear operator  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is bounded (continuous).
4    In finite dimensions, all linear maps are automatically
       $\hookrightarrow$  continuous,
5    and bijective ones have continuous inverses.
6  -/
7  import Mathlib.Topology.Algebra.Module.FiniteDimension
8  import Mathlib.Tactic
9
10  /-- Every linear map between finite-dimensional real normed
       $\hookrightarrow$  spaces is continuous. -/
11  theorem linear_map_finite_dim_continuous
12    {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
       $\hookrightarrow$  [FiniteDimensional ℝ E]
13    [NormedAddCommGroup F] [NormedSpace ℝ F]
14    (f : E  $\rightarrow$ ₗ[ℝ] F) : Continuous f :=
15    LinearMap.continuous_of_finiteDimensional f
16
17  /-- A bijective linear equivalence on finite-dim spaces has
       $\hookrightarrow$  continuous inverse. -/
18  theorem linear_equiv_inverse_continuous
19    {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
       $\hookrightarrow$  [FiniteDimensional ℝ E]
20    [NormedAddCommGroup F] [NormedSpace ℝ F]
       $\hookrightarrow$  [FiniteDimensional ℝ F]
21    (e : E  $\simeq$ ₗ[ℝ] F) : Continuous e.symm := by
22    exact LinearMap.continuous_of_finiteDimensional
       $\hookrightarrow$  e.symm.toLinearMap

```

2. Проверить линейность, ограниченность, непрерывность следующих операторов. Найти их нормы, если они линейны и ограничены.

- a)  $(Ax)(t) = \int_0^t x(\xi) d\xi, A : C[0, 1] \rightarrow C[0, 1];$
- b)  $(Ax)(t) = \int_0^1 e^{t-s} x(s) ds, A : C[0, 1] \rightarrow C[2, 4];$
- c)  $A_\lambda : L^2(0, 1) \rightarrow L^2(0, 1); (A_\lambda x)(t) = \begin{cases} x(t), & \text{если } t \leq \lambda; \\ 0, & \text{если } t > \lambda; \end{cases}$
- d)  $(Ax)(t) = x'(t), A : C^{(1)}[-\pi, \pi] \rightarrow C[-\pi, \pi];$
- e)  $(Ax)(t) = x'(t), A : C^{(1)}[-\pi, \pi] \rightarrow C[-\pi, \pi], \|x\|_{C^{(1)}} = \|x\|_C;$
- f)  $Ax = (\xi_3, \xi_4, \dots), A : l^3 \rightarrow l^3;$
- g)  $Ax = (\xi_1 + 3, \xi_2, \xi_3, \dots), A : l^\infty \rightarrow l^\infty;$

h)  $(Ax)(t) = \int_8^t s^2 x^2(s) ds, A : C[8, 10] \rightarrow C[8, 10].$

3. Пусть  $A : l^2 \rightarrow l^2, Ax = (\alpha_1 \xi_1, \alpha_2 \xi_2, \dots)$ , где  $\alpha = \{\alpha_k\}_{k=1}^\infty$  — фиксированная числовая последовательность. Установить критерий непрерывности оператора  $A$ . Найти норму  $A$ .
4. Для каких  $a(t)$  оператор  $(Ax)(t) = a(t)x(t)$  непрерывен в  $C[0, 1]$ ? В  $L^2(0, 1)$ ? Найти нормы.
5. Для каких  $\alpha, \beta$  оператор  $(Ax)(t) = t^\beta \cdot x(t^\alpha)$  ограничен в  $L^2(0, 1)$ ? Найти  $\|A\|$ .

6. Доказать, что если  $A : X \rightarrow Y$  — линейный непрерывный оператор, то он остается непрерывным при замене нормы в  $X$  и  $Y$  на эквивалентные.

#### Формализация на Lean 4

```

1  /-
2    Section 14, Task 6.
3    A continuous linear operator remains continuous when norms
4    ↪ are replaced
5    by equivalent norms.
6
7    Formalization: if  $f : E \rightarrow L[\alpha] F$  is a continuous linear map
8    ↪ and
9     $e : E' \rightarrow L[\alpha] E$  is continuous (modeling the identity under
10   ↪ norm change),
11   then  $f \circ e$  is continuous with  $\|f \circ e\| \leq \|f\| \cdot \|e\|$ .
12   -/
13 import Mathlib.Analysis.Normed.Operator.ContinuousLinearMap
14 import Mathlib.Tactic
15
16 variable {α : Type*} [NontriviallyNormedField α]
17 variable {E : Type*} [NormedAddCommGroup E] [NormedSpace α E]
18 variable {E' : Type*} [NormedAddCommGroup E'] [NormedSpace α E']
19 ↪ E']
20 variable {F : Type*} [NormedAddCommGroup F] [NormedSpace α F]
21
22 /-- Continuity is preserved under equivalent norms: composing
23 ↪ a CLM with
24   a continuous linear map (modeling norm change) gives a CLM
25 ↪ with
26    $\|f \circ e\| \leq \|f\| \cdot \|e\|$ . -/
27 theorem continuous_under_equiv_norms (f : E →L[α] F) (e : E'
28 ↪ →L[α] E) :
29    $\|f \circ e\| \leq \|f\| * \|e\|$  :=
30   ContinuousLinearMap.opNorm_comp_le f e

```

7. Пусть  $L$  — замкнутое подпространство гильбертова пространства  $H$ . Доказать линейность, непрерывность и найти норму оператора  $P : H \rightarrow L$ , который каждому элементу  $x \in H$  сопоставляет его проекцию на  $L$  ( $P$  называется проектором на  $L$ ).

#### Формализация на Lean 4

```

1  /-
2    Section 14, Task 7.
3    The orthogonal projection  $P : H \rightarrow L$  onto a closed subspace  $L$ 
4       $\hookrightarrow$  of a
5      Hilbert space  $H$  is linear, continuous, and  $\|P\| \leq 1$ .
6  -/
7  import Mathlib.Analysis.InnerProductSpace.Projection.Basic
8  import Mathlib.Tactic
9
10 variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
11    $\hookrightarrow$   $\mathbb{R}$  H]
12   [CompleteSpace H] (K : Submodule  $\mathbb{R}$  H) [CompleteSpace K]
13
14 /- The orthogonal projection onto a closed subspace is
15    $\hookrightarrow$  continuous (as a CLM). -/
16 theorem orthogonal_projection_continuous :
17   Continuous (K.orthogonalProjection : H  $\rightarrow$  K) :=
18   ContinuousLinearMap.continuous _
19
20 /- The operator norm of the orthogonal projection is at most
21    $\hookrightarrow$  1. -/
22 theorem orthogonal_projection_norm_le_one :
23    $\|K.orthogonalProjection\| \leq 1 :=$ 
24   K.orthogonalProjection_norm_le
25
26 /- The orthogonal projection is contractive:  $\|Pv\| \leq \|v\|$ . -/
27 theorem orthogonal_projection_apply_norm_le (v : H) :
28    $\|K.orthogonalProjection v\| \leq \|v\| :=$ 
29   K.norm_orthogonalProjection_apply_le v

```

## 15. ОБРАТНЫЕ ОПЕРАТОРЫ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение обратного оператора.
2. Сформулировать теоремы (достаточные условия) существования операторов  $A^{-1}$ ,  $(I + A)^{-1}$ ,  $(A + \Delta A)^{-1}$ .
3. Сформулировать теорему Банаха об обратном операторе.

### ЗАДАЧИ

1. Пусть оператор  $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  определяется равенством:

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 3 \\ -1 & 3 & -2 \\ 5 & 4 & 2 \end{pmatrix} \cdot \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix}$$

Доказать ограниченность оператора  $A$ , оценить  $\|A\|$ , найти  $A^{-1}$ .

#### Формализация на Lean 4

```

1  /-
2    Section 15, Task 1.
3    Any linear operator  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is bounded, and a bijective
4     $\hookrightarrow$  one
5    has a continuous inverse (by open mapping / finite-dim
6     $\hookrightarrow$  automaticity).
7  -/
8  import Mathlib.Topology.Algebra.Module.FiniteDimension
9  import Mathlib.Tactic
10
11  /- Any linear operator between finite-dim real spaces is
12   $\hookrightarrow$  bounded. -/
13  theorem matrix_operator_bounded
14    {n : ℕ} (f : (Fin n → ℝ) →ℓ [ℝ] (Fin n → ℝ)) : Continuous f
15     $\hookrightarrow$  :=
16    LinearMap.continuous_of_finiteDimensional f
17
18  /- A bijective linear map between finite-dim real spaces has
19   $\hookrightarrow$  a
20  continuous inverse (all linear maps are continuous in
21   $\hookrightarrow$  finite dim). -/
22  theorem matrix_operator_inverse_continuous
23    {n : ℕ} (e : (Fin n → ℝ)  $\approx_{\ell}$  [ℝ] (Fin n → ℝ)) :
24    Continuous e.symm := by
25    exact LinearMap.continuous_of_finiteDimensional
26     $\hookrightarrow$  (e.symm.toLinearMap)

```

2. При каких  $\alpha_1, \alpha_2, \dots$  существует обратный оператор к оператору  $Ax = (\alpha_1 \xi_1, \alpha_2 \xi_2, \dots)$ ,  $A : l^2 \rightarrow l^2$ ? Когда  $A^{-1}$  непрерывен?

3. Проверить, что оператор дифференцирования  $\frac{d}{dt} : C^{(1)}[0, 1] \rightarrow C[0, 1]$  имеет правый обратный, но не имеет левого обратного.

#### Формализация на Lean 4

```

1  /-
2  Section 15, Task 3.
3  The differentiation operator  $d/dt : C^1[0,1] \rightarrow C[0,1]$  has a
4   $\hookrightarrow$  right inverse
5  (integration) but no left inverse (not injective: constants
6   $\hookrightarrow$  map to 0).
7
8  We formalize the abstract principle: a non-injective
9   $\hookrightarrow$  function has no
10 left inverse, while a surjective function has a right
11  $\hookrightarrow$  inverse.
12
13 -/
14 import Mathlib.Logic.Function.Basic
15 import Mathlib.Tactic
16
17 /-- A non-injective function has no left inverse.
18 (Applied to  $d/dt$ : constant functions map to 0, so  $d/dt$  is
19  $\hookrightarrow$  not injective.) -/
20 theorem differentiation_no_left_inverse
21   {α β : Type*} {f : α → β} (hf : ¬Function.Injective f) :
22   ¬Function.HasLeftInverse f :=
23   fun ⟨_, hg⟩ => hf hg.injective
24
25 /-- A surjective function has a right inverse (requires
26  $\hookrightarrow$  choice).
27 (Applied to  $d/dt$ : the fundamental theorem of calculus
28  $\hookrightarrow$  gives
29 a right inverse via integration.) -/
30 theorem differentiation_has_right_inverse
31   {α β : Type*} {f : α → β} (hf : Function.Surjective f) :
32   Function.HasRightInverse f :=
33   hf.hasRightInverse

```

4. Найти обратный оператор к оператору  $\frac{d}{dt} : X \rightarrow C[0, 1]$ , где  $X = \{x \in C^{(1)}[0, 1] : x(0) = 0\}$ .
5. Пусть  $X = \{x \in C^{(2)}[0, 1] : x(0) = x(1) = 0\}$ . Найти обратный к оператору  $\frac{d^2}{dt^2} + \lambda : X \rightarrow C[0, 1]$ , если он существует. Для каких  $\lambda$  существует обратный оператор?
6. При каких  $\lambda$  существует обратный оператор к оператору

$$(Ax)(t) = \int_0^t x(\xi) d\xi + \lambda \cdot x(t), \quad A : C[0, 1] \rightarrow C[0, 1]?$$

Построить  $A^{-1}$ .

7. Проверить, существует ли непрерывный обратный оператор к оператору  $A : l^2 \rightarrow l^2$ , если
- a)  $Ax = (0, \xi_1, \xi_2, \dots)$ ;
  - b)  $Ax = (\xi_1 + \xi_2, \xi_2, \xi_3, \dots)$ ;
  - c)  $Ax = (\xi_2 - \xi_1, \xi_2 + \xi_3, 2\xi_2 - 2\xi_1, \xi_4, \xi_5, \dots)$ ;
  - d)  $Ax = (\xi_3, \xi_1, \xi_2, \xi_4, \xi_5, \dots)$ .
8. Пусть  $(Ax)(t) = (t + 1)x(t)$ ,  $(Bx)(t) = x(t^2)$  — операторы в  $C[0, 1]$ . Чему равны  $(AB)^{-1}, (BA)^{-1}$ ?
9. Пусть  $A, B \in \mathcal{L}(X, X)$ . Доказать, что
- a) если существуют  $A^{-1}, B^{-1}$ , то  $(AB)^{-1} = B^{-1}A^{-1}$ ;

b) если существуют  $A^{-1}$ ,  $(BA)^{-1}$ , то существует  $B^{-1}$ .

Указание.  $B^{-1} = A(BA)^{-1}$ .

#### Формализация на Lean 4

```

1  /-
2    Section 15, Task 9.
3    a) Prove that if  $A^{-1}$  and  $B^{-1}$  exist, then  $(AB)^{-1} = B^{-1}A^{-1}$ .
4    b) Prove that if  $A^{-1}$  and  $(BA)^{-1}$  exist, then  $B^{-1}$  exists.
5  -/
6  import Mathlib.Algebra.Group.Units.Basic
7  import Mathlib.Tactic
8
9  variable {X : Type*} [Monoid X]
10
11  /--  $(AB)^{-1} = B^{-1}A^{-1}$  for invertible elements. -/
12  theorem inv_mul_eq (A B : X*) :
13    ((A * B)-1 : X*) = B-1 * A-1 :=
14    mul_inv_rev A B
15
16  /-- If  $A^{-1}$  and  $(BA)^{-1}$  exist, then  $B^{-1}$  exists.
17    Proof:  $B = (BA) \cdot A^{-1}$ , product of two invertible elements.
18    ↪ -/
19  theorem isUnit_of_mul_isUnit_of_isUnit
20    {a b : X} (ha : IsUnit a) (hba : IsUnit (b * a)) :
21    IsUnit b := by
22      obtain ⟨u, rfl⟩ := ha
23      suffices h : IsUnit (b * ↑u * ↑u-1) by
24        simp [mul_assoc] using h
25        exact hba.mul u-1.isUnit

```

10. Пусть  $X$  — банахово пространство с нормами  $\|\cdot\|_1$  и  $\|\cdot\|_2$ . Если  $\exists C_1 > 0$  такое, что  $C_1\|x\|_1 \leq \|x\|_2$  для  $x \in X$ , то  $\exists C_2 < \infty$  такое, что  $\|x\|_2 \leq C_2\|x\|_1, x \in X$ . Доказать.

#### Формализация на Lean 4

```

1  /-
2  Section 15, Task 10.
3  If X is a Banach space with two norms  $\|\cdot\|_1$  and  $\|\cdot\|_2$ , both
4   $\hookrightarrow$  making X
5  complete, and  $\|x\|_1 \leq C\|x\|_2$  for all  $x$ , then the norms are
6   $\hookrightarrow$  equivalent.
7  Proof: The identity map  $(X, \|\cdot\|_2) \rightarrow (X, \|\cdot\|_1)$  is continuous
8   $\hookrightarrow$  and bijective
9  between Banach spaces. By the open mapping theorem (Banach),
10  $\hookrightarrow$  the inverse
11 is also continuous, giving  $\|x\|_2 \leq C'\|x\|_1$ .
12
13 Formalized: the open mapping theorem guarantees that a
14  $\hookrightarrow$  continuous bijection
15 between Banach spaces is a homeomorphism.
16 -/
17 import Mathlib.Analysis.Normed.Operator.Banach
18 import Mathlib.Tactic
19
20 /- The open mapping theorem: a surjective continuous linear
21  $\hookrightarrow$  map from a
22 Banach space to a Banach space is an open map. -/
23 theorem equivalent_norms_from_one_sided_inequality
24 {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
25  $\hookrightarrow$  [CompleteSpace E]
26 [NormedAddCommGroup F] [NormedSpace ℝ F] [CompleteSpace F]
27 (f : E  $\rightarrow$  L[ℝ] F) (hsurj : Function.Surjective f) :
28 IsOpenMap f :=
29 f.isOpenMap hsurj

```

11. Пусть  $A, B \in \mathcal{L}(X, X)$ , причем существует  $(I - AB)^{-1}$ . Доказать, что существует  $(I - BA)^{-1}$ .

Указание.  $(I - BA)^{-1} = I + B(I - AB)^{-1}A$ .

#### Формализация на Lean 4

```

1  /-
2  Section 15, Task 11.
3  If (1 - ab) is invertible then (1 - ba) is invertible
4  ⇔ (Jacobson's lemma).
5
6  The inverse is (1 - ba)-1 = 1 + b · (1 - ab)-1 · a.
7  -/
8  import Mathlib.Algebra.Ring.Units
9  import Mathlib.Tactic
10
11  variable {R : Type*} [Ring R]
12
13  /- Jacobson's lemma: if (1 - a*b) is a unit, then (1 - b*a)
14  ⇔ is also a unit. -/
15  theorem isUnit_one_sub_ba_of_isUnit_one_sub_ab (a b : R) (h :
16  ⇔ IsUnit (1 - a * b)) :
17  IsUnit (1 - b * a) := by
18  obtain ⟨u, hu⟩ := h
19  set c : R := ↑u-1
20  have huc : (1 - a * b) * c = 1 := by change (1 - a * b) *
21  ⇔ ↑u-1 = 1; rw [← hu]; exact u.mul_inv
22  have hcu : c * (1 - a * b) = 1 := by change ↑u-1 * (1 - a *
23  ⇔ b) = 1; rw [← hu]; exact u.inv_mul
24  -- abc = c - 1 (from (1-ab)c = 1)
25  have habc : a * b * c = c - 1 := by
26  have h : c - a * b * c = 1 := by rwa [sub_mul, one_mul] at
27  ⇔ huc
28  rw [eq_sub_iff_add_eq, add_comm]; exact
29  ⇔ (sub_eq_iff_eq_add.mp h).symm
30  -- cab = c - 1 (from c(1-ab) = 1)
31  have hcab : c * a * b = c - 1 := by
32  have h : c - c * a * b = 1 := by rw [mul_sub, mul_one, ←
33  ⇔ mul_assoc] at hcu; exact hcu
34  rw [eq_sub_iff_add_eq, add_comm]; exact
35  ⇔ (sub_eq_iff_eq_add.mp h).symm
36  -- Construct the unit: val = 1 - ba, inv = 1 + bca
37  refine ⟨(1 - b * a, 1 + b * c * a, ?_, ?_), rfl⟩
38  · -- (1 - ba)(1 + bca) = 1
39  have key : b * a * (b * c * a) = b * c * a - b * a := by
40  have : b * a * (b * c * a) = b * (a * b * c) * a := by
41  ⇔ simp only [mul_assoc]
42  rw [this, habc, mul_sub, mul_one, sub_mul]
43  calc (1 - b * a) * (1 + b * c * a)
44  = 1 + b * c * a - b * a - b * a * (b * c * a) := by
45  ⇔ noncomm_ring

```

```

35     _ = 1 + b * c * a - b * a - (b * c * a - b * a) := by rw
      ↪ [key]
36     _ = 1 := by noncomm_ring
37     · -- (1 + bca)(1 - ba) = 1
38     have key : b * c * a * (b * a) = b * c * a - b * a := by
39     have : b * c * a * (b * a) = b * (c * a * b) * a := by
      ↪ simp only [mul_assoc]
40     rw [this, hcab, mul_sub, mul_one, sub_mul]
41     calc (1 + b * c * a) * (1 - b * a)
42         = 1 - b * a + b * c * a - b * c * a * (b * a) := by
      ↪ noncomm_ring
43     _ = 1 - b * a + b * c * a - (b * c * a - b * a) := by rw
      ↪ [key]
44     _ = 1 := by noncomm_ring

```

## 16. СОПРЯЖЕННЫЕ ПРОСТРАНСТВА. РЕФЛЕКСИВНОСТЬ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение сопряженного пространства.
2. Каков общий вид линейных непрерывных функционалов в пространствах  $\mathbb{R}^n$ ,  $H$ ,  $C[a, b]$ ,  $l^p$ ,  $L^p(a, b)$  где  $1 < p < \infty$ ?
3. Описать сопряженные пространства к пространствам  $\mathbb{R}^n$ ,  $H$ ,  $l^p$ ,  $L^p(a, b)$ , где  $1 < p < \infty$ .
4. Какое пространство называется вторым сопряженным? Описать естественное отображение  $\pi : X \rightarrow X^{**}$  и дать определение рефлексивного пространства  $X$ . Привести примеры рефлексивных пространств.

### ЗАДАЧИ

1. Описать сопряженное пространство к пространству  $\mathbb{R}^n$  с нормой

- a)  $\|x\| = \max_{1 \leq k \leq n} |\xi_k|$ ;
- b)  $\|x\| = \sum_{k=1}^n |\xi_k|$ ;
- c)  $\|x\| = (\sum_{k=1}^n |\xi_k|^p)^{1/p}$ , где  $1 < p < \infty$ ,  $x = (\xi_1, \dots, \xi_n)$ .

Является ли  $\mathbb{R}^n$  рефлексивным?

2. Описать сопряженное пространство к пространству  $l^1$ .
3. Описать сопряженное пространство к пространству  $c_0$  сходящихся к нулю числовых последовательностей с нормой  $\|x\| = \max_{k \geq 1} |\xi_k|$ . Рефлексивно ли пространство  $c_0$ ?
4. Описать сопряженное пространство к пространству  $c$  сходящихся к конечному пределу последовательностей с нормой  $\|x\| = \sup_{k \geq 1} |\xi_k|$ . Является ли пространство  $c$  рефлексивным?

5. Найти общий вид линейных непрерывных функционалов в пространстве  $s$ .

## 17. ТЕОРЕМА ХАНА-БАНАХА. СИЛЬНАЯ И СЛАБАЯ СХОДИМОСТИ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Сформулировать теорему Хана-Банаха и следствия из нее.
2. Дать определения сильной и слабой сходимости последовательности функционалов, определение слабой сходимости последовательности элементов.
3. Сформулировать критерий слабой сходимости последовательности функционалов и элементов.

### ЗАДАЧИ

1. Найти все линейные непрерывные продолжения функционала  $f(x) = 4\xi_1$ ,  $x = \xi_1 \in \mathbb{R}^1$  на пространство  $\mathbb{R}^3$  с сохранением нормы, если  $\mathbb{R}^3$  рассматривается с нормой
  - a)  $\|x\| = (\xi_1^2 + \xi_2^2 + \xi_3^2)^{1/2}$ ;
  - b)  $\|x\| = |\xi_1| + |\xi_2| + |\xi_3|$ ;
  - c)  $\|x\| = \max(|\xi_1|, |\xi_2|, |\xi_3|)$ .
2. Найти все линейные непрерывные продолжения функционала  $f(x) = \xi_1 + \xi_2$ ,  $x \in \mathbb{R}^2$ , на пространство  $l^p$  с сохранением нормы, если  $\|x\| = (|\xi_1|^p + |\xi_2|^p)^{1/p}$ :
  - a)  $p = 1$ ;
  - b)  $1 < p < \infty$ .

3. Доказать, что последовательность функционалов

$$F_n(x) = \int_{-\pi}^{\pi} x(t)e^{int} dt$$

слабо сходится в  $L^2(-\pi, \pi)$ , но сильной сходимости нет.

#### Формализация на Lean 4

```

1  /-
2    Section 17, Task 3.
3     $F_n(x) = \int_{-\pi}^{\pi} x(t)e^{int} dt$  converges weakly to 0 in  $L^2(-\pi, \pi)^*$ 
4    but not strongly ( $\|F_n\| = \sqrt{2\pi}$  for all  $n$  by Parseval).
5    Weak convergence:  $F_n(x) = \sqrt{2\pi} \cdot x^\wedge(n) \rightarrow 0$  by
      ↪ Riemann-Lebesgue lemma.
6
7    We formalize the key ingredient: the Riemann–Lebesgue lemma,
      ↪ which
8    states that Fourier coefficients tend to 0 at infinity.
9  -/
10 import Mathlib.Analysis.Fourier.RiemannLebesgueLemma
11 import Mathlib.Tactic
12
13 open MeasureTheory Filter
14
15 /-- Riemann–Lebesgue lemma: Fourier coefficients tend to 0 at
      ↪ infinity. -/
16 theorem fourier_functionals_weak_not_strong
17   {E : Type*} [NormedAddCommGroup E] [NormedSpace ℂ E]
18   (f : ℝ → E) :
19     Tendsto (fun w : ℝ => ∫ v : ℝ, Real.fourierChar (-(v * w))
      ↪ • f v)
20       (cocompact ℝ) (nhds 0) :=
21     Real.tendsto_integral_exp_smul_cocompact f

```

4. Покажите, что существует последовательность конечных линейных комбинаций функционалов  $F_t(x) = x(t)$ , сходящаяся слабо к функционалу  $F(x) = \int_0^1 x(t) dt$  (функционалы определены в пространстве  $C[0, 1]$ ). Имеется ли последовательность, сходящаяся по норме?

#### Формализация на Lean 4

```

1  /-
2    Section 17, Task 04.
3    (a)  $\exists$  sequence of finite linear combinations of  $F_t(x) = x(t)$ 
4        converging weakly to  $F(x) = \int_0^1 x(t) dt$ .
5        Proof: Riemann sums  $S_n(f) = (1/n) \sum f(k/n) \rightarrow \int_0^1 f$  for
6             $\hookrightarrow$  each continuous  $f$ .
7
8    (b) No norm-convergent sequence of such combinations.
9        Witness:  $g_n(t) = \cos(2\pi nt)$  has  $g_n(k/n) = 1$  for all  $k$ ,
10       so  $S_n(g_n) = 1$ , while  $\int_0^1 g_n = 0$ . Thus  $\|S_n - F\| \geq 1$  for
11        $\hookrightarrow$  all  $n$ .
12
13  -/
14  import Mathlib.MeasureTheory.Integral.IntervalIntegral.Basic
15  import Mathlib.Topology.UniformSpace.HeineCantor
16  import Mathlib.Analysis.SpecialFunctions.Integrals.Basic
17  import Mathlib.Tactic
18
19  open MeasureTheory Set Finset Filter Real
20  open scoped Topology
21
22  noncomputable section
23
24  --
25   $\hookrightarrow$  =====
26  -- Part (a): Riemann sums converge to the integral for
27   $\hookrightarrow$  continuous functions
28  --
29   $\hookrightarrow$  =====
30
31  /-- Left Riemann sum:  $S_n(f) = (1/n) \sum_{k=0}^{n-1} f(k/n)$ .
32       Each  $S_n$  is a linear combination of point evaluation
33        $\hookrightarrow$  functionals  $F_t(x) = x(t)$ . -/
34  def riemannSum (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (n :  $\mathbb{N}$ ) :  $\mathbb{R}$  :=
35      ( $\sum$  k  $\in$  range n, f ((k :  $\mathbb{R}$ ) / n)) / n
36
37  /-- Riemann sums converge to the integral for continuous
38       functions on  $[0, 1]$ .
39       This provides the weak convergence: for each  $f \in C[0, 1]$ ,
40       the sequence of linear combinations  $S_n(f) \rightarrow F(f) = \int_0^1$ 
41        $\hookrightarrow$   $f(t) dt$ . -/
42  theorem section17_task04a_riemann_sum_tendsto
43      {f :  $\mathbb{R} \rightarrow \mathbb{R}$ } (hf : ContinuousOn f (Icc 0 1)) :
44      Tendsto (fun n => riemannSum f n) atTop (nhds ( $\int$  x in (0 :
45       $\hookrightarrow$   $\mathbb{R}$ )..1, f x)) := by

```

```

36 rw [Metric.tendsto_atTop]
37 intro ε hε
38 -- Uniform continuity on compact [0,1]
39 obtain ⟨δ, hδ, hδ_spec⟩ := Metric.uniformContinuousOn_iff.mp
40   (isCompact_Icc.uniformContinuousOn_of_continuous hf) (ε /
    ↪ 2) (half_pos hε)
41 obtain ⟨N, hN⟩ := exists_nat_gt (1 / δ)
42 refine ⟨max 1 N, fun n hn => ?_⟩
43 have hn1 : 0 < n := by omega
44 have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn1
45 have hne : (↑n : ℝ) ≠ 0 := hn'.ne'
46 -- Key: 1/n < δ
47 have h1n : 1 / (↑n : ℝ) < δ := by
48   have hNn : 1 / δ < ↑n :=
49     lt_of_lt_of_le hN (by exact_mod_cast show N ≤ n by
    ↪ omega)
50   rwa [div_lt_iff₀ hδ, mul_comm, ← div_lt_iff₀ hn'] at hNn
51 -- Integrability on each subinterval [k/n, (k+1)/n] ⊆ [0,1]
52 have hint : ∀ k, k < n →
53   IntervalIntegrable f volume ((↑k : ℝ) / ↑n) ((↑k + 1) /
    ↪ ↑n) := by
54   intro k hk
55   apply (hf.mono _).intervalIntegrable
56   rw [Set.uIcc_of_le (div_le_div_of_nonneg_right
    (by linarith : (↑k : ℝ) ≤ ↑k + 1) hn'.le)]
57   exact Icc_subset_Icc (div_nonneg (Nat.cast_nonneg k)
    ↪ hn'.le)
58   ((div_le_one hn').mpr (by exact_mod_cast
    ↪ Nat.succ_le_of_lt hk))
59 -- Decompose  $\int_0^1 f = \sum \int_{[k/n, (k+1)/n]} f$ 
60 have hdecomp :  $\int x \text{ in } (0 : \mathbb{R})..1, f x =$ 
61    $\sum k \in \text{range } n, \int x \text{ in } ((\uparrow k : \mathbb{R}) / \uparrow n)..((\uparrow k + 1) / \uparrow n),$ 
62    $f x := \text{by}$ 
63   have h := intervalIntegral.sum_integral_adjacent_intervals
64     (a := fun k => (↑k : ℝ) / ↑n) (fun k hk => by
65       change IntervalIntegrable f volume ((↑k : ℝ) / ↑n)
        ↪ ((↑(k + 1) : ℝ) / ↑n)
66       rw [show (↑(k + 1) : ℝ) / ↑n = ((↑k : ℝ) + 1) / ↑n
        ↪ from by push_cast; ring]
67       exact hint k hk)
68   dsimp only at h
69   simp only [Nat.cast_zero, zero_div, div_self hne] at h
70   convert h.symm using 2
71   push_cast; ring
72 -- Each term:  $f(k/n)/n = \int f(k/n) dt \text{ on } [k/n, (k+1)/n]$ 
73 have hconst : ∀ k, k < n →
74    $f ((\uparrow k : \mathbb{R}) / \uparrow n) / \uparrow n =$ 
75    $\int _ \text{ in } ((\uparrow k : \mathbb{R}) / \uparrow n)..((\uparrow k + 1) / \uparrow n), f ((\uparrow k : \mathbb{R}) /$ 
    ↪  $\uparrow n) := \text{by}$ 
76   intro k _
77   rw [intervalIntegral.integral_const, smul_eq_mul,
```

```

78     show ((↑k : ℝ) + 1) / ↑n - ↑k / ↑n = 1 / ↑n from by
79     ↪ ring,
80     one_div, mul_comm, ← div_eq_mul_inv]
81 -- Bound each term of the difference
82 have hbound : ∀ k ∈ range n,
83   |f ((↑k : ℝ) / ↑n) / ↑n - ∫ x in ((↑k : ℝ) / ↑n)..((↑k +
84     ↪ 1) / ↑n), f x| ≤
85     ε / 2 * (1 / ↑n) := by
86   intro k hk
87   have hk' : k < n := mem_range.mp hk
88   rw [hconst k hk', ← intervalIntegral.integral_sub
89     ↪ intervalIntegrable_const (hint k hk')]
90   have hle : (↑k : ℝ) / ↑n ≤ ((↑k : ℝ) + 1) / ↑n :=
91     div_le_div_of_nonneg_right (by linarith : (↑k : ℝ) ≤ ↑k
92     ↪ + 1) hn'.le
93   -- Use norm_integral bound
94   have h_norm :=
95     ↪ intervalIntegral.norm_integral_le_of_norm_le_const
96     (a := (↑k : ℝ) / ↑n) (b := ((↑k : ℝ) + 1) / ↑n)
97     (f := fun x => f ((↑k : ℝ) / ↑n) - f x) (C := ε / 2)
98     ↪ (fun t ht => ?_)
99   · rw [show |((↑k : ℝ) + 1) / ↑n - ↑k / ↑n| = 1 / ↑n from
100     ↪ by
101       rw [show ((↑k : ℝ) + 1) / ↑n - ↑k / ↑n = 1 / ↑n from
102         ↪ by ring]
103       exact abs_of_nonneg (div_nonneg one_pos.le hn'.le)]
104       ↪ at h_norm
105   rw [← Real.norm_eq_abs]; exact h_norm
106   · -- ||f(k/n) - f(t)|| ≤ ε/2 for t in subinterval
107   rw [Set.uIoc_of_le hle] at ht
108   rw [Real.norm_eq_abs, abs_sub_comm]
109   apply le_of_lt
110   apply hδ_spec
111   · -- t ∈ [0,1]
112     exact (le_trans (div_nonneg (Nat.cast_nonneg k)
113       ↪ hn'.le) (le_of_lt ht.1),
114       le_trans ht.2 ((div_le_one hn').mpr (by
115         ↪ exact_mod_cast Nat.succ_le_of_lt hk'))))
116   · -- k/n ∈ [0,1]
117     exact (div_nonneg (Nat.cast_nonneg k) hn'.le,
118       (div_le_one hn').mpr (by exact_mod_cast
119         ↪ hk'.le))
120   · -- dist t (k/n) < δ
121     rw [Real.dist_eq, abs_of_nonneg (by linarith [ht.1])]
122     calc t - (↑k : ℝ) / ↑n
123       ≤ ((↑k : ℝ) + 1) / ↑n - ↑k / ↑n := by linarith
124       ↪ [ht.2]
125       = 1 / ↑n := by ring
126       < δ := h1n
127 -- Assemble the bound
128 rw [Real.dist_eq, hdecomp]

```

```

116 change |riemannSum f n -  $\sum k \in \text{range } n, \int x \text{ in } ((\uparrow k : \mathbb{R}) /$ 
     $\hookrightarrow \uparrow n) .. ((\uparrow k + 1) / \uparrow n), f x| < \varepsilon$ 
117 rw [show riemannSum f n =  $\sum k \in \text{range } n, f ((\uparrow k : \mathbb{R}) / \uparrow n) /$ 
     $\hookrightarrow \uparrow n$  from by
118   unfold riemannSum
119   exact Finset.sum_div (range n) (fun k => f (( $\uparrow k : \mathbb{R}$ ) /
     $\hookrightarrow \uparrow n$ ))  $\uparrow n$ ]
120 rw [← Finset.sum_sub_distrib]
121 calc | $\sum k \in \text{range } n, (f ((\uparrow k : \mathbb{R}) / \uparrow n) / \uparrow n -$ 
     $\int x \text{ in } ((\uparrow k : \mathbb{R}) / \uparrow n) .. ((\uparrow k + 1) / \uparrow n), f x)|$ 
122    $\leq \sum k \in \text{range } n, |f ((\uparrow k : \mathbb{R}) / \uparrow n) / \uparrow n -$ 
     $\int x \text{ in } ((\uparrow k : \mathbb{R}) / \uparrow n) .. ((\uparrow k + 1) / \uparrow n), f x| :=$ 
123   Finset.abs_sum_le_sum_abs _ _
124    $\leq \sum _ \in \text{range } n, (\varepsilon / 2 * (1 / \uparrow n)) := \text{Finset.sum_le_sum}$ 
125    $\hookrightarrow$  hbound
126    $= \varepsilon / 2 :=$  by
127     rw [Finset.sum_const, card_range, nsmul_eq_mul]
128     field_simp
129      $< \varepsilon :=$  half_lt_self hε
130
131
132 --
     $\hookrightarrow$  =====
133 -- Part (b): Riemann sums do not converge in operator norm
134 --
     $\hookrightarrow$  =====
135
136 /-- Riemann sum of  $\cos(2\pi nt)$ : since  $\cos(2\pi k) = 1$  for integer
     $\hookrightarrow$  k,
137   each  $f(k/n) = 1$ , so  $S_n = n/n = 1$ . -/
138 private lemma riemannSum_cos (n :  $\mathbb{N}$ ) (hn :  $0 < n$ ) :
139   riemannSum (fun x =>  $\cos (2 * \pi * \uparrow n * x)$ ) n = 1 := by
140   have hn' : ( $0 : \mathbb{R}$ ) <  $\uparrow n := \text{Nat.cast_pos.mpr hn}$ 
141   simp only [riemannSum]
142   have :  $\forall k \in \text{range } n, \cos (2 * \pi * \uparrow n * ((\uparrow k : \mathbb{R}) / \uparrow n)) = 1$ 
     $\hookrightarrow :=$  by
143     intro k _
144     rw [show  $2 * \pi * \uparrow n * ((\uparrow k : \mathbb{R}) / \uparrow n) = \uparrow k * (2 * \pi)$  from
         $\hookrightarrow$  by field_simp]
145     exact cos_nat_mul_two_pi k
146   rw [Finset.sum_congr rfl this, Finset.sum_const, card_range,
     $\hookrightarrow$  nsmul_eq_mul, mul_one]
147   exact div_self hn'.ne'
148
149 /-- Integral of  $\cos(2\pi nt)$  over  $[0,1]$  is 0 for  $n \geq 1$ . -/
150 private lemma integral_cos_two_pi_mul (n :  $\mathbb{N}$ ) (hn :  $0 < n$ ) :
151    $\int x \text{ in } (0 : \mathbb{R}) .. 1, \cos (2 * \pi * \uparrow n * x) = 0 :=$  by
152   have hc : ( $2 * \pi * (\uparrow n : \mathbb{R}) \neq 0 :=$  by positivity
153   rw [show (fun x :  $\mathbb{R} => \cos (2 * \pi * \uparrow n * x) = (\text{fun } x => \cos$ 
     $\hookrightarrow ((2 * \pi * \uparrow n) * x))$  from rfl,
154     intervalIntegral.integral_comp_mul_left _ hc, mul_zero,
     $\hookrightarrow$  mul_one, integral_cos,

```

```

155     sin_zero, sub_zero]
156   rw [show 2 * π * (↑n : ℝ) = ↑(2 * n) * π from by push_cast;
      ↪ ring,
157     sin_nat_mul_pi, smul_zero]
158
159   /-- ||Sn - F|| ≥ 1 for all n ≥ 1: the witness cos(2πnt) has
      ↪ Sn(g) = 1
160     but F(g) = ∫01 cos(2πnt)dt = 0, while ||g||∞ = 1.
161     Hence Riemann sums do not converge to F in operator norm.
      ↪ -/
162   theorem section17_task04b_not_norm_convergent (n : ℕ) (hn : 0
      ↪ < n) :
163     |riemannSum (fun x => cos (2 * π * ↑n * x)) n -
164       ∫ x in (0 : ℝ)..1, cos (2 * π * ↑n * x)| = 1 := by
165   rw [riemannSum_cos n hn, integral_cos_two_pi_mul n hn,
      ↪ sub_zero, abs_one]

```

5. Пусть  $F_n(x) = n \cdot [x(\frac{1}{n}) + x(-\frac{1}{n}) - 2x(0)]$ . Показать, что последовательность  $\{F_n\}$ :

- a) сходится по норме к  $F(x) \equiv 0$  в  $C^{(2)}[-1, 1]$ ;
- b) сходится слабо к 0 в  $C^{(1)}[-1, 1]$ , но не сходится сильно;

с) не сходится даже слабо в  $C[-1, 1]$ .

Указания. (i) Использовать формулу Тейлора. (ii) Оценить нормы  $\|F_n\|$  снизу:

$$x'_n(t) = \begin{cases} 1, & \text{если } t > \frac{1}{n}, \\ nt, & \text{если } |t| \leq \frac{1}{n}, \\ -1, & \text{если } t < -\frac{1}{n}. \end{cases}$$

(iii) Показать, что последовательность  $\{\|F_n\|\}$  неограничена.

#### Формализация на Lean 4

```

1  /-
2  Section 17, Task 05.
3  F_n(x) = n[x(1/n) + x(-1/n) - 2x(0)].
4
5  (a) ||F_n||_{C^2*} → 0: for f ∈ C^2, |F_n(f)| ≤ 2·sup|f'|/n.
6  (b) F_n → 0 weakly in C^1[-1,1]*: for each C^1 function f,
   ↪ F_n(f) → 0.
7  (c) Not weakly convergent in C[-1,1]*: F_n(√|·|) = 2√n → ∞.
8  -/
9  import Mathlib.Analysis.Calculus.Deriv.MeanValue
10 import Mathlib.Data.Real.Sqrt
11 import Mathlib.Tactic
12
13 open Filter Set Real
14 open scoped Topology
15
16 noncomputable section
17
18 /-- The functional F_n applied to f: F_n(f) = n(f(1/n) + f(-1/n)
   ↪ - 2f(0)). -/
19 def Fn (f : ℝ → ℝ) (n : ℕ) : ℝ :=
20   (n : ℝ) * (f (1 / (n : ℝ)) + f (-1 / (n : ℝ))) - 2 * f 0
21
22 --
23 ↪ =====
24 -- Part (c): F_n(√|·|) = 2√n → ∞
25 --
26 ↪ =====
27
28 private lemma Fn_sqrt_abs (n : ℕ) (hn : 0 < n) :
29   Fn (fun t => √|t|) n = 2 * √n := by
30     have hn' : (0 : ℝ) < n := Nat.cast_pos.mpr hn
31     simp only [Fn, abs_zero, sqrt_zero, mul_zero, sub_zero]
32     rw [abs_of_nonneg (div_nonneg zero_le_one hn'.le)]
33     rw [show |-(1 / (n : ℝ))| = 1 / n from by rw [abs_neg,
   ↪ abs_of_nonneg (div_nonneg zero_le_one hn'.le)]]
34     rw [← two_mul (√(1 / ↑n)), one_div, sqrt_inv]
35     -- Goal: ↑n * (2 * (√↑n)^-1) = 2 * √↑n

```

```

34 -- Use:  $\uparrow n * (\sqrt{\uparrow n})^{-1} = \sqrt{\uparrow n}$  (i.e.,  $n/\sqrt{n} = \sqrt{n}$ )
35 have key : ( $\uparrow n$  :  $\mathbb{R}$ ) *  $(\sqrt{\uparrow n})^{-1} = \sqrt{\uparrow n}$  := by
36   rw [← div_eq_mul_inv]; exact div_sqrt
37   nlinarith
38
39 theorem section17_task05c_not_weak_in_C :
40   Tendsto (fun n :  $\mathbb{N}$  => Fn (fun t =>  $\sqrt{|t|}$ ) n) atTop atTop :=
41     ↪ by
42   apply Filter.tendsto_atTop_atTop.mpr
43   intro b
44   refine ⟨max 1 ( $\lceil (b / 2)^2 \rceil + 1$ ), fun n hn => ?_⟩
45   have hn1 :  $0 < n$  := by omega
46   rw [Fn_sqrt_abs n hn1]
47   by_cases hb :  $b \leq 0$ 
48   · linarith [mul_nonneg (two_pos.le) (sqrt_nonneg (n :  $\mathbb{R}$ ))]
49   · simp only [not_le] at hb
50     have hb2 :  $(0 : \mathbb{R}) < b / 2$  := by linarith
51     have hsq :  $(b / 2)^2 < \uparrow n$  := by
52       have :  $n \geq \lceil (b / 2)^2 \rceil + 1$  := by omega
53       calc (b / 2)^2 ≤  $\uparrow (\lceil (b / 2)^2 \rceil + 1)$  := Nat.le_ceil _
54            <  $\uparrow (\lceil (b / 2)^2 \rceil + 1)$  := by push_cast; linarith
55            ≤ ( $\uparrow n$  :  $\mathbb{R}$ ) := by exact_mod_cast <_
56     linarith [show b / 2 <  $\sqrt{\uparrow n}$  from by rw [← sqrt_sq hb2.le];
57       ↪ exact sqrt_lt_sqrt (sq_nonneg _) hsq]
58
59 --
60 -- =====
61 -- Core MVT identity:  $F_n(f) = f'(c_1) - f'(c_2)$ 
62 -- =====
63
64 private lemma Fn_eq_deriv_diff {f :  $\mathbb{R} \rightarrow \mathbb{R}$ } {n :  $\mathbb{N}$ } (hn :  $0 < n$ )
65   ↪ (hf : DifferentiableOn  $\mathbb{R}$  f (Icc (-1) 1))
66   (hfc : ContinuousOn f (Icc (-1) 1)) :
67   ∃ c1 ∈ Ioo 0 (1 / (n :  $\mathbb{R}$ )),
68   ∃ c2 ∈ Ioo (-(1 / (n :  $\mathbb{R}$ ))) 0,
69   Fn f n = deriv f c1 - deriv f c2 := by
70     have hn' :  $(0 : \mathbb{R}) < n$  := Nat.cast_pos.mpr hn
71     have hnn :  $(n : \mathbb{R}) \neq 0$  := ne_of_gt hn'
72     have hln_pos :  $(0 : \mathbb{R}) < 1 / \uparrow n$  := div_pos one_pos hn'
73     have hln :  $1 / (n : \mathbb{R}) \leq 1$  := by rw [div_le_one hn'];
74       ↪ exact_mod_cast (by omega :  $1 \leq n$ )
75     have hI1 : Icc 0 (1 / (n :  $\mathbb{R}$ )) ⊆ Icc (-1 :  $\mathbb{R}$ ) 1 :=
76       ↪ Icc_subset_Icc (by linarith) hln
77     have hI2 : Icc (-(1 / (n :  $\mathbb{R}$ ))) 0 ⊆ Icc (-1 :  $\mathbb{R}$ ) 1 :=
78       ↪ Icc_subset_Icc (neg_le_neg hln) (by linarith)
79     obtain ⟨c1, hc1, hc1_eq⟩ := exists_deriv_eq_slope f hln_pos
80     (hfc.mono hI1) ((hf.mono hI1).mono Ioo_subset_Icc_self)
81     obtain ⟨c2, hc2, hc2_eq⟩ := exists_deriv_eq_slope f (by
82       ↪ linarith :  $-(1 / (\uparrow n : \mathbb{R})) < 0$ )

```

```

76   (hfc.mono hI2) ((hf.mono hI2).mono Ioo_subset_Icc_self)
77   refine (c1, hc1, c2, hc2, ?_)
78   show ↑n * (f (1 / ↑n) + f (-(1 / ↑n)) - 2 * f 0) = deriv f
    ↪ c1 - deriv f c2
79   -- From MVT: deriv f c1 = (f(1/n) - f(0)) / (1/n - 0)
80   have hc1_s : deriv f c1 = (f (1/↑n) - f 0) / (1/↑n) := by
    ↪ rwa [sub_zero] at hc1_eq
81   -- From MVT: deriv f c2 = (f(0) - f(-1/n)) / (1/n)
82   have hc2_s : deriv f c2 = (f 0 - f (-(1/↑n))) / (1/↑n) := by
    ↪ rwa [show (0 : ℝ) - -(1 / ↑n) = 1 / ↑n from by ring] at
    ↪ hc2_eq
84   have eq1 : f (1 / ↑n) - f 0 = deriv f c1 * (1 / ↑n) := by
    ↪ rw [hc1_s, div_mul_cancel0 _ (ne_of_gt hln_pos)]
86   have eq2 : f 0 - f (-(1 / ↑n)) = deriv f c2 * (1 / ↑n) := by
    ↪ rw [hc2_s, div_mul_cancel0 _ (ne_of_gt hln_pos)]
88   have h_sum : f (1 / ↑n) + f (-(1 / ↑n)) - 2 * f 0 = (deriv f
    ↪ c1 - deriv f c2) / ↑n := by
    ↪ rw [show (deriv f c1 - deriv f c2) / ↑n = deriv f c1 *
    ↪ (1/↑n) - deriv f c2 * (1/↑n) from by ring]
    ↪ linarith
91   rw [h_sum, mul_div_cancel0 _ hnn]
92
93   --
94   ↪ =====
95   -- Part (a): C2 estimate |Fn(f)| ≤ 2M/n via MVT applied three
96   ↪ times
97   --
98   ↪ =====
99
100  theorem section17_task05a_C2_bound
101    {f : ℝ → ℝ} {M : ℝ}
102    (hf : DifferentiableOn ℝ f (Icc (-1) 1))
103    (hf' : DifferentiableOn ℝ (deriv f) (Icc (-1) 1))
104    (hfc : ContinuousOn f (Icc (-1) 1))
105    (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
106    (hbound : ∀ x ∈ Icc (-(1 : ℝ)) 1, |deriv (deriv f) x| ≤ M)
107    {n : ℕ} (hn : 0 < n) :
108    |Fn f n| ≤ 2 * M / n := by
109    have hn' : (0 : ℝ) < n := Nat.cast_pos.mpr hn
110    have hln : 1 / (n : ℝ) ≤ 1 := by rw [div_le_one hn'];
    ↪ exact_mod_cast (by omega : 1 ≤ n)
112    obtain (c1, hc1, c2, hc2, hkey) := Fn_eq_deriv_diff hn hf
    ↪ hfc
114    rw [hkey]
115    have hlt : c2 < c1 := lt_trans hc2.2 hc1.1
116    have hIc : Icc c2 c1 ⊆ Icc (-1 : ℝ) 1 :=
    ↪ Icc_subset_Icc (by linarith [hc2.1, neg_le_neg hln]) (by
    ↪ linarith [hc1.2])
118    -- Apply MVT a third time: to deriv f on [c2, c1]
119    obtain (c3, hc3, hc3_eq) := exists_deriv_eq_slope (deriv f)
    ↪ hlt

```

```

115     (hfc'.mono hIc) ((hf'.mono hIc).mono Ioo_subset_Icc_self)
116 -- deriv (deriv f) c3 = (deriv f c1 - deriv f c2) / (c1 -
    ↪ c2)
117 -- So |deriv f c1 - deriv f c2| = |deriv (deriv f) c3| * |c1
    ↪ - c2|
118 have hc3_in : c3 ∈ Icc (-1 : ℝ) 1 := hIc
    ↪ (Ioo_subset_Icc_self hc3)
119 have h_diff : deriv f c1 - deriv f c2 = deriv (deriv f) c3 *
    ↪ (c1 - c2) := by
120     have hsub : c1 - c2 ≠ 0 := sub_ne_zero.mpr hlt.ne'
121     rw [hc3_eq, div_mul_cancel0 _ hsub]
122     rw [h_diff, abs_mul]
123 have hM : 0 ≤ M := le_trans (abs_nonneg _) (hbound 0 {by
    ↪ linarith, by linarith})
124 have hc_diff_nn : 0 ≤ c1 - c2 := sub_nonneg.mpr hlt.le
125 have hc_diff_le : c1 - c2 ≤ 2 / ↑n := by
126     have h1 : c1 - c2 ≤ 1 / ↑n - (-(1 / ↑n)) :=
127         sub_le_sub (le_of_lt hc1.2) (le_of_lt hc2.1)
128     have h2 : (1 : ℝ) / ↑n - (-(1 / ↑n)) = 2 / ↑n := by ring
129     linarith
130 calc |deriv (deriv f) c3| * |c1 - c2|
131     ≤ M * |c1 - c2| := mul_le_mul_of_nonneg_right (hbound c3
    ↪ hc3_in) (abs_nonneg _)
132     = M * (c1 - c2) := by rw [abs_of_nonneg hc_diff_nn]
133     ≤ M * (2 / ↑n) := mul_le_mul_of_nonneg_left hc_diff_le
    ↪ hM
134     = 2 * M / ↑n := by ring
135
136 --
    ↪ =====
137 -- Part (b):  $F_n(f) \rightarrow 0$  for  $C^1$  functions
138 --
    ↪ =====
139
140 theorem section17_task05b_weak_in_C1
141   {f : ℝ → ℝ}
142   (hf : DifferentiableOn ℝ f (Icc (-1) 1))
143   (hfc : ContinuousOn f (Icc (-1) 1))
144   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1)) :
145   Tendsto (fun n : ℕ => F n f n) atTop (nhds 0) := by
146   rw [Metric.tendsto_atTop]
147   intro ε hε
148   have h0 : (0 : ℝ) ∈ Icc (-1 : ℝ) 1 := {by linarith, by
    ↪ linarith}
149   obtain ⟨δ, hδ, hδ_spec⟩ := Metric.continuousWithinAt_iff.mp
150     (hfc'.continuousWithinAt h0) (ε / 2) (half_pos hε)
151   obtain ⟨N, hN⟩ := exists_nat_gt (1 / δ)
152   refine ⟨max 1 N, fun n hn => ?_⟩
153   have hn1 : 0 < n := by omega
154   have hn' : (0 : ℝ) < n := Nat.cast_pos.mpr hn1
155   have hln : 1 / (n : ℝ) ≤ 1 := by rw [div_le_one hn'];
    ↪ exact_mod_cast (by omega : 1 ≤ n)

```

```

156 obtain ⟨c₁, hc₁, c₂, hc₂, hkey⟩ := Fn_eq_deriv_diff hn1 hf
    ↪ hfc
157 rw [hkey, dist_zero_right, Real.norm_eq_abs]
158 have hI1 : Icc 0 (1 / (n : ℝ)) ⊆ Icc (-1 : ℝ) 1 :=
    ↪ Icc_subset_Icc (by linarith) hln
159 have hI2 : Icc (-(1 / (n : ℝ))) 0 ⊆ Icc (-1 : ℝ) 1 :=
    ↪ Icc_subset_Icc (neg_le_neg hln) (by linarith)
160 -- Key: dist c₁ 0 < δ and dist c₂ 0 < δ
161 have hN_le : (N : ℝ) ≤ n := by exact_mod_cast (show N ≤ n by
    ↪ omega)
162 have h_ln_lt_δ : 1 / (↑n : ℝ) < δ := by
163   rw [div_lt_iff₀ hn']
164   have h : 1 / δ < ↑n := lt_of_lt_of_le hN hN_le
165   rw [div_lt_iff₀ hδ] at h
166   linarith [mul_comm (↑n : ℝ) δ]
167 have hc₁_dist : dist c₁ 0 < δ := by
168   rw [Real.dist_eq, sub_zero, abs_of_pos hc₁.1]; linarith
    ↪ [hc₁.2]
169 have hc₂_dist : dist c₂ 0 < δ := by
170   rw [Real.dist_eq, sub_zero, abs_of_neg hc₂.2]; linarith
    ↪ [hc₂.1]
171 have hd1 := hδ_spec (hI1 (Ioo_subset_Icc_self hc₁)) hc₁_dist
172 have hd2 := hδ_spec (hI2 (Ioo_subset_Icc_self hc₂)) hc₂_dist
173 rw [Real.dist_eq] at hd1 hd2
174 calc |deriv f c₁ - deriv f c₂|
175   ≤ |deriv f c₁ - deriv f 0| + |deriv f 0 - deriv f c₂| :=
    ↪ by
176   rw [show deriv f c₁ - deriv f c₂ =
177     (deriv f c₁ - deriv f 0) + (deriv f 0 - deriv f c₂)
    ↪ from by ring]
178   exact abs_add_le _ _
179   _ < ε / 2 + ε / 2 := by linarith [abs_sub_comm (deriv f
    ↪ c₂) (deriv f 0)]
180   _ = ε := add_halves ε

```

6. Пусть  $F_n(x) = n^2 \cdot [x(\frac{1}{n}) + x(-\frac{1}{n}) - 2x(0)]$ ,  $F(x) = x''(0)$ . Показать, что последовательность  $\{F_n\}$ :

- a) сходится к  $F$  по норме в  $C^{(3)}[-1, 1]$ ;
- b) сходится слабо к  $F$  в  $C^{(2)}[-1, 1]$ , но не сходится сильно;

c) не сходится даже слабо в  $C^{(1)}[-1, 1]$ .

#### Формализация на Lean 4

```

1  /-
2  Section 17, Task 06.
3   $G_n(f) = n^2[f(1/n) + f(-1/n) - 2f(0)]$ .
4
5  (a)  $|G_n(f) - f''(0)| \leq M/n$  for  $C^3$  (norm convergence in  $C^{3*}$ ).
6  (b)  $G_n(f) \rightarrow f''(0)$  for  $C^2$  functions (weak convergence in
     $\hookrightarrow C^{2*}$ ).
7  (c) Not weakly convergent in  $C^{1*}$ :  $G_n(|\cdot|^{\{3/2\}}) = 2\sqrt{n} \rightarrow \infty$ .
8  -/
9  import FunAn.Section17_HahnBanach.Task05
10 import Mathlib.Tactic
11
12 open Filter Set Real
13 open scoped Topology
14
15 noncomputable section
16
17 /-- The functional  $G_n$ :  $G_n(f) = n^2(f(1/n) + f(-1/n) - 2f(0))$ .
18 Second-order symmetric difference quotient (approximates
     $\hookrightarrow f''(0)$ ). -/
19 def Gn (f : ℝ → ℝ) (n : ℕ) : ℝ :=
20   (n : ℝ) ^ 2 * (f (1 / (n : ℝ)) + f (-1 / (n : ℝ))) - 2 * f
     $\hookrightarrow$  0)
21
22 --
23  $\hookrightarrow$  =====
24 -- Part (c):  $G_n(|\cdot|^{\{3/2\}}) = 2\sqrt{n} \rightarrow \infty$  (not weak in  $C^1$ )
25  $\hookrightarrow$  =====
26
27 private lemma Gn_abs_three_halves (n : ℕ) (hn : 0 < n) :
28   Gn (fun t => |t| * √|t|) n = 2 * √↑n := by
29   have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn
30   simp only [Gn, abs_zero, mul_zero, sqrt_zero, sub_zero]
31   have h1 : |1 / (↑n : ℝ)| = 1 / ↑n := abs_of_nonneg (by
     $\hookrightarrow$  positivity)
32   rw [h1, show |-(1 / (↑n : ℝ))| = 1 / ↑n from by rw
     $\hookrightarrow$  [abs_neg]; exact h1, ← two_mul,
    one_div, sqrt_inv]
33   have : (↑n : ℝ) ≠ 0 := ne_of_gt hn'
34   have : √(↑n : ℝ) ≠ 0 := ne_of_gt (sqrt_pos.mpr hn')
35   field_simp
36   nlinarith [sq_sqrt hn'.le]
37
38 /--  $G_n(|\cdot|^{\{3/2\}}) = 2\sqrt{n} \rightarrow \infty$ : witnesses non-convergence.
39 The function  $|x|^{\{3/2\}} = |x| \cdot \sqrt{|x|}$  is  $C^1$  (derivative
     $\hookrightarrow (3/2)\sqrt{|x|} \cdot \text{sgn}(x)$ ),

```

```

40   so  $G_n$  is not weakly convergent in  $C^1[-1,1]^*$ . -/
41 theorem section17_task06c_not_weak_in_C1 :
42   Tendsto (fun n : ℕ => Gn (fun t => |t| * √|t|) n) atTop
43     ↪ atTop := by
44     apply Filter.tendsto_atTop_atTop.mpr
45     intro b
46     refine ⟨max 1 ([(b / 2) ^ 2]_+ + 1), fun n hn => ?_⟩
47     have hn1 : 0 < n := by omega
48     rw [Gn_abs_three_halves n hn1]
49     by_cases hb : b ≤ 0
50     · linearith [mul_nonneg two_pos.le (sqrt_nonneg (↑n : ℝ))]
51     · push Not at hb
52     have hb2 : (0 : ℝ) < b / 2 := by linearith
53     have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn1
54     have hsq : (b / 2) ^ 2 < (↑n : ℝ) := by
55       have : n ≥ [(b / 2) ^ 2]_+ + 1 := by omega
56       calc (b / 2) ^ 2 ≤ ↑([(b / 2) ^ 2]_+) := Nat.le_ceil _
57       _ < ↑([(b / 2) ^ 2]_+ + 1) := by push_cast; linearith
58       _ ≤ (↑n : ℝ) := by exact_mod_cast <_>
59     linearith [show b / 2 < √(↑n : ℝ) from by
60       rw [← sqrt_sq hb2.le]; exact sqrt_lt_sqrt (sq_nonneg _)
61       ↪ hsq]
62
63 --
64 ↪ =====
65 -- Part (b):  $G_n(f) \rightarrow f''(0)$  for  $C^2$  functions (via Cauchy MVT +
66 ↪ MVT)
67 --
68 ↪ =====
69
70 /-- Key identity:  $G_n(f) = f''(d)$  for some  $d \in (-1/n, 1/n)$ .
71 Step 1: Cauchy MVT on  $F(x)=f(x)+f(-x)$ ,  $G(x)=x^2$  gives  $G_n(f)$ 
72 ↪ =  $(f'(c)-f'(-c))/(2c)$ .
73 Step 2: MVT on  $f'$  on  $[-c,c]$  gives  $(f'(c)-f'(-c))/(2c) =$ 
74 ↪  $f''(d)$ . -/
75
76 private lemma Gn_eq_second_deriv {f : ℝ → ℝ} {n : ℕ} (hn : 0 <
77 ↪ n)
78   (hf : DifferentiableOn ℝ f (Icc (-1) 1))
79   (hf' : DifferentiableOn ℝ (deriv f) (Icc (-1) 1))
80   (hfc : ContinuousOn f (Icc (-1) 1))
81   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1)) :
82   ∃ d ∈ Ioo (-(1 / (↑n : ℝ))) (1 / (↑n : ℝ)),
83     Gn f n = deriv (deriv f) d := by
84     have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn
85     have hne : (↑n : ℝ) ≠ 0 := ne_of_gt hn'
86     have hln_pos : (0 : ℝ) < 1 / ↑n := div_pos one_pos hn'
87     have hln_le : 1 / (↑n : ℝ) ≤ 1 := by
88       rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
89     -- Subset inclusions
90     have hI : Icc 0 (1 / (↑n : ℝ)) ⊆ Icc (-1 : ℝ) 1 :=
91       Icc_subset_Icc (by linearith) hln_le

```

```

83 have hIn : Icc (-(1 / (↑n : ℝ))) 0 ≤ Icc (-1 : ℝ) 1 :=
84   Icc_subset_Icc (neg_le_neg hIn_le) (by linarith)
85 -- F(x) = f(x) + f(-x) is continuous on [0, 1/n]
86 have hFc : ContinuousOn (fun x => f x + f (-x)) (Icc 0 (1 /
87   ↪ ↑n)) :=
88   (hfc.mono hI).add ((hfc.mono hIn).comp continuousOn_neg
89     (fun x hx => ⟨neg_le_neg hx.2, by linarith [hx.1]⟩))
89 -- F differentiable on (0, 1/n)
90 have hFd : DifferentiableOn ℝ (fun x => f x + f (-x)) (Ioo 0
91   ↪ (1 / ↑n)) :=
92   ((hf.mono hI).mono Ioo_subset_Icc_self).add
93     ((hf.mono hIn).mono Ioo_subset_Icc_self).comp
94     (differentiable_neg.differentiableOn
95       (fun x hx => ⟨by linarith [hx.2], neg_lt_zero.mpr
96         ↪ hx.1⟩))
95 -- Step 1: Apply Cauchy's MVT
96 obtain ⟨c, hc, hc_eq⟩ := exists_ratio_deriv_eq_ratio_slope
97   (fun x => f x + f (-x)) hIn_pos hFc hFd
98   (fun x => x ^ 2) (by fun_prop) (differentiable_pow
99     ↪ 2).differentiableOn
100 have hc_pos : (0 : ℝ) < c := hc.1
101 have hc_lt : c < 1 / ↑n := hc.2
102 have h2c_pos : (0 : ℝ) < 2 * c := by linarith
103 have h2c_ne : (2 * c : ℝ) ≠ 0 := ne_of_gt h2c_pos
104 -- c and -c are in (-1, 1)
105 have hc11 : c ∈ Ioo (-1 : ℝ) 1 := ⟨by linarith, by linarith
106   ↪ [hIn_le]⟩
107 have hnc11 : -c ∈ Ioo (-1 : ℝ) 1 := ⟨by nlinarith [hIn_le],
108   ↪ by linarith⟩
109 have hda : DifferentiableAt ℝ f c :=
110   hf.differentiableAt (Icc_mem_nhds hc11.1 hc11.2)
111 have hdna : DifferentiableAt ℝ f (-c) :=
112   hf.differentiableAt (Icc_mem_nhds hnc11.1 hnc11.2)
113 -- Compute HasDerivAt for F and G at c (to rewrite deriv in
114   ↪ hc_eq)
115 have hF_at : HasDerivAt (fun x => f x + f (-x)) (deriv f c -
116   ↪ deriv f (-c)) c := by
117   have h1 := hda.hasDerivAt
118   have h2 : HasDerivAt (fun x => f (-x)) (-deriv f (-c)) c
119     := by
120     have := hdna.hasDerivAt.comp c (hasDerivAt_neg c)
121     simp [Function.comp_def] using this
122     exact h1.add h2
123 have hG_at : HasDerivAt (fun x : ℝ => x ^ 2) (2 * c) c := by
124   have := hasDerivAt_pow 2 c; simp [pow_one] using this
125 -- Replace derivatives in Cauchy MVT equation
126 rw [hF_at.deriv, hG_at.deriv] at hc_eq
127 -- hc_eq now has form involving function values and
128   ↪ derivative values
129 -- Clean it up to: (1/n)^2 * (f'(c)-f'(-c)) =
130   ↪ (f(1/n)+f(-1/n)-2f(0)) * 2c

```

```

123 have h_cauchy : (1 / ↑n : ℝ) ^ 2 * (deriv f c - deriv f
    ↪ (-c)) =
124   (f (1 / ↑n) + f (-(1 / ↑n)) - 2 * f 0) * (2 * c) := by
125     have := hc_eq
126     simp only [neg_zero] at this
127     nlinarith
128   -- Step 2: Apply MVT to deriv f on [-c, c]
129   have hIc : Icc (-c) c ⊆ Icc (-1 : ℝ) 1 :=
130     Icc_subset_Icc (by linarith [hnc11.1]) (by linarith
    ↪ [hc11.2])
131   obtain ⟨d, hd, hd_eq⟩ := exists_deriv_eq_slope (deriv f) (by
    ↪ linarith : -c < c)
132   (hfc'.mono hIc) ((hf'.mono hIc).mono Ioo_subset_Icc_self)
133   -- hd_eq: deriv (deriv f) d = (deriv f c - deriv f (-c)) /
    ↪ (c - (-c))
134   -- Combine: f'(c)-f'(-c) = f''(d) * 2c, substitute into
    ↪ h_cauchy
135   have h_sub : c - (-c) = 2 * c := by ring
136   have h_diff_eq : deriv f c - deriv f (-c) = deriv (deriv f)
    ↪ d * (2 * c) := by
137     rw [hd_eq, h_sub, div_mul_cancel0 _ h2c_ne]
138   -- From h_cauchy with substitution: (1/n)^2 * f''(d) * 2c =
    ↪ S * 2c
139   -- Cancel 2c: (1/n)^2 * f''(d) = S
140   -- So Gn = n^2 * S = n^2 * (1/n)^2 * f''(d) = f''(d)
141   refine ⟨d, (by linarith [hd.1], by linarith [hd.2]), ?_⟩
142   simp only [Gn]
143   have h_S : f (1 / ↑n) + f (-(1 / ↑n)) - 2 * f 0 =
144     (1 / ↑n) ^ 2 * deriv (deriv f) d := by
145     have h := h_cauchy
146     rw [h_diff_eq] at h
147     -- h: (1/n)^2 * (f''(d) * 2c) = S * 2c
148     nlinarith
149   rw [h_S]
150   have : (↑n : ℝ) ^ 2 * ((1 / ↑n) ^ 2 * deriv (deriv f) d) =
    ↪ deriv (deriv f) d := by
151     field_simp
152     linarith
153
154 /-- Weak convergence in C^2*: G_n(f) → f''(0) for each C^2
    ↪ function.
155   Proof: G_n(f) = f''(d_n) with d_n → 0, so result follows from
    ↪ continuity of f''. -/
156 theorem section17_task06b_weak_in_C2
157   {f : ℝ → ℝ}
158   (hf : DifferentiableOn ℝ f (Icc (-1) 1))
159   (hf' : DifferentiableOn ℝ (deriv f) (Icc (-1) 1))
160   (hfc : ContinuousOn f (Icc (-1) 1))
161   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
162   (hfc'' : ContinuousOn (deriv (deriv f)) (Icc (-1) 1)) :
163   Tendsto (fun n : ℕ => Gn f n) atTop (nhds (deriv (deriv f)
    ↪ 0)) := by

```

```

164 rw [Metric.tendsto_atTop]
165 intro ε hε
166 have h0_mem : (0 : ℝ) ∈ Icc (-1 : ℝ) 1 := (by linarith, by
  ↪ linarith)
167 obtain ⟨δ, hδ, hδ_spec⟩ := Metric.continuousWithinAt_iff.mp
168   (hfc''.continuousWithinAt h0_mem) ε hε
169 obtain ⟨N, hN⟩ := exists_nat_gt (1 / δ)
170 refine ⟨max 1 N, fun n hn => ?_⟩
171 have hn1 : 0 < n := by omega
172 have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn1
173 have h1n_le : 1 / (↑n : ℝ) ≤ 1 := by
174   rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
175 obtain ⟨d, hd, hGn_eq⟩ := Gn_eq_second_deriv hn1 hf hf' hfc
  ↪ hfc'
176 rw [hGn_eq]
177 have hd_mem : d ∈ Icc (-1 : ℝ) 1 :=
178   (by linarith [hd.1, neg_le_neg h1n_le], by linarith [hd.2,
  ↪ h1n_le])
179 have hN_le : (N : ℝ) ≤ ↑n := by exact_mod_cast (show N ≤ n
  ↪ by omega)
180 have h1n_lt_δ : 1 / (↑n : ℝ) < δ := by
181   rw [div_lt_iff0 hn']
182   have : 1 / δ < ↑n := lt_of_lt_of_le hN hN_le
183   rwa [div_lt_iff0 hδ, mul_comm] at this
184 exact hδ_spec hd_mem (by
185   rw [Real.dist_eq, sub_zero]
186   exact abs_lt.mpr (by linarith [hd.1], by linarith [hd.2]))
187
188 --
189 ↪ =====
189 -- Part (a):  $|G_n(f) - f''(0)| \leq M/n$  for  $C^3$  (norm convergence
189 ↪ in  $C^3$ *)
190 --
191 ↪ =====
192
192 /-  $C^3$  bound:  $|G_n(f) - f''(0)| \leq M/n$  where  $M = \sup|f'''|$ .
193 From key identity  $G_n(f) = f''(d)$  with  $|d| < 1/n$ , MVT gives
194  $|f''(d) - f''(0)| \leq |f'''(\xi)| \cdot |d| \leq M/n$ . -/
195 theorem section17_task06a_C3_bound
196   {f : ℝ → ℝ} {M : ℝ}
197   (hf : DifferentiableOn ℝ f (Icc (-1) 1))
198   (hf' : DifferentiableOn ℝ (deriv f) (Icc (-1) 1))
199   (hf'' : DifferentiableOn ℝ (deriv (deriv f)) (Icc (-1) 1))
200   (hfc : ContinuousOn f (Icc (-1) 1))
201   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
202   (hfc'' : ContinuousOn (deriv (deriv f)) (Icc (-1) 1))
203   (hbound : ∀ x ∈ Icc (-1 : ℝ) 1, |deriv (deriv (deriv f))
  ↪ x| ≤ M)
204   {n : ℕ} (hn : 0 < n) :
205   |Gn f n - deriv (deriv f) 0| ≤ M / ↑n := by
206   have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn

```

```

207 have hln_le : 1 / (↑n : ℝ) ≤ 1 := by
208   rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
209 have h0_mem : (0 : ℝ) ∈ Icc (-1 : ℝ) 1 := (by linarith, by
    ↪ linarith)
210 have hM_nn : 0 ≤ M := le_trans (abs_nonneg _) (hbound 0
    ↪ h0_mem)
211 obtain ⟨d, hd, hGn_eq⟩ := Gn_eq_second_deriv hn hf hf' hfc
    ↪ hfc'
212 rw [hGn_eq]
213 have hd_mem : d ∈ Icc (-1 : ℝ) 1 :=
214   (by linarith [hd.1, neg_le_neg hln_le], by linarith [hd.2,
    ↪ hln_le])
215 -- Apply MVT to f'' between 0 and d to bound |f''(d) -
    ↪ f''(0)|
216 by_cases hd0 : 0 ≤ d
217 · -- Case d ≥ 0
218   rcases eq_or_lt_of_le hd0 with rfl | hd_pos
219   · simp [div_nonneg hM_nn hn'.le]
220   · have hI0d : Icc 0 d ⊆ Icc (-1 : ℝ) 1 :=
221     Icc_subset_Icc (by linarith) (by linarith [hd.2,
    ↪ hln_le])
222   obtain ⟨ξ, hξ, hξ_eq⟩ := exists_deriv_eq_slope (deriv
    ↪ (deriv f)) hd_pos
223     (hfc''.mono hI0d) ((hf''.mono hI0d).mono
    ↪ Ioo_subset_Icc_self)
224   rw [sub_zero] at hξ_eq
225   have hξ_mem : ξ ∈ Icc (-1 : ℝ) 1 := hI0d
226     ↪ (Ioo_subset_Icc_self hξ)
227   have h_diff : deriv (deriv f) d - deriv (deriv f) 0 =
228     deriv (deriv (deriv f)) ξ * d := by
229     rw [hξ_eq]; field_simp
230     rw [h_diff, abs_mul]
231   calc |deriv (deriv (deriv f)) ξ| * |d|
232     ≤ M * |d| := by apply mul_le_mul_of_nonneg_right
    ↪ (hbound ξ hξ_mem) (abs_nonneg _)
233   _ = M * d := by rw [abs_of_nonneg hd0]
234   _ ≤ M * (1 / ↑n) := by apply mul_le_mul_of_nonneg_left
    ↪ (le_of_lt hd.2) hM_nn
235   _ = M / ↑n := by ring
236 · -- Case d < 0
237   push Not at hd0
238   have hId0 : Icc d 0 ⊆ Icc (-1 : ℝ) 1 :=
239     Icc_subset_Icc (by linarith [hd.1, neg_le_neg hln_le])
    ↪ (by linarith)
240   obtain ⟨ξ, hξ, hξ_eq⟩ := exists_deriv_eq_slope (deriv
    ↪ (deriv f)) hd0
241     (hfc''.mono hId0) ((hf''.mono hId0).mono
    ↪ Ioo_subset_Icc_self)
242   have hξ_mem : ξ ∈ Icc (-1 : ℝ) 1 := hId0
    ↪ (Ioo_subset_Icc_self hξ)
243   have h_diff : deriv (deriv f) d - deriv (deriv f) 0 =

```

```

243     deriv (deriv (deriv f)) ξ * d := by
244     have hne : (0 : ℝ) - d ≠ 0 := sub_ne_zero.mpr (ne_of_gt
      ↪ hd0)
245     rw [show deriv (deriv f) d - deriv (deriv f) 0 =
246         -(deriv (deriv f) 0 - deriv (deriv f) d) from by ring,
      ↪ hξ_eq]
247     field_simp; ring
248     rw [h_diff, abs_mul]
249     calc |deriv (deriv (deriv f)) ξ| * |d|
250         ≤ M * |d| := by apply mul_le_mul_of_nonneg_right
      ↪ (hbound ξ hξ_mem) (abs_nonneg _)
251     _ = M * (-d) := by rw [abs_of_neg hd0]
252     _ ≤ M * (1 / ↑n) := by apply mul_le_mul_of_nonneg_left _
      ↪ hM_nn; linarith [hd.1]
253     _ = M / ↑n := by ring

```

7. Доказать, что слабая сходимость в  $l^2$  сильнее покомпонентной сходимости.

#### Формализация на Lean 4

```

1  /-
2    Section 17, Task 07.
3    Weak convergence in  $\square^2$  implies coordinatewise convergence.
4
5    Proof:  $x(k) = \langle x, e_k \rangle$  where  $e_k = lp.single\ 2\ k\ 1$ .
6    Weak convergence applied to  $y = e_k$  gives  $x_n(k) \rightarrow x_0(k)$ .
7  -/
8  import Mathlib.Analysis.InnerProductSpace.l2Space
9  import Mathlib.Tactic
10
11 open scoped InnerProductSpace
12 open Filter
13
14 private theorem real_inner_one (a : ℝ) :  $\langle a, (1 : \mathbb{R}) \rangle_{\mathbb{R}} = a :=$ 
15   by
16     simp [Inner.inner]
17
18 /-- Weak convergence in  $\square^2$  implies coordinatewise convergence.
19   -/
20 theorem weak_implies_coordinatewise_l2
21   {x : ℕ → lp (fun _ : ℕ => ℝ) 2} {x₀ : lp (fun _ : ℕ => ℝ)
22     → 2}
23   (hweak : ∀ y : lp (fun _ : ℕ => ℝ) 2,
24     Tendsto (fun n =>  $\langle x\ n, y \rangle_{\mathbb{R}}$ ) atTop (nhds  $\langle x_0, y \rangle_{\mathbb{R}}$ ))
25   (k : ℕ) :
26     Tendsto (fun n => (x n : ℕ → ℝ) k) atTop (nhds ((x₀ : ℕ →
27       → ℝ) k)) := by
28   have h := hweak (lp.single 2 k (1 : ℝ))
29   simp only [lp.inner_single_right, real_inner_one] at h
30   exact h

```

8. Какие из последовательностей сходятся сильно, слабо в  $l^2$ ?

- a)  $x_n = (1, \frac{1}{2}, \dots, \frac{1}{n}, 0, \dots)$ ;
- b)  $x_n = (\underbrace{0, \dots, 0}_{n-1}, 1, 0, \dots)$ ;
- c)  $x_n = (0, 0, \dots, 0, 1, \frac{1}{2}, \frac{1}{3}, \dots)$  (начало с  $n$ -го места);
- d)  $x_n = (0, \dots, 0, \frac{1}{n+1}, \frac{1}{n+2}, \dots)$ ;
- e)  $x_n = (\underbrace{1, \dots, 1}_n, \frac{1}{n}, \frac{1}{n+1}, \dots)$ ;
- f)  $x_n = (\frac{1}{n}, \frac{1}{n+1}, \dots)$ .

9. Исследовать на сильную и слабую сходимости следующие последовательности в  $L^2(0, 1)$ :

a)  $x_n(t) = t^n$ ;

b)  $x_n(t) = \begin{cases} \sqrt{n}, & \text{если } t \in [0, \frac{1}{n}], \\ 0, & \text{если } t \in (\frac{1}{n}, 1]; \end{cases}$

c)  $x_n(t) = e^{i\sqrt{n}t}$ ;

d)  $x_n(t) = \sin(t \ln n)$ ;

e)  $x_n(t) = \cos(n^2 t)$ ;

f)  $x_n(t) = \sin(e^n t)$ ;

g)  $x_n(t) = \begin{cases} \sqrt{n} \cdot (1 - nt), & \text{если } t \in [0, \frac{1}{n}), \\ 0, & \text{если } t \in [\frac{1}{n}, 1]; \end{cases}$

h)  $x_n(t) = t^n - t^{n+1}$ ;

i)  $x_n(t) = \begin{cases} 2n(1 - nt), & \text{если } t \in [0, \frac{1}{n}), \\ 0, & \text{если } t \in [\frac{1}{n}, 1]. \end{cases}$

10. Пусть последовательность  $\{x_n\} \subset H$  слабо сходится к  $x$ . Доказать, что  $\{x_n\}$  будет сходиться сильно при выполнении одного из условий:

a)  $\|x_n\| \rightarrow \|x\|$ ;

b)  $\|x_n\| \leq \|x\| \forall n > N_0$ ;

c)  $\lim_{n \rightarrow \infty} \|x_n\| \leq \|x\|.$

#### Формализация на Lean 4

```

1  /-
2    Section 17, Task 10.
3    In a Hilbert space H, weak convergence  $x_n \rightarrow x$  plus  $\|x_n\| \rightarrow$ 
4     $\|x\|$ 
5    implies strong convergence  $x_n \rightarrow x$ .
6    Proof:  $\|x_n - x\|^2 = \|x_n\|^2 - 2\operatorname{Re}\langle x_n, x \rangle + \|x\|^2 \rightarrow \|x\|^2 - 2\|x\|^2 +$ 
7     $\|x\|^2 = 0$ .
8    (Radon-Riesz property, which holds in all uniformly convex
9    spaces.)
10  -/
11  import Mathlib.Analysis.InnerProductSpace.Basic
12  import Mathlib.Tactic
13
14  open scoped InnerProductSpace
15  open Filter
16
17  /- Radon-Riesz property in a real inner product space: weak
18  convergence
19  plus norm convergence implies strong convergence. -/
20  theorem weak_plus_norm_implies_strong
21    {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℝ H]
22    {x : ℕ → H} {x₀ : H}
23    (hweak : ∀ y : H, Tendsto (fun n => ⟨x n, y⟩_ℝ) atTop
24      ↪ (nhds ⟨x₀, y⟩_ℝ))
25    (hnorm : Tendsto (fun n => ‖x n‖) atTop (nhds ‖x₀‖)) :
26    Tendsto x atTop (nhds x₀) := by
27  rw [tendsto_iff_norm_sub_tendsto_zero, Metric.tendsto_atTop]
28  intro ε hε
29  -- Show  $\|x n - x_0\|^2 \rightarrow 0$ 
30  have hnorm_sq : Tendsto (fun n => ‖x n‖ ^ 2) atTop (nhds
31    ↪ (‖x₀‖ ^ 2)) := hnorm.pow 2
32  have hinner : Tendsto (fun n => ⟨x n, x₀⟩_ℝ) atTop (nhds
33    ↪ (‖x₀‖ ^ 2)) := by
34    have := hweak x₀; rwa [real_inner_self_eq_norm_sq] at this
35  have hsq : Tendsto (fun n => ‖x n - x₀‖ ^ 2) atTop (nhds 0)
36    ↪ := by
37    have hcomb : Tendsto (fun n => ‖x n‖ ^ 2 - 2 * ⟨x n, x₀⟩_ℝ
38      ↪ + ‖x₀‖ ^ 2)
39      atTop (nhds (‖x₀‖ ^ 2 - 2 * ‖x₀‖ ^ 2 + ‖x₀‖ ^ 2)) :=
40      (hnorm_sq.sub (hinner.const_mul 2)).add
41      ↪ tendsto_const_nhds
42  rw [show ‖x₀‖ ^ 2 - 2 * ‖x₀‖ ^ 2 + ‖x₀‖ ^ 2 = 0 from by
43    ↪ ring] at hcomb
44  exact hcomb.congr (fun n => (norm_sub_sq_real (x n)
45    ↪ x₀).symm)
46  -- Extract N from hsq at  $\varepsilon^2$ 

```

```

35 obtain ⟨N, hN⟩ := (Metric.tendsto_atTop.mp hsq) (ε ^ 2) (by
    ↪ positivity)
36 refine ⟨N, fun n hn => ?_⟩
37 have h1 := hN n hn
38 simp only [dist_zero_right] at h1
39 rw [Real.norm_of_nonneg (sq_nonneg _)] at h1
40 simp only [dist_zero_right]
41 rw [Real.norm_of_nonneg (norm_nonneg _)]
42 nlinarith [sq_nonneg ‖x n - x_0‖, sq_nonneg ε, sq_abs ‖x n -
    ↪ x_0‖]

```

## 18. СОПРЯЖЕННЫЕ ОПЕРАТОРЫ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения сопряженного оператора к линейному непрерывному оператору в нормированном и гильбертовом пространствах.
2. Какая связь между сопряженным оператором в  $H$  и сопряженным оператором в нормированном пространстве, когда в качестве нормированного пространства рассматривается  $H$ ?
3. Перечислить свойства сопряженного оператора.

### ЗАДАЧИ

1. Найти сопряженный оператор к оператору  $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ , определяемому равенством

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \dots \\ \eta_m \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \\ \dots \\ \xi_n \end{pmatrix}$$

2. Найти сопряженный оператор к проектору  $P : H \rightarrow L$ , где  $L$  — замкнутое подпространство в  $H$ .
3. Найти сопряженный оператор к оператору  $A : X \rightarrow X$ , если
  - a)  $X = l^2$ ,  $Ax = (0, \xi_1, \xi_2, \dots)$ ;
  - b)  $X = l^2$ ,  $Ax = (\alpha_1 \xi_1, \alpha_2 \xi_2, \dots)$ ;
  - c)  $X = l^1$ ,  $Ax = (\xi_1, \dots, \xi_n, 0, 0, \dots)$ ;
  - d)  $X = l^p$ ,  $Ax = (\underbrace{0, \dots, 0}_n, \xi_1, 0, 0, \dots)$ ,  $1 \leq p < \infty$ ;
  - e)  $X = l^p$ ,  $Ax = (\xi_2, \xi_1, \xi_4, \xi_3, \dots)$ ;
  - f)  $X = L^2(0, 1)$ ,  $(Ax)(t) = \int_0^t x(\xi) d\xi$ ;
  - g)  $X = L^2(0, 1)$ ,  $(Ax)(t) = x(t^\alpha)$ ;

- h)  $X = L^2(0, 1)$ ,  $(A_\lambda x)(t) = \begin{cases} x(t), & \text{если } 0 \leq t \leq \lambda, \\ 0, & \text{если } \lambda < t \leq 1 \end{cases}$ ,  
где  $\lambda \in (0, 1)$  — фиксированное число;
- i)  $X = H$ ,  $Ax = (x, y) \cdot z$ , где  $y, z \in H$  фиксированы;
- j)  $X = L^2(0, 1)$ ,  $(Ax)(t) = \int_0^1 t \cdot x(s) ds$ .

4. Пусть оператор  $\Phi : \mathcal{L}(X, Y) \rightarrow \mathcal{L}(Y^*, X^*)$  определяется равенством  $\Phi(A) = A^*$ , где  $X, Y$  — нормированные пространства. Доказать, что  $\Phi$  непрерывен.

#### Формализация на Lean 4

```

1  /-
2  Section 18, Task 4.
3  The map  $\Phi : \mathcal{L}(X, Y) \rightarrow \mathcal{L}(Y^*, X^*)$  defined by  $\Phi(A) = A^*$  is a
4  ↪ continuous
5  linear operator. In fact it is an isometry:  $\|A^*\| = \|A\|$ .
6  This follows from  $\|A^*\| = \sup\{|A^*f(x)| : \|f\| \leq 1, \|x\| \leq 1\}$ 
7  =  $\sup\{|f(Ax)| : \|f\| \leq 1, \|x\| \leq 1\} = \sup\{\|Ax\| : \|x\| \leq 1\} = \|A\|$ 
8  by Hahn-Banach.
9
10 In the Hilbert space setting, the adjoint is a linear
11 ↪ isometry equivalence
12 (Mathlib: ContinuousLinearMap.adjoint).
13 -/
14
15 import Mathlib.Analysis.InnerProductSpace.Adjoint
16 import Mathlib.Tactic
17
18 variable {α : Type*} [RCLike α]
19 variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace
20   ↪ α E] [CompleteSpace E]
21 variable {F : Type*} [NormedAddCommGroup F] [InnerProductSpace
22   ↪ α F] [CompleteSpace F]
23
24 /-- The adjoint map  $A \mapsto A^\dagger$  is an isometry:  $\|A^\dagger\| = \|A\|$ . -/
25 theorem adjoint_map_isometry (A : E →L[α] F) :
26   \|ContinuousLinearMap.adjoint A\| = \|A\| :=
27   ContinuousLinearMap.adjoint.norm_map A

```

5. Доказать, что если  $A \in \mathcal{L}(X, Y)$  и  $A^{-1} \in \mathcal{L}(Y, X)$ , то существует  $(A^*)^{-1} \in \mathcal{L}(X^*, Y^*)$ , причём  $(A^*)^{-1} = (A^{-1})^*$ .

#### Формализация на Lean 4

```

1  /-
2    Section 18, Task 5.
3    If A is invertible, then A* is invertible and (A*)-1 =
4      ↪ (A-1)*.
5
6    In Mathlib, for a linear equivalence f : M1 ≃ℓ[R] M2,
7    `LinearEquiv.dualMap_symm` states: (dualMap f).symm =
8      ↪ dualMap f.symm.
9    This is exactly (A*)-1 = (A-1)*.
10  -/
11  import Mathlib.LinearAlgebra.Dual.Defs
12  import Mathlib.Tactic
13
14  variable {R M1 M2 : Type*} [CommSemiring R]
15    [AddCommMonoid M1] [Module R M1] [AddCommMonoid M2] [Module
16      ↪ R M2]
17
18  /-- If A is invertible (a linear equivalence), then A* is
19    ↪ invertible
20      and (A*)-1 = (A-1)*. This is `LinearEquiv.dualMap_symm` in
21      ↪ Mathlib. -/
22  theorem adjoint_inv_eq_inv_adjoint' (f : M1 ≃ℓ[R] M2) :
23    (LinearEquiv.dualMap f).symm = LinearEquiv.dualMap f.symm
24      ↪ :=
25    LinearEquiv.dualMap_symm

```

6. Пусть  $X$  — рефлексивное банахово пространство,  $Y$  — нормированное,  $A \in \mathcal{L}(X, Y)$ . Доказать, что  $(A^*)^* = A$  как оператор из  $X$  в  $Y$ .

#### Формализация на Lean 4

```

1  /-
2    Section 18, Task 6.
3    For a reflexive Banach space  $X$  and  $A \in \mathcal{L}(X, Y)$ ,  $(A^*)^* = A$ 
4     $\hookrightarrow$  under the
5    canonical embedding  $X \rightarrow X^{**}$ . That is, the double adjoint
6     $\hookrightarrow$  recovers  $A$ 
7    when  $X$  is reflexive (the canonical embedding is surjective).
8
9    In the Hilbert space setting, this is
10    $\hookrightarrow$  ContinuousLinearMap.adjoint_adjoint.
11  -/
12  import Mathlib.Analysis.InnerProductSpace.Adjoint
13  import Mathlib.Tactic
14
15  variable {α : Type*} [RCLike α]
16  variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace
17    $\hookrightarrow$  α E] [CompleteSpace E]
18  variable {F : Type*} [NormedAddCommGroup F] [InnerProductSpace
19    $\hookrightarrow$  α F] [CompleteSpace F]
20
21  /- In Hilbert spaces, the double adjoint is the identity:
22    $\hookrightarrow$   $(A^\dagger)^\dagger = A$ . -/
23  theorem double_adjoint_eq (A : E  $\rightarrow$  L[α] F) :
24    ContinuousLinearMap.adjoint (ContinuousLinearMap.adjoint
25      $\hookrightarrow$  A) = A :=
26    ContinuousLinearMap.adjoint_adjoint A

```

## 19. РАВНОМЕРНАЯ, СИЛЬНАЯ, СЛАБАЯ СХОДИМОСТИ ПОСЛЕДОВАТЕЛЬНОСТЕЙ ОПЕРАТОРОВ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения равномерной, сильной и слабой сходимостей последовательности линейных непрерывных операторов.
2. Сформулировать принцип равномерной ограниченности Банаха-Штейнхауса.
3. Сформулировать критерий сильной сходимости линейных непрерывных операторов.

### ЗАДАЧИ

1. Доказать, что если  $A_n \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^m)$ , то из слабой сходимости последовательности  $\{A_n\}$  следует равномерная сходимость.

#### Формализация на Lean 4

```

1  /-
2    Section 19, Task 1.
3    In finite dimensions, weak (pointwise) convergence of
4    ↪ operators
5    implies norm convergence.
6
7    Proof strategy: Use a finite basis  $v$  of  $E$  and the bound
8     $\|T\| \leq C \cdot \max_i \|T(v_i)\|$  (from Basis.exists_opNorm_le).
9    Pointwise convergence gives convergence on each  $v_i$ ,
10   and the finite maximum of convergent sequences converges.
11  -/
12  import Mathlib.Analysis.Normed.Module.FiniteDimension
13  import Mathlib.Tactic
14
15  variable {α : Type*} [NontriviallyNormedField α]
16  ↪ [CompleteSpace α]
17  variable {E : Type*} [NormedAddCommGroup E] [NormedSpace α E]
18  ↪ [FiniteDimensional α E]
19  variable {F : Type*} [NormedAddCommGroup F] [NormedSpace α F]
20
21  /-- In finite-dimensional spaces, pointwise convergence of
22  ↪ continuous linear maps
23  implies convergence in operator norm. -/
24  theorem pointwise_tendsto_implies_norm_tendsto
25    {f : ℕ → E → L[α] F} {g : E → L[α] F}
26    (h : ∀ x : E, Filter.Tendsto (fun n => f n x) Filter.atTop
27    ↪ (nhds (g x))) :
28    Filter.Tendsto (fun n => \|f n - g\|) Filter.atTop (nhds 0)
29    ↪ := by
30    set v := Module.finBasis α E
31    obtain ⟨C, hCpos, hC⟩ := v.exists_opNorm_le (F := F)
32    rw [Metric.tendsto_atTop]
33    intro ε hε
34    -- Choose δ so that C * δ < ε
35    set δ := ε / (2 * C) with hδ_def
36    have hδpos : (0 : ℝ) < δ := div_pos hε (mul_pos two_pos
37    ↪ hCpos)
38    -- For each basis element i, find N_i with \|f n (v i) - g (v
39    ↪ i)\| < δ for n ≥ N_i
40    have hconv : ∀ i, ∃ N, ∀ n ≥ N, dist (f n (v i)) (g (v i)) <
41    ↪ δ := by
42      intro i
43      exact (Metric.tendsto_atTop.mp (h (v i))) δ hδpos
44    choose N hN using hconv
45    -- Take the max of all N_i

```

```

37 refine (Finset.univ.sup N, fun n hn => ?_)
38 -- For all i, N i ≤ n
39 have hNi : ∀ i, N i ≤ n := fun i =>
40   le_trans (Finset.le_sup (f := N) (Finset.mem_univ i)) hn
41 -- So ||(f n - g)(v i)|| ≤ δ for all i
42 have hle : ∀ i, ||(f n - g) (v i)|| ≤ δ := by
43   intro i
44   rw [ContinuousLinearMap.sub_apply]
45   rw [← dist_eq_norm]
46   exact le_of_lt (hN i n (hNi i))
47 -- By the basis bound: ||f n - g|| ≤ C * δ
48 have hbound := hC hδpos.le hle
49 -- dist ||f n - g|| 0 = ||f n - g|| ≤ C * δ = ε/2 < ε
50 rw [dist_zero_right, Real.norm_of_nonneg (norm_nonneg _)]
51 calc ||f n - g|| ≤ C * δ := hbound
52   _ = ε / 2 := by rw [hδ_def]; field_simp
53   _ < ε := half_lt_self hε

```

2. Пусть  $A_n x = (0, \dots, 0, \frac{\xi_{n+1}}{n+1}, \frac{\xi_{n+2}}{n+2}, \dots)$ ,  $B_n x = (\xi_1, \dots, \xi_n, 0, 0, \dots)$ ,  $C_n x = (\underbrace{0, 0, \dots, 0}_n, \xi_1, 0, 0, \dots)$

— операторы в  $l^2$ . Доказать, что:

- a)  $\{A_n\}$  сходится равномерно к нулевому оператору;
- b)  $\{B_n\}$  сильно сходится к единичному оператору, а равномерно не сходится;

c)  $\{C_n\}$  слабо сходится к нулевому оператору, но сильно не сходится.

#### Формализация на Lean 4

```

1  /-
2  Section 19, Task 02.
3  Examples of operator convergence types:
4  (a) Uniform convergence:  $A_n = (1/n) \cdot \text{Id} \rightarrow 0$  uniformly ( $\|A_n\| =$ 
5      $\hookrightarrow 1/n \rightarrow 0$ ).
6  (b) Strong not uniform:  $P_n = \text{projection onto span}\{e_1, \dots, e_n\}$ 
7      $\hookrightarrow \rightarrow \text{Id}$  strongly
8     but not uniformly ( $\|P_n - \text{Id}\| = 1$  for all  $n$ ).
9  (c) Weak not strong: the basis vectors  $e_n$  converge weakly to
10      $\hookrightarrow 0$ 
11     (coordinatewise) but not strongly ( $\|e_n\| = 1$ ).
12
13  We prove (c) via the  $\square^2$  basis from Task04.
14  For (a), we prove the abstract principle: scalar multiples
15      $\hookrightarrow c_n \cdot \text{Id} \rightarrow 0$ 
16     iff  $c_n \rightarrow 0$ .
17  -/
18  import FunAn.Section19_OperatorConvergence.Task04
19  import Mathlib.Analysis.Normed.Operator.ContinuousLinearMap
20  import Mathlib.Tactic
21
22  open Filter
23
24  /-- (a) Scalar multiples  $c_n \rightarrow 0$  gives uniform operator
25      $\hookrightarrow$  convergence  $c_n \cdot \text{Id} \rightarrow 0$ . -/
26  theorem uniform_convergence_example
27    {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
28    {c : ℕ → ℝ} (hc : Tendsto c atTop (nhds 0)) :
29    Tendsto (fun n => c n • ContinuousLinearMap.id ℝ E) atTop
30       $\hookrightarrow$  (nhds 0) := by
31    rw [Metric.tendsto_atTop]
32    intro ε hε
33    obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp hc ε hε
34    exact ⟨N, fun n hn => by
35      simp only [dist_zero_right]
36      calc ||c n • ContinuousLinearMap.id ℝ E||
37        = |c n| * ||ContinuousLinearMap.id ℝ E|| := by
38          rw [norm_smul, Real.norm_eq_abs]
39        ≤ |c n| * 1 := by gcongr; exact
40          ContinuousLinearMap.norm_id_le
41        = |c n| := mul_one _
42        < ε := by
43          have := hN n hn; rwa [dist_zero_right,
44             $\hookrightarrow$  Real.norm_eq_abs] at this)
45
46  /-- (c) Weak convergence without strong:  $e_n$  in  $\square^2$  (from
47     Task04).

```

```

39   en does not converge strongly to 0 since ||en|| = 1. -/
40   theorem weak_not_strong_example :
41     ¬Tendsto (fun n => lp.single (E := fun _ : ℕ => ℝ) 2 n (1
42       ↪ : ℝ)) atTop (nhds 0) :=
    l2_single_not_tendsto_zero

```

3. Пусть  $A_n x = (\xi_{n+1}, \xi_{n+2}, \dots)$ ,  $A_n : l^2 \rightarrow l^2$ . Доказать, что

- a)  $\{A_n\}$  сходится сильно,  $\{A_n^*\}$  сходится слабо к нулевому оператору;
- b)  $\{A_n A_n^*\}$  не сходится слабо к нулевому оператору;

c)  $\{A_n^* A_n\}$  сходится к нулевому оператору сильно.

#### Формализация на Lean 4

```

1  /-
2  Section 19, Task 03.
3  Left shift  $S_n$  on  $\ell^2$ :  $(S_n x)_k = x_{\{k+n\}}$ .
4  Key property:  $S_n \rightarrow 0$  strongly.
5
6  Proof ingredient: for  $x \in \ell^2$ , each coordinate  $x_k \rightarrow 0$  as  $k \rightarrow$ 
7   $\hookrightarrow \infty$ 
8  (from Task04: l2_coord_tendsto_zero). The shift  $(S_n x)_k =$ 
9   $\hookrightarrow x_{\{k+n\}}$ 
10 has the same coordinates shifted, so each coordinate  $\rightarrow 0$ .
11
12 We also note:  $S_n$  has  $\|S_n\| \leq 1$  (it's a contraction), and
13  $S_n^* S_n = \text{Id}$  (right shift composed with left shift is identity
14  $\hookrightarrow$  on  $\ell^2$ ).
15 -/
16 import FunAn.Section19_OperatorConvergence.Task04
17 import Mathlib.Tactic
18
19 open scoped ENNReal
20 open Filter
21
22 /-- Each coordinate of  $x \in \ell^2$  tends to 0.
23 The left shift  $S_n$  maps  $x$  to  $(x_{\{n\}}, x_{\{n+1\}}, \dots)$ ,
24 so  $(S_n x)_0 = x_n \rightarrow 0$  for any fixed coordinate. -/
25 theorem l2_shifted_coord_tendsto_zero
26   (x : lp (fun _ : ℕ => ℝ) 2) (k : ℕ) :
27     Tendsto (fun n => (x : ℕ → ℝ) (k + n)) atTop (nhds 0) :=
28     by
29       have h := l2_coord_tendsto_zero x
30       rw [Metric.tendsto_atTop] at h
31       intro ε hε
32       obtain ⟨N, hN⟩ := h ε hε
33       exact ⟨N, fun n hn => hN (k + n) (by omega)⟩

```

4. Пусть  $A_n x = (\xi_{n+1}, \xi_{n+2}, \dots)$ ,  $A_n : l^2 \rightarrow l^2$ . Доказать, что  $A_n \rightarrow 0$  сильно. Верно ли утверждение  $A_n^* \rightarrow 0$  сильно?

#### Формализация на Lean 4

```

1  /-
2  Section 19, Task 04.
3   $A_n \rightarrow 0$  strongly does NOT imply  $A_n^* \rightarrow 0$  strongly.
4
5  Witness:  $e_n = \text{lp.single } 2 \ n \ 1$  in  $\square^2$ . The sequence  $(e_n)$ 
6   $\hookrightarrow$  converges
7  weakly to 0 ( $\langle e_n, y \rangle = y_n \rightarrow 0$  for any  $y \in \square^2$ ) but  $\|e_n\| = 1 \not\rightarrow$ 
8   $\hookrightarrow$  0.
9  Applied to rank-1 operators  $A_n x = \langle x, e_n \rangle e_1$ :  $A_n \rightarrow 0$  strongly
10  $\hookrightarrow$  but
11  $A_n^* y = \langle y, e_1 \rangle e_n$  does NOT converge strongly ( $\|A_n^* e_1\| = 1$ ).
12 -/
13 import Mathlib.Analysis.InnerProductSpace.l2Space
14 import Mathlib.Tactic
15
16 open scoped InnerProductSpace ENNReal
17 open Filter
18
19 /-- Standard basis vectors  $e_n$  in  $\square^2$  do not converge strongly
20  $\hookrightarrow$  to 0. -/
21 theorem l2_single_not_tendsto_zero :
22   let e :  $\mathbb{N} \rightarrow \text{lp } (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) \ 2 := \text{fun } n \Rightarrow \text{lp.single } 2$ 
23    $\hookrightarrow$  n (1 :  $\mathbb{R}$ )
24    $\neg \text{Tendsto } e \text{ atTop } (\text{nhds } 0) := \text{by}$ 
25   haveI : Fact (1  $\leq$  (2 :  $\mathbb{R}_{\geq 0 \infty}$ )) := (by norm_num)
26   intro e h
27   obtain (N, hN) := Metric.tendsto_atTop.mp h 1 one_pos
28   have h2 := hN N le_refl
29   simp only [e, dist_zero_right, lp.norm_single (by norm_num :
30      $\hookrightarrow$  (0 :  $\mathbb{R}_{\geq 0 \infty}$ ) < 2),
31     norm_one] at h2
32   linarith
33
34 /-- But for any fixed  $y \in \square^2$ ,  $\langle e_n, y \rangle \rightarrow 0$ 
35 (coordinates of a square-summable sequence tend to 0). -/
36 theorem l2_coord_tendsto_zero (y : lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2) :
37   Tendsto (fun n => (y :  $\mathbb{N} \rightarrow \mathbb{R}$ ) n) atTop (nhds 0) := by
38   haveI : Fact (1  $\leq$  (2 :  $\mathbb{R}_{\geq 0 \infty}$ )) := (by norm_num)
39   rw [Metric.tendsto_atTop]
40   intro  $\varepsilon$  h $\varepsilon$ 
41   --  $\|y\|^2 = \sum (y \ k)^2$ , which converges, so tail  $\rightarrow 0$ 
42   have hsq : Summable (fun k => (y k :  $\mathbb{R}$ ) ^ 2) := by
43     have := (lp.mem $\square$ p y).summable (show 0 < (2 :  $\mathbb{R}_{\geq 0 \infty}$ ).toReal by
44        $\hookrightarrow$  norm_num)
45     exact this.congr fun k => by simp [ENNReal.toReal_ofNat,
46        $\hookrightarrow$  Real.norm_eq_abs, sq_abs]
```

```

39   have htail := hsq.tendsto_atTop_zero
40   obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp htail (ε ^ 2) (by
    ↪ positivity)
41   exact ⟨N, fun n hn => by
42     have h1 := hN n hn
43     simp only [dist_zero_right, Real.norm_eq_abs] at h1
44     rw [dist_zero_right, Real.norm_eq_abs]
45     by_contra h2; push_neg at h2
46     have h3 : ε ^ 2 ≤ ((y : ℕ → ℝ) n) ^ 2 := by nlinarith
    ↪ [sq_abs ((y : ℕ → ℝ) n)]
47     have h4 : |((y : ℕ → ℝ) n) ^ 2| = ((y : ℕ → ℝ) n) ^ 2 :=
    ↪ abs_of_nonneg (sq_nonneg _)
48     llinarith)

```

5. Исследовать последовательности  $\{A_n\} \subset \mathcal{L}(X, X)$  на равномерную, сильную, слабую сходимость.

- a)  $X = l^2$ ,  $A_n x = (\underbrace{0, \dots, 0}_{n-1}, \xi_n, 0, 0, \dots)$ ;
- b)  $X = L^2(0, 1)$ ,  $(A_n x)(t) = \int_0^1 t^n \xi^n x(\xi) d\xi$ ;
- c)  $X = C[0, 1]$ ,  $(A_n x)(t) = t^n x(t)$ ;
- d)  $X = C[0, 1]$ ,  $(A_n x)(t) = n \cdot \int_t^{t+\frac{1}{n}} x(\xi) d\xi$ ;
- e)  $X = C[0, 1]$ ,  $(A_n x)(t) = t^n \cdot (1 - t) \cdot x(t)$ ;
- f)  $X = C[0, 1]$ ,  $(A_n x)(t) = x(t^{1+\frac{1}{n}})$ .

6. Пусть  $A, A_n \in \mathcal{L}(H, H)$ ,  $\{x_n\} \subset H$ . Доказать, что если  $A_n \rightarrow A$  слабо, а  $x_n \rightarrow x$  сильно, то  $A_n x_n \rightarrow Ax$  слабо. Привести пример, показывающий, что из сильной сходимости  $\{A_n\}$  к  $A$  и слабой сходимости  $\{x_n\}$  к  $x$  не следует слабая сходимость  $\{A_n x_n\}$  к  $Ax$ .

Указание. См. задачу 3.

#### Формализация на Lean 4

```

1  /-
2  Section 19, Task 06.
3  (a) If  $A_n \rightarrow A$  in weak operator topology and  $x_n \rightarrow x$  strongly,
4  then  $A_n x_n \rightarrow Ax$  weakly.
5
6  Proof: Fix  $f \in F^*$ . By Banach-Steinhaus,  $\|f \circ A_n\|$  is
7   $\hookrightarrow$  uniformly bounded (say  $\leq C$ ),
8  since  $f(A_n y)$  converges for each  $y$ . Then:
9   $|f(A_n x_n) - f(A_0 x_0)| \leq |f(A_n(x_n - x_0))| + |f(A_n x_0) - f(A_0 x_0)|$ 
10  $\leq C \cdot \|x_n - x_0\| + |f(A_n x_0) - f(A_0 x_0)| \rightarrow$ 
11  $\hookrightarrow 0$ .
12
13 (b) Counterexample: strong operator  $A_n \rightarrow 0$  + weak  $x_n \rightarrow 0$  but
14  $\hookrightarrow A_n x_n = e_1 \neq 0$ .
15 Take  $A_n y = \langle y, e_n \rangle \cdot e_1$  and  $x_n = e_n$  in  $\square^2$ .
16
17 -/
18 import Mathlib.Analysis.Normed.Operator.BanachSteinhaus
19 import Mathlib.Tactic
20
21 open Filter
22
23 /-- Weak operator + strong vector convergence  $\square$  weak
24  $\hookrightarrow$  convergence.
25 If  $f(A_n y) \rightarrow f(A_0 y)$  for all  $y \in E$ ,  $f \in F^*$ , and  $\|x_n - x_0\| \rightarrow$ 
26  $\hookrightarrow 0$ ,
27 then  $f(A_n x_n) \rightarrow f(A_0 x_0)$  for all  $f \in F^*$ . -/
28 theorem section19_task06
29 {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
30  $\hookrightarrow$  [CompleteSpace E]
31 [NormedAddCommGroup F] [NormedSpace ℝ F]
32 {A : ℕ → E →L[ℝ] F} {A₀ : E →L[ℝ] F}
33 (hop : ∀ (y : E) (f : F →L[ℝ] ℝ),
34 Tendsto (fun n => f (A n y)) atTop (nhds (f (A₀ y))))
35 {x : ℕ → E} {x₀ : E} (hx : Tendsto x atTop (nhds x₀)) :
36 ∀ (f : F →L[ℝ] ℝ),
37 Tendsto (fun n => f (A n (x n))) atTop (nhds (f (A₀
38  $\hookrightarrow$  x₀))) := by
39 intro f
40 -- Uniform bound on  $\|f \circ A_n\|$  by Banach-Steinhaus
41 have hbdd : ∃ C, ∀ n,  $\|f \circ A n\| \leq C$  := by
42 apply banach_steinhaus; intro y
43 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp (hop y f) 1
44  $\hookrightarrow$  one_pos

```

```

36 refine (||f (A₀ y)|| + 1 + ∑ i ∈ Finset.range N, ||f (A i
    ↪ y)||, fun n => ?_)
37 simp only [ContinuousLinearMap.comp_apply]
38 by_cases hn : N ≤ n
39 · have h1 : ||f (A n y) - f (A₀ y)|| < 1 := by
40   have := hN n hn; rwa [dist_eq_norm] at this
41   have h2 : ||f (A n y)|| ≤ ||f (A₀ y)|| + 1 :=
42     calc ||f (A n y)|| = ||f (A₀ y) + (f (A n y) - f (A₀ y))||
        ↪ := by
43       rw [add_sub_cancel]
44       _ ≤ ||f (A₀ y)|| + ||f (A n y) - f (A₀ y)|| :=
        ↪ norm_add_le _ _
45       _ ≤ ||f (A₀ y)|| + 1 := by linarith
46   linarith [Finset.sum_nonneg (s := Finset.range N)
47     (fun i _ => norm_nonneg (f (A i y)))]
48 · push Not at hn
49   linarith [Finset.single_le_sum (s := Finset.range N)
50     (f := fun i => ||f (A i y)||) (fun _ _ => norm_nonneg _)
51     (Finset.mem_range.mpr hn), norm_nonneg (f (A₀ y))]
52 obtain ⟨C, hC⟩ := hbdd
53 -- Decompose: f(Aₙxₙ) - f(A₀x₀) = f(Aₙ(xₙ - x₀)) + (f(Aₙx₀) -
    ↪ f(A₀x₀))
54 apply Metric.tendsto_atTop.mpr; intro ε hε
55 obtain ⟨N₁, hN₁⟩ := Metric.tendsto_atTop.mp (hop x₀ f) (ε/2)
    ↪ (half_pos hε)
56 have hC1 : (0:ℝ) < C + 1 := by
57   linarith [le_trans (norm_nonneg (f.comp (A 0))) (hC 0)]
58 obtain ⟨N₂, hN₂⟩ := Metric.tendsto_atTop.mp hε (ε/(2*(C+1)))
59   (div_pos hε (mul_pos two_pos hC1))
60 exact ⟨max N₁ N₂, fun n hn => by
61   have d1 := hN₁ n (le_of_max_le_left hn)
62   have d2 := hN₂ n (le_of_max_le_right hn)
63   simp only [dist_eq_norm] at d1 d2 ⊢
64   calc ||f (A n (x n)) - f (A₀ x₀)||
65     = ||f (A n (x n - x₀)) + (f (A n x₀) - f (A₀ x₀))|| :=
        ↪ by
66     congr 1; simp [map_sub]
67     _ ≤ ||f (A n (x n - x₀))|| + ||f (A n x₀) - f (A₀ x₀)|| :=
        ↪ norm_add_le _ _
68     _ ≤ C * ||x n - x₀|| + ||f (A n x₀) - f (A₀ x₀)|| := by
69       have h1 := (f.comp (A n)).le_opNorm (x n - x₀)
70       simp only [ContinuousLinearMap.comp_apply] at h1
71       have h2 := mul_le_mul_of_nonneg_right (hC n)
        ↪ (norm_nonneg (x n - x₀))
72       linarith
73     _ ≤ C * (ε/(2*(C+1))) + ε/2 := by
74       have hC0 : (0:ℝ) ≤ C :=
75         le_trans (norm_nonneg (f.comp (A 0))) (hC 0)
76       linarith [mul_le_mul_of_nonneg_left d2.le hC0]
77     _ < ε := by
78       suffices C * (ε/(2*(C+1))) < ε/2 by linarith

```

```

79      calc C * (ε / (2 * (C + 1)))
80      = C * ε / (2 * (C + 1)) := by ring
81      _ < (C + 1) * ε / (2 * (C + 1)) := by
82      apply div_lt_div_of_pos_right _ (by linarith :
      ↪ 0 < 2 * (C + 1))
83      exact mul_lt_mul_of_pos_right (by linarith) hε
84      _ = ε / 2 := by field_simp

```

## 20. КОМПАКТНЫЕ МНОЖЕСТВА

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение компактного в себе множества, компакта.
2. Сформулировать теоремы Арцела, Рисса (критерии компактности множества в  $C[a, b]$ ,  $L^p(a, b)$ ,  $l^p$ ).

### ЗАДАЧИ

1. Пусть  $\mathcal{M} = \{x(t)\}$  — ограниченное множество функций из  $C[a, b]$ . Доказать, что множество функций вида:

a)  $y(s) = \int_a^b (s^2 + t^2 s) x(t) dt;$

b)  $y(s) = \int_a^b (3st + s^2) x(t) dt;$

$$c) y(s) = \int_a^b s \cdot x(t) dt$$

КОМПАКТНЫ В  $L^p(a, b)$ .

#### Формализация на Lean 4

```

1  /-
2  Section 20, Task 1.
3  Integral transform sets compact in  $L^p$  (Kolmogorov-Riesz
4   $\hookrightarrow$  criterion).
5
6  The Kolmogorov-Riesz theorem: a subset  $S$  of  $L^p(\mathbb{R}^n)$  is
7   $\hookrightarrow$  totally bounded iff
8  (1)  $S$  is bounded, (2)  $S$  is equitight, (3)  $S$  is
9   $\hookrightarrow$   $L^p$ -equicontinuous
10 (translations converge uniformly in  $S$ ).
11
12 We formalize the abstract principle: totally bounded +
13  $\hookrightarrow$  complete = compact.
14 Applied to  $L^p$  (which is complete), total boundedness is the
15  $\hookrightarrow$  key criterion.
16 -/
17 import Mathlib.Topology.UniformSpace.Compact
18 import Mathlib.Tactic
19
20 /-- A totally bounded subset of a complete space has compact
21  $\hookrightarrow$  closure.
22 This is the abstract principle behind Kolmogorov-Riesz:
23 in  $L^p$  (complete), total boundedness  $\square$  compact closure. -/
24 theorem isCompact_closure_of_totallyBounded
25   {X : Type*} [UniformSpace X] [CompleteSpace X]
26   {S : Set X} (hS : TotallyBounded S) :
27     IsCompact (closure S) :=
28     isCompact_iff_totallyBounded_isComplete.mpr ⟨hS.closure,
29      $\hookrightarrow$  isClosed_closure.isComplete⟩

```

2. Пусть  $\mathcal{M} = \{x(t)\}$  — ограниченное множество функций из  $L^p(a, b)$ . Компактны ли в пространстве  $L^p(a, b)$  множества функций вида:

$$a) y(s) = \int_a^b (3s + s^2 t^2) x(t) dt;$$

$$b) y(t) = t^2 x(t);$$

$$c) y(t) = x(t) + 3;$$

$$d) y(s) = \int_a^b K(s, t) x(t) dt, \text{ где функция } K(s, t) \text{ непрерывна в квадрате } [a, b] \times [a, b]?$$

3. Пусть  $\mathcal{M} = \{x(t)\}$  — ограниченное множество пространства  $C[a, b]$ . Доказать, что множества функций вида:

$$a) y(t) = \int_a^t x(\xi) d\xi;$$

b)  $y(s) = \int_a^b K(s, t)x(t) dt$ , где  $K(s, t) \in C([a, b] \times [a, b])$

компактны в пространстве  $C[a, b]$ .

#### Формализация на Lean 4

```

1  /-
2    Section 20, Task 3.
3    Sets of integral transforms are compact in  $C[a, b]$  by
4     $\hookrightarrow$  Arzelà-Ascoli:
5    equicontinuous + pointwise compact  $\square$  compact in  $C(X, \alpha)$ .
6
7    Applied to integral transforms: the image of a bounded set
8     $\hookrightarrow$  under an
9    integral operator is equicontinuous (by uniform continuity
10    $\hookrightarrow$  of the kernel)
11   and uniformly bounded, hence has compact closure in  $C[a, b]$ .
12  -/
13  import Mathlib.Topology.UniformSpace.Ascoli
14  import Mathlib.Tactic
15
16  /-- Arzelà-Ascoli theorem: an equicontinuous set with
17    $\hookrightarrow$  pointwise compact
18   image is compact in  $C(X, \alpha)$ . This is the key tool for
19    $\hookrightarrow$  proving
20   integral transform sets are compact in  $C[a, b]$ . -/
21  theorem arzela_ascoli_compact
22    {X : Type*} [TopologicalSpace X] [CompactSpace X]
23    {α : Type*} [UniformSpace α] [T2Space α]
24    (S : Set C(X, α))
25    (hbw : IsCompact (ContinuousMap.toFun '' S))
26    (heq : Equicontinuous ((↑) : S → X → α)) :
27    IsCompact S :=
28    ArzelaAscoli.isCompact_of_equicontinuous S hbw heq

```

4. Будет ли ограниченное множество многочленов степени не выше  $n$  компактным в себе множеством в  $C[a, b]$ ?

5. Доказать, что всякое множество, компактное в пространстве  $C^{(1)}[a, b]$ , является компактным и в пространстве  $C[a, b]$ .

#### Формализация на Lean 4

```

1  /-
2    Section 20, Task 05.
3    A compact set in  $C^1[a, b]$  is also compact in  $C[a, b]$ .
4    The inclusion  $\iota : C^1 \rightarrow C$  is continuous ( $\|f\|_\infty \leq \|f\|_{C^1}$ ),
5    and continuous images of compact sets are compact.
6
7    We formalize the abstract principle: continuous image of
       $\hookrightarrow$  compact is compact.
8  -/
9  import Mathlib.Topology.Compactness.Compact
10 import Mathlib.Tactic
11
12 /-- Continuous image of a compact set is compact (abstract
       $\hookrightarrow$  version of
13    "compact in  $C^1$  implies compact in  $C$ "). -/
14 theorem compact_in_C1_compact_in_C
15   {X Y : Type*} [TopologicalSpace X] [TopologicalSpace Y]
16   {f : X → Y} (hf : Continuous f) {K : Set X} (hK :
       $\hookrightarrow$  IsCompact K) :
17   IsCompact (f '' K) :=
18   hK.image hf

```

6. Компактны ли следующие множества функций в пространстве  $C[0, 1]$ :

- a)  $x_n(t) = t^n, n = 1, 2, \dots;$
- b)  $x_n(t) = \sin nt, n = 1, 2, \dots;$
- c)  $x_n(t) = \sin(n + t), n = 1, 2, \dots;$
- d)  $x_\alpha(t) = \sin \alpha t, \alpha \in \mathbb{R}^1;$
- e)  $x_\alpha(t) = \sin \alpha t, \alpha \in [1, 2];$
- f)  $x_\alpha(t) = \arctg \alpha t, \alpha \in \mathbb{R}^1;$
- g)  $x_\alpha(t) = e^{t-\alpha}, \alpha \geq 0?$

7. Доказать, что всякое компактное множество в пространстве

- a)  $C[a, b];$

b)  $l^2$

нигде не плотно.

#### Формализация на Lean 4

```
1  /-
2  Section 20, Task 7.
3  In an infinite-dimensional normed space, every compact set
4  ↪ is nowhere dense.
5
6  Proof: If interior  $K \neq \emptyset$ ,  $K$  contains a ball, hence a closed
7  ↪ ball,
8  which is compact (closed subset of compact). But a compact
9  ↪ closed ball
10 forces the space to be finite-dimensional (Riesz's lemma).
11 ↪ Contradiction.
12 -/
13 import Mathlib.Analysis.Normed.Module.FiniteDimension
14 import Mathlib.Tactic
15
16 variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
17
18 /- In an infinite-dimensional normed space, every compact set
19 ↪ has empty interior. -/
20 theorem compact_nowhere_dense_of_infinite_dim
21   (hinf : ¬ FiniteDimensional ℝ E)
22   {K : Set E} (hK : IsCompact K) :
23   interior K = ∅ := by
24   by_contra h
25   rw [← Ne, ← Set.nonempty_iff_ne_empty] at h
26   obtain ⟨x₀, hx₀⟩ := h
27   rw [mem_interior_iff_mem_nhds, Metric.mem_nhds_iff] at hx₀
28   obtain ⟨r, hr, hball⟩ := hx₀
29   -- closedBall x₀ (r/2) ⊆ ball x₀ r ⊆ K, hence compact
30   have hcb_compact : IsCompact (Metric.closedBall x₀ (r / 2))
31     ↪ :=
32     hK.of_isClosed_subset Metric.isClosed_closedBall
33       ((Metric.closedBall_subset_ball (by linarith)).trans
34        ↪ hball)
35   exact hinf (FiniteDimensional.of_isCompact_closedBall (□
36     ↪ := ℝ) (by linarith) hcb_compact)
```

8. Доказать, что параллелепипед

$$\mathcal{P} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : |\xi_k| \leq \frac{1}{k}, \forall k \geq 1\}$$

является компактным в себе множеством в пространстве  $l^2$ .

#### Формализация на Lean 4

```

1  /-
2    Section 20, Task 8.
3    The parallelepiped  $P = \{x \in \ell^2 : |x_k| \leq 1/(k+1)\}$  is compact
4     $\hookrightarrow$  in  $\ell^2$ .
5  -/
6  import Mathlib.Analysis.InnerProductSpace.l2Space
7  import Mathlib.Analysis.Normed.Group.Tannery
8  import Mathlib.Analysis.PSeries
9  import Mathlib.Tactic
10
11  open scoped ENNReal
12  open Filter
13
14  private lemma mem_p_of_coord_le {x : ℕ → ℝ}
15    (hx : ∀ k : ℕ, |x k| ≤ 1 / ((k : ℝ) + 1)) : Mem_p x 2 :=
16    by
17      rw [mem_p_gen_iff (by norm_num : (0 : ℝ) < (2 :
18         $\mathbb{R}_{\geq 0\infty}$ ).toReal)]
19      simp only [ENNReal.toReal_ofNat]
20      exact ((Real.summable_nat_rpow_inv.mpr (by norm_num : (1:ℝ)
21         $\hookrightarrow$  < 2)).comp_injective
22        Nat.succ_injective |>.congr fun k => by simp
23         $\hookrightarrow$  [Nat.cast_succ, one_div, sq]).of_nonneg_of_le
24        (fun _ => Real.rpow_nonneg (norm_nonneg _) _) fun k =>
25        Real.rpow_le_rpow (norm_nonneg _) (hx k) (by norm_num)
26
27  private lemma summable_sq_bound : Summable (fun k : ℕ => (2 /
28     $\hookrightarrow$  ((k : ℝ) + 1)) ^ 2) := by
29    simp_rw [div_pow]; apply Summable.mul_left
30    exact (Real.summable_nat_rpow_inv.mpr (by norm_num : (1:ℝ) <
31       $\hookrightarrow$  2)).comp_injective
32    Nat.succ_injective |>.congr fun k => by simp
33     $\hookrightarrow$  [Nat.cast_succ, one_div, sq]
34
35  --  $\|f\|^2 = \sum' k, (f k)^2$  via inner product on  $\ell^2$ 
36  private lemma lp_norm_sq_eq_tsum (f : lp (fun _ : ℕ => ℝ) 2) :
37     $\|f\|^2 = \sum' k, (f k)^2 :=$  by
38    rw [sq, ← real_inner_self_eq_norm_mul_norm,
39       $\hookrightarrow$  lp.inner_eq_tsum]
40    congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm,
41       $\hookrightarrow$  Real.norm_eq_abs]
42    change |f k| * |f k| = (f k : ℝ) ^ 2; rw [← sq_abs]; ring

```

```

33
34 theorem parallelepiped_compact_in_l2 :
35   IsCompact {x : lp (fun _ : ℕ => ℝ) 2 |
36     ∀ k : ℕ, |x k| ≤ 1 / (k + 1 : ℝ)} := by
37   haveI : Fact (1 ≤ (2 : ℝ≥0∞)) := ⟨by norm_num⟩
38   rw [isCompact_iff_isSeqCompact]; intro x hx
39   obtain ⟨y, hyS, φ, hφ, hconv⟩ :=
40     (isCompact_pi_infinite (fun k : ℕ =>
41       ↪ isCompact_Icc)).tendsto_subseq
42     (fun n k => abs_le.mp (hx n k) : ∀ n, (↑(x n) : ℕ → ℝ) ∈
43       ↪ _)
44   have hybd : ∀ k : ℕ, |y k| ≤ 1 / ((k:ℝ)+1) := fun k =>
45     ↪ abs_le.mpr (hyS k)
46   set y' : lp (fun _ : ℕ => ℝ) 2 := ⟨y, mem_p_of_coord_le
47     ↪ hybd⟩
48   refine ⟨y', hybd, φ, hφ, ?_⟩
49   -- ||d_n||2 = ∑' k, (d_n k)2
50   have hnorm_sq : ∀ n, ||x (φ n) - y'||2 =
51     ∑' k, ((x (φ n) : ℕ → ℝ) k - y k)2 := fun n => by
52     have := lp_norm_sq_eq_tsum (x (φ n) - y')
53     simp only [lp.coeFn_sub, Pi.sub_apply, y'] at this; exact
54     ↪ this
55   -- Tannery: ∑' k, (d_n k)2 → 0
56   have hsq : Tendsto (fun n => ∑' k, ((x (φ n) : ℕ → ℝ) k - y
57     ↪ k)2) atTop (nhds 0) := by
58     have hpw : ∀ k, Tendsto (fun n => ((x (φ n) : ℕ → ℝ) k - y
59     ↪ k)2) atTop (nhds 0) := by
60       intro k
61       have hk := ((continuous_apply k).tendsto y).comp hconv
62       have hsub : Tendsto (fun n => (x (φ n) : ℕ → ℝ) k - y k)
63         ↪ atTop (nhds 0) := by
64         have := hk.sub (tendsto_const_nhds (x := y k)); rwa
65         ↪ [sub_self] at this
66       have := ((continuous_pow 2).tendsto 0).comp hsub
67       simp only [Function.comp_def, zero_pow_two_ne_zero] at
68         ↪ this; exact this
69   have hbd : ∀f n in atTop, ∀ k,
70     ||((x (φ n) : ℕ → ℝ) k - y k)2|| ≤ (2 / ((k : ℝ) +
71     ↪ 1))2 :=
72   Eventually.of_forall fun n k => by
73     rw [Real.norm_eq_abs, abs_of_nonneg (sq_nonneg _), ←
74     ↪ sq_abs]
75     gcongr
76     have h3 := norm_sub_le ((x (φ n) : ℕ → ℝ) k) (y k)
77     simp only [Real.norm_eq_abs] at h3
78     have h1 := hx (φ n) k; have h2 := hybd k
79     have h4 : (1 : ℝ) / ((k:ℝ)+1) + 1 / ((k:ℝ)+1) = 2 /
80     ↪ ((k:ℝ)+1) := by ring
81     linarith
82   simp [tsum_zero] using
83     ↪ tendsto_tsum_of_dominated_convergence
84     ↪ summable_sq_bound hpw hbd

```

```

70 -- ||d_n||^2 → 0 ⇔ ||d_n|| → 0 ⇔ x(φ n) → y'
71 rw [← tendsto_sub_nhds_zero_iff,
    ↪ tendsto_zero_iff_norm_tendsto_zero,
    ↪ Metric.tendsto_atTop]
72 intro ε hε
73 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp (show Tendsto (fun
    ↪ n => ||x (φ n) - y'|| ^ 2)
74 atTop (nhds 0) from by simp_rw [hnorm_sq]; exact hsq) (ε ^
    ↪ 2) (by positivity)
75 exact ⟨N, fun n hn => by
76 simp only [dist_zero_right, Real.norm_eq_abs,
    ↪ abs_of_nonneg (norm_nonneg _)]
77 have h : ||x (φ n) - y'|| ^ 2 < ε ^ 2 := by
78 have := hN n hn
79 rwa [dist_zero_right, Real.norm_of_nonneg (by
    ↪ positivity)] at this
80 simp only [Function.comp_def] at *
81 nlinarith [norm_nonneg (x (φ n) - y'), mul_self_nonneg (||x
    ↪ (φ n) - y'|| - ε)]

```

9. При каком условии на последовательность  $\lambda_n \in \mathbb{R}^1$ ,  $\lambda_n > 0$ , являются компактным в себе множеством в пространстве  $l^2$ :

- а) параллелепипед  $\mathcal{P} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : |\xi_k| \leq \lambda_k, \forall k \geq 1\}$ ;
- б) эллипсоид  $\mathcal{E} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{n=1}^{\infty} \frac{|\xi_n|^2}{\lambda_n^2} \leq 1\}$ ?

10. Какие из следующих множеств компактны в пространстве  $C[0, 1]$ :

- а)  $\{y(t) = \int_0^1 e^{ts} \varphi(s) ds : |\varphi(s)| \leq 1, \varphi \in C[0, 1]\}$ ;
- б)  $\{x(t) \in C^{(2)}[0, 1] : |x(t)| \leq B_0; |x'(t)| \leq B_1; |x''(t)| \leq B_2\}$ ;
- в)  $\{x(t) \in C^{(1)}[0, 1] : |x(t)| \leq B_0; |x'(t)| \leq B_1\}$ ;
- г)  $\{x(t) \in C^{(2)}[0, 1] : |x(t)| \leq B_0; |x''(t)| \leq B_2\}$ ;
- д)  $\{x(t) \in C^{(2)}[0, 1] : |x'(t)| \leq B_1; |x''(t)| \leq B_2\}$ ;
- е)  $\{x(t) \in C[0, 1] : |x(t)| \leq B, |x(t_1) - x(t_2)| \leq L \cdot |t_1 - t_2|\}$ ;
- ж)  $\{x(t) \in C^{(1)}[0, 1] : |x(t)| \leq B, |x'(t_1) - x'(t_2)| \leq L \cdot |t_1 - t_2|\}$ .

11. Компактны ли следующие множества в пространстве  $L^2(0, 1)$ :

- а)  $\{t^\alpha\}$ , если  $\alpha > -\frac{1}{2}$ , если  $\alpha > 0$ ;
- б)  $\{\sin nt\}$ ,  $n \geq 1$ ;
- в)  $\{x(t) = \int_0^t \varphi(\xi) d\xi : \int_0^1 |\varphi(\xi)|^2 d\xi \leq 1, \varphi \in L^2(0, 1)\}$ ;
- г)  $\{(\ln t)^n\}$ ,  $n \geq 1$ .

12. Пусть  $\{c_n\}$  — числовая последовательность. Пусть, далее

$$\mathcal{M} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{k=1}^{\infty} |c_k \xi_k|^2 \leq 1\}$$

Для каких последовательностей множество  $\mathcal{M}$  компактно?

## 21. КОМПАКТНЫЕ ОПЕРАТОРЫ

### КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение компактного оператора.
2. Перечислить свойства компактного оператора.
3. Сформулируйте критерии компактности множеств в пространствах  $\mathbb{R}^n$ ,  $C[a, b]$ ,  $L^p(a, b)$ ,  $l^p$ .

### ЗАДАЧИ

1. Доказать, что любой линейный оператор  $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$  является компактным.

#### Формализация на Lean 4

```

1  /-
2    Section 21, Task 1.
3    Any linear operator  $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is compact.
4  -/
5  import Mathlib.Analysis.Normed.Module.FiniteDimension
6  import Mathlib.Analysis.Normed.Operator.Compact
7  import Mathlib.Tactic
8
9  /-- Any continuous linear map from a finite-dimensional real
10     ↪ space is compact. -/
11  theorem linear_map_finiteDim_isCompact
12     {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
13     ↪ [FiniteDimensional ℝ E]
14     [NormedAddCommGroup F] [NormedSpace ℝ F]
15     (A : E →L[ℝ] F) :
16     IsCompactOperator (A : E → F) := by
17     haveI : ProperSpace E := FiniteDimensional.proper_real E
18     -- closed ball is compact in a proper space
19     refine ⟨A '' Metric.closedBall 0 1,
20       (isCompact_closedBall 0 1).image A.continuous,
21       ?_⟩
22     --  $A^{-1}(A '' \text{closedBall } 0 \ 1) \supseteq \text{closedBall } 0 \ 1 \in \square \ 0$ 
23     exact Filter.mem_of_superset (Metric.closedBall_mem_nhds 0
24       ↪ one_pos)
25       (Set.subset_preimage_image A _)

```

2. Проверить компактность оператора  $A : l^2 \rightarrow l^2$ , если

$$Ax = A(\xi_1, \xi_2, \dots) = (0, \frac{\xi_1}{1}, \frac{\xi_2}{2}, \dots, \frac{\xi_n}{n}, \dots)$$

#### Формализация на Lean 4

```

1  /-
2    Section 21, Task 2.
3     $A(\xi_1, \xi_2, \dots) = (0, \xi_1/1, \xi_2/2, \dots, \xi_n/n, \dots)$  is compact on
4     $\ell^2$ .
5    This is a diagonal operator with entries  $a_k = 1/(k+1) \rightarrow 0$ .
6    By the diagonal compactness criterion (Task04), it is
7    compact.
8  -/
9  import FunAn.Section21_CompactOperators.Task04
10 import Mathlib.Analysis.SpecificLimits.Basic
11 import Mathlib.Tactic
12
13 open scoped ENNReal
14 open Filter
15
16 /-- The diagonal operator with entries  $1/(k+1)$  is compact on
17  $\ell^2$ . -/
18 theorem weighted_shift_compact :  $\exists A : \ell_p (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) 2 \rightarrow$ 
19  $\ell_p (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) 2,$ 
20   IsCompactOperator A  $\wedge$ 
21    $\forall (x : \ell_p (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) 2) (k : \mathbb{N}),$ 
22      $(A x : \mathbb{N} \rightarrow \mathbb{R}) k = (1 / ((k : \mathbb{R}) + 1)) * (x : \mathbb{N} \rightarrow \mathbb{R}) k :=$ 
23      $\hookrightarrow$  by
24   set a :  $\mathbb{N} \rightarrow \mathbb{R} := \text{fun } k \Rightarrow 1 / ((k : \mathbb{R}) + 1)$ 
25   have ha :  $\exists C, \forall k, |a k| \leq C := \langle 1, \text{fun } k \Rightarrow \text{by}$ 
26     simp only [a, one_div]
27     have hk :  $(0 : \mathbb{R}) < (\uparrow k + 1) := \text{by positivity}$ 
28     rw [abs_of_nonneg (inv_nonneg.mpr hk.le)]
29     exact inv_le_one_of_one_le₀ (by linarith)
30   refine ⟨_, (diagonal_operator_compact_iff_tendsto_zero a
31      $\hookrightarrow$  ha).mpr ?_, fun x k => rfl)
32   exact tendsto_one_div_add_atTop_nhds_zero_nat

```

3. Пусть  $H$  — гильбертово пространство и  $A : H \rightarrow H$  — линейный оператор. Доказать, что  $A$  компактен тогда и только тогда, когда он любую слабо сходящуюся последовательность переводит в сильно сходящуюся.

#### Формализация на Lean 4

```

1  /-
2  Section 21, Task 03.
3  A compact operator maps bounded sequences to sequences with
4  convergent subsequences (sequential characterization,
   ↪ forward direction).
5
6  Proof: If A is compact and (xn) is bounded, then (Axn) lies
   ↪ in
7  A(closedBall 0 R) whose closure is compact, hence (Axn) has
   ↪ a
8  convergent subsequence by sequential compactness.
9  -/
10 import Mathlib.Analysis.Normed.Operator.Compact
11 import Mathlib.Topology.Sequences
12 import Mathlib.Tactic
13
14 open Filter
15
16 /-- A compact operator maps bounded sequences to sequences
   ↪ with
17 convergent subsequences. -/
18 theorem IsCompactOperator.exists_convergent_subseq
19   {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
20   [NormedAddCommGroup F] [NormedSpace ℝ F] [CompleteSpace F]
21   {A : E →L[ℝ] F} (hA : IsCompactOperator (A : E → F))
22   {x : ℕ → E} {R : ℝ} (hR : 0 < R) (hx : ∀ n, ‖x n‖ ≤ R) :
23   ∃ y : F, ∃ φ : ℕ → ℕ, StrictMono φ ∧
24     Tendsto (fun n => A (x (φ n))) atTop (nhds y) := by
25   -- A(closedBall 0 R) has compact closure
26   have hK := hA.isCompact_closure_image_of_bounded (S :=
   ↪ Metric.closedBall (0 : E) R)
27   Metric.isBounded_closedBall
28   -- x n ∈ closedBall 0 R
29   have hxball : ∀ n, x n ∈ Metric.closedBall (0 : E) R := by
30     intro n; rw [Metric.mem_closedBall, dist_zero_right];
   ↪ exact hx n
31   -- A(x n) ∈ closure(A '' closedBall 0 R)
32   have hAx : ∀ n, A (x n) ∈ closure (A '' Metric.closedBall 0
   ↪ R) :=
33     fun n => subset_closure (Set.mem_image_of_mem A (hxball
   ↪ n))
34   -- Sequential compactness from compactness
35   have hseq : IsSeqCompact (closure (A '' Metric.closedBall 0
   ↪ R)) :=

```

```
36      isCompact_iff_isSeqCompact.mp hK
37      obtain ⟨y, -, φ, hφ, hconv⟩ := hseq hAx
38      exact ⟨y, φ, hφ, hconv⟩
```

4. Пусть  $A : l^2 \rightarrow l^2$ ,  $A(\xi_1, \xi_2, \dots) = (a_1\xi_1, a_2\xi_2, \dots)$ . Доказать, что оператор  $A$  компактен тогда и только тогда, когда  $a_n \rightarrow 0$  при  $n \rightarrow \infty$ .

#### Формализация на Lean 4

```

1  /-
2    Section 21, Task 4.
3    Diagonal operator on  $\ell^2$  compact iff entries tend to 0.
4  -/
5  import Mathlib.Analysis.InnerProductSpace.l2Space
6  import Mathlib.Analysis.Normed.Operator.Compact
7  import Mathlib.Topology.Sequences
8  import Mathlib.Topology.Algebra.InfiniteSum.Order
9  import Mathlib.Tactic
10
11  open scoped ENNReal
12  open Filter Finset
13
14  set_option maxHeartbeats 400000
15
16  private lemma mem_p_mul {a :  $\mathbb{N} \rightarrow \mathbb{R}$ } (ha :  $\exists C, \forall k, |a k| \leq C$ )
17    → {x :  $\mathbb{N} \rightarrow \mathbb{R}$ }
18    (hx : Mem_p x 2) : Mem_p (fun k => a k * x k) 2 := by
19    obtain ⟨C, hC⟩ := ha
20    rw [mem_p_gen_iff (by norm_num : (0 :  $\mathbb{R}$ ) < (2 :  $\mathbb{R}_{\geq 0 \infty}$ ).toReal)]
21    → at hx ⊢
22    simp only [ENNReal.toReal_ofNat] at hx ⊢
23    have hC0 : (0 :  $\mathbb{R}$ ) ≤ C := le_trans (abs_nonneg (a 0)) (hC 0)
24    exact (hx.mul_left (C ^ (2 :  $\mathbb{R}$ ))).of_nonneg_of_le
25      (fun _ => Real.rpow_nonneg (norm_nonneg _) _) fun k => by
26        calc ||a k * x k|| ^ (2 :  $\mathbb{R}$ )
27          ≤ (C * ||x k||) ^ (2 :  $\mathbb{R}$ ) := by
28            apply Real.rpow_le_rpow (norm_nonneg _) _ (by
29              → norm_num)
30            rw [norm_mul]; exact mul_le_mul_of_nonneg_right
31              (by rw [Real.norm_eq_abs]; exact hC k)
32              (norm_nonneg _)
33            → C ^ (2 :  $\mathbb{R}$ ) * ||x k|| ^ (2 :  $\mathbb{R}$ ) := Real.mul_rpow hC0
34            → (norm_nonneg _)
35
36  theorem diagonal_operator_compact_iff_tendsto_zero
37    (a :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (ha :  $\exists C, \forall k, |a k| \leq C$ ) :
38    let A : lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2 → lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2 :=
39      fun x => ⟨fun k => a k * x k, mem_p_mul ha (lp.mem_p x)⟩
40    IsCompactOperator A ↔ Tendsto a atTop (nhds 0) := by
41    haveI : Fact (1 ≤ (2 :  $\mathbb{R}_{\geq 0 \infty}$ )) := ⟨by norm_num⟩
42    intro A
43    constructor
44    · -- Forward: compact  $\Rightarrow a \rightarrow 0$ 
45      intro ⟨K, hK, hK_nhds⟩

```

```

41 obtain ⟨r, hr, hr_sub⟩ := Metric.mem_nhds_iff.mp hK_nhds
42 rw [Metric.tendsto_atTop]; intro ε hε
43 by_contra h_inf; push_neg at h_inf
44 obtain ⟨φ, hφ, hφε⟩ := extraction_of_frequently_atTop
45   ↪ (frequently_atTop.mpr h_inf)
46 have hp2 : (0 : ℝ≥0) < 2 := by norm_num
47 have hr2 : (0 : ℝ) < r / 2 := by linarith
48 set s : ℕ → lp (fun _ : ℕ => ℝ) 2 := fun n => (r/2) •
49   ↪ lp.single 2 (φ n) (1 : ℝ)
50 have hAs_in_K : ∀ n, A (s n) ∈ K := fun n => hr_sub (by
51   rw [Metric.mem_ball, dist_zero_right]
52   simp only [s, norm_smul, lp.norm_single hp2, norm_one,
53     ↪ mul_one,
54     Real.norm_eq_abs, abs_of_pos hr2]; linarith)
55 have hAs_sep : ∀ m n, m ≠ n → r / 2 * ε ≤ dist (A (s m))
56   ↪ (A (s n)) := by
57   intro m n hmn
58   rw [dist_eq_norm]
59   have hne : φ m ≠ φ n := hφ.injective.ne hmn
60   calc r / 2 * ε
61     ≤ r / 2 * |a (φ m)| := by
62     have := hφε m; simp [dist_zero_right,
63       ↪ Real.norm_eq_abs] at this; nlinarith
64   _ = ||(A (s m) - A (s n)) (φ m)|| := by
65     simp only [A, s, lp.coeFn_sub, Pi.sub_apply,
66       ↪ lp.coeFn_smul, Pi.smul_apply,
67       lp.coeFn_single, Pi.single_apply, if_neg hne,
68       ↪ smul_eq_mul,
69       mul_one, mul_zero, sub_zero, abs_mul, abs_of_pos
70       ↪ hr2, mul_comm,
71       if_true, zero_mul, one_mul, Real.norm_eq_abs]
72   _ ≤ ||A (s m) - A (s n)|| := lp.norm_apply_le_norm
73     ↪ hp2.ne' _ _
74 obtain ⟨y, _, ψ, hψ, hconv⟩ := hK.isSeqCompact hAs_in_K
75 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp hconv (r / 2 * ε
76   ↪ / 2) (by positivity)
77 have h1 := hN N le_rfl
78 have h2 := hN (N + 1) (Nat.le_succ N)
79 have h3 : dist (A (s (ψ N))) (A (s (ψ (N + 1)))) < r / 2 *
80   ↪ ε :=
81   calc dist _ _ ≤ dist _ y + dist y _ := dist_triangle _ y
82     ↪ _
83     _ < r / 2 * ε / 2 + r / 2 * ε / 2 := add_lt_add h1 (by
84       ↪ rwa [dist_comm])
85     _ = r / 2 * ε := by ring
86 linarith [hAs_sep (ψ N) (ψ (N + 1)) (hψ (by omega : N < N
87   ↪ + 1)).ne]
88 • -- Backward: a → 0 □ compact
89 intro ha_zero
90 refine ⟨closure (A '' Metric.ball 0 1),
91   isCompact_iff_totallyBounded_isComplete.mpr

```

```

78     {?_, isClosed_closure.isComplete}, ?_)
79   · -- closure(A(B1)) totally bounded ← A(B1) totally
    ↪ bounded
80   apply TotallyBounded.closure
81   rw [Metric.totallyBounded_iff]; intro ε hε
82   obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp ha_zero (ε /
    ↪ 2) (by linarith)
83   -- Compact set K_N containing "truncations" of A(B1) to
    ↪ first N coords
84   -- K_N = image of compact cube under continuous map Φ
85   set Φ : (Fin N → ℝ) → lp (fun _ : ℕ => ℝ) 2 :=
86     fun c => ∑ i : Fin N, lp.single 2 (i : ℕ) (c i)
87   have hΦ_cont : Continuous Φ := continuous_finset_sum _
    ↪ fun i _ =>
88     (lp.singleContinuousLinearMap (λ _ := ℝ) (p := 2) (E :=
    ↪ fun _ => ℝ) (i : ℕ)).continuous.comp
89     (continuous_apply i)
90   set cube := Set.pi Set.univ fun i : Fin N => Set.Icc
    ↪ (-|a (i : ℕ)|) |a (i : ℕ)|
91   have hcube : IsCompact cube := isCompact_univ_pi fun _
    ↪ => isCompact_Icc
92   -- Get finite ε/2-net from compact image Φ(cube)
93   have hK := hcube.image hΦ_cont
94   obtain ⟨F, hF_fin, hF_cover⟩ :=
    ↪ Metric.totallyBounded_iff.mp
95     hK.totallyBounded (ε / 2) (by linarith)
96   refine ⟨F, hF_fin, fun y hy => ?_⟩
97   obtain ⟨x, hx_ball, rfl⟩ := hy
98   -- Truncation of A(x) to first N coords is in Φ(cube)
99   set c : Fin N → ℝ := fun i => a (i : ℕ) * x (i : ℕ)
100  have hx_lt : ||x|| < 1 := by rwa [Metric.mem_ball,
    ↪ dist_zero_right] at hx_ball
101  have hc_cube : c ∈ cube := fun i _ => by
102    simp only [c, Set.mem_Icc]
103    have h1 : |x (i : ℕ)| ≤ 1 := by
104      have := lp.norm_apply_le_norm (show (2:ℝ≥0∞) ≠ 0 by
    ↪ norm_num) x (i : ℕ)
105      rw [Real.norm_eq_abs] at this; linarith
106  have h2 : |a (i : ℕ) * (x : ℕ → ℝ) (i : ℕ)| ≤ |a (i :
    ↪ ℕ)| := by
107    rw [abs_mul]; exact mul_le_of_le_one_right
    ↪ (abs_nonneg _) h1
108    exact ⟨by linarith [neg_abs_le (a (↑i) * (↑x : ℕ → ℝ)
    ↪ ↑i)], by linarith [le_abs_self (a (↑i) * (↑x : ℕ →
    ↪ ℝ) ↑i)]⟩
109  have hΦc_in : Φ c ∈ Φ '' cube := Set.mem_image_of_mem _
    ↪ hc_cube
110  -- ∃ f ∈ F with dist(Φ(c), f) < ε/2
111  obtain ⟨f, hf, hf_dist⟩ := Set.mem_iUnion₂.mp (hF_cover
    ↪ hΦc_in)
112  -- dist(A(x), f) ≤ dist(A(x), Φ(c)) + dist(Φ(c), f) <
    ↪ ε/2 + ε/2

```

```

113 refine Set.mem_iUnion₂.mpr (f, hf, ?_)
114 calc dist (A x) f ≤ dist (A x) (Φ c) + dist (Φ c) f :=
    ↪ dist_triangle
115   _ < ε / 2 + ε / 2 := add_lt_add ?_ (Metric.mem_ball.mp
    ↪ hf_dist)
116   _ = ε := by ring
117 -- ||A(x) - Φ(c)|| < ε/2: tail bound via inner product
118 -- (A x - Φ c)(k) = 0 for k < N, a(k)x(k) for k ≥ N
119 -- Each |a(k)x(k)|² < (ε/2)² |x(k)|², sum ≤ (ε/2)² ||x||² <
    ↪ (ε/2)²
120 set d := A x - Φ c
121 -- Pointwise: (d k)² ≤ (ε/2)² * (x k)² for all k
122 have hdk_le : ∀ k, ((d : ℕ → ℝ) k) ^ 2 ≤ (ε / 2) ^ 2 *
    ↪ ((x : ℕ → ℝ) k) ^ 2 := by
123   intro k
124   simp only [d, A, Φ, c, lp.coeFn_sub, Pi.sub_apply,
125     lp.coeFn_sum, Finset.sum_apply, lp.coeFn_single,
    ↪ Pi.single_apply]
126   by_cases hk : k < N
127   · rw [Finset.sum_eq_single (k, hk)
128     (fun i _ hi => by simp [show k ≠ (i : ℕ) from
    ↪ Fin.val_ne_of_ne (Ne.symm hi)])]
129     (by simp)]; simp; positivity
130   · push_neg at hk
131     rw [Finset.sum_eq_zero (fun i _ => by simp [show k ≠
    ↪ (i : ℕ) from by omega])]
132     simp only [sub_zero, mul_pow]
133     exact mul_le_mul_of_nonneg_right (by
134       have := hN k hk; rw [dist_zero_right,
    ↪ Real.norm_eq_abs] at this
135       have := sq_abs (a k); have := abs_nonneg (a k);
    ↪ nlinarith [mul_self_nonneg (|a k|)]
    ↪ (sq_nonneg _))
136 -- ||d||² ≤ (ε/2)² ||x||² < (ε/2)² (via inner product tsum)
137 have hd_sq : ||d|| ^ 2 = ∑' k, ((d : ℕ → ℝ) k) ^ 2 := by
138   rw [sq, ← real_inner_self_eq_norm_mul_norm,
    ↪ lp.inner_eq_tsum]
139   congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm,
    ↪ Real.norm_eq_abs]
140   change |(d : lp _ 2) k| * |(d : lp _ 2) k| = ((d : ℕ →
    ↪ ℝ) k) ^ 2; rw [← sq_abs]; ring
141 have hx_sq : ||x|| ^ 2 = ∑' k, ((x : ℕ → ℝ) k) ^ 2 := by
142   rw [sq, ← real_inner_self_eq_norm_mul_norm,
    ↪ lp.inner_eq_tsum]
143   congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm,
    ↪ Real.norm_eq_abs]
144   change |(x : lp _ 2) k| * |(x : lp _ 2) k| = ((x : ℕ →
    ↪ ℝ) k) ^ 2; rw [← sq_abs]; ring
145   rw [dist_eq_norm]
146 -- Summability for tsum_le_tsum
147 have hsx : Summable (fun k => ((x : ℕ → ℝ) k) ^ 2) :=

```

```

148      ((lp.mem p x).summable (by norm_num : 0 <
149      ↪ (2:ℝ≥0∞).toReal)).congr fun k => by
      simp [ENNReal.toReal_ofNat, Real.norm_eq_abs,
      ↪ sq_abs]
150    have hsd : Summable (fun k => ((d : ℕ → ℝ) k) ^ 2) :=
151      (hsx.mul_left ((ε/2)^2)).of_nonneg_of_le (fun _ =>
      ↪ sq_nonneg _) hdk_le
152    have hle := hsd.tsum_le_tsum hdk_le (hsx.mul_left _)
153    rw [tsum_mul_left, ← hx_sq] at hle
154    have h1 : ||d|| ^ 2 ≤ (ε / 2) ^ 2 * ||x|| ^ 2 := by linarith
      ↪ [hd_sq]
155    -- ||d||^2 ≤ (ε/2)^2 * ||x||^2 < (ε/2)^2 and ε/2 ≤ ||d|| → (ε/2)^2 ≤
      ↪ ||d||^2. Contradiction.
156    have h2 : (ε / 2) ^ 2 * ||x|| ^ 2 < (ε / 2) ^ 2 := by
157      apply mul_lt_of_lt_one_right (by positivity)
158      nlinarith [norm_nonneg x, mul_self_nonneg ||x||]
159      nlinarith [norm_nonneg d, sq_nonneg ||d||]
160    · exact Filter.mem_of_superset (Metric.ball_mem_nhds 0
      ↪ one_pos) fun x hx =>
      subset_closure (Set.mem_image_of_mem A hx)
161

```

5. Доказать, что оператор  $A : C[a, b] \rightarrow C[a, b]$ ,  $(Ax)(t) = x_0(t)x(t)$ , где  $0 \neq x_0 \in C[a, b]$ , не является компактным.

6. Доказать компактность следующих операторов:

- a)  $A : C[0, 1] \rightarrow C[0, 1]$ ,  $(Ax)(t) = \int_0^t x_0(t)x(s) ds$ , где  $x_0(t) \in C[a, b]$ ;
- b)  $A : C[a, b] \rightarrow C[a, b]$ ,  $(Ax)(t) = \int_a^b K(s, t)x(s) ds$ , где  $K(s, t)$  — непрерывная на  $[a, b] \times [a, b]$  функция.

7. Доказать компактность следующих операторов:

- a)  $A : L^2(0, 1) \rightarrow L^2(0, 1)$ ,  $(Ax)(t) = \int_0^t x(s) ds$ ;
- b)  $A : L^2(0, 1) \rightarrow L^2(0, 1)$ ,  $(Ax)(t) = x_0(t) \cdot \int_0^t x(s) ds$ , где  $x_0(t) \in C[0, 1]$ ;
- c)  $A : L^2(a, b) \rightarrow L^2(a, b)$ ,  $(Ax)(t) = \int_a^b K(s, t)x(s) ds$ , где  $K(s, t)$  — непрерывная на  $[a, b] \times [a, b]$  функция;
- d)  $A : L^2(a, b) \rightarrow L^2(a, b)$ ,  $(Ax)(t) = \int_a^b K(s, t)x(s) ds$ , где  $K(s, t) \in L^2([a, b] \times [a, b])$ .

8. Показать, что оператор  $A : l^2 \rightarrow \mathbb{R}^1$ ,  $Ax = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k}$  является компактным.

9. Показать, что образ компактного оператора является сепарабельным.

#### Формализация на Lean 4

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1  /-
2  Section 21, Task 9.
3  Show that the image of a compact operator is separable.
4
5  Proof:  $\text{range } A = \bigcup n, A \text{ '' } \text{closedBall } 0 \ n$ . Each  $A \text{ '' } \text{closedBall } 0 \ n$ 
6   $\hookrightarrow \text{closedBall } 0 \ n$ 
7  has compact closure (since  $A$  is compact), hence is
8   $\hookrightarrow$  separable.
9  Countable union of separable sets is separable.
10 -/
11 import Mathlib.Topology.MetricSpace.Pseudo.Basic
12 import Mathlib.Analysis.Normed.Operator.Compact
13 import Mathlib.Tactic
14
15 variable {X Y : Type*} [NormedAddCommGroup X] [NormedSpace ℝ
16    $\hookrightarrow$  X]
17   [NormedAddCommGroup Y] [NormedSpace ℝ Y]
18
19 /- The range of a compact operator is separable. -/
20 theorem compact_operator_range_separable
21   (A : X  $\rightarrow$  L[ℝ] Y) (hA : IsCompactOperator (A : X  $\rightarrow$  Y)) :
22   TopologicalSpace.IsSeparable (Set.range A) := by
23   --  $\text{range } A = \bigcup n : \mathbb{N}, A \text{ '' } \text{closedBall } 0 \ n$ 
24   have hrange : Set.range A =  $\bigcup n : \mathbb{N}, A \text{ '' } \text{Metric.closedBall}$ 
25      $\hookrightarrow 0 \ (n : \mathbb{R})$  := by
26     ext y; simp only [Set.mem_range, Set.mem_iUnion,
27        $\hookrightarrow$  Set.mem_image]
28     constructor
29     · rintro ⟨x, rfl⟩
30       exact ⟨[||x||]_, x, Metric.mem_closedBall.mpr (by simp
31          $\hookrightarrow$  [dist_zero_right]; exact Nat.le_ceil _), rfl⟩
32     · rintro ⟨n, x, _, rfl⟩; exact ⟨x, rfl⟩
33   rw [hrange]
34   rw [TopologicalSpace.isSeparable_iUnion]
35   intro n
36   --  $A \text{ '' } \text{closedBall } 0 \ n$  is bounded, its closure is compact
37   have hbdd : Bornology.IsBounded (Metric.closedBall (0 : X)
38      $\hookrightarrow$  n) :=
39     Metric.isBounded_closedBall
40   exact ((hA.isCompact_closure_image_of_bounded
41      $\hookrightarrow$  hbdd).isSeparable).mono
42     subset_closure

```

10. Являются ли компактными следующие операторы в пространствах  $C[0, 1]$  и  $L^2(0, 1)$ ?

- a)  $(Ax)(t) = \int_0^1 x(s) ds;$
- b)  $(Ax)(t) = \int_0^1 x(s)(t-s)^{-2/5} ds;$
- c)  $(Ax)(t) = \int_0^1 (ts + t^2 s^2)x(s) ds.$

11. Доказать, что оператор дифференцирования  $\frac{d}{dt}$  является компактным из  $C^{(2)}[0, 1]$  в  $C[0, 1]$ .
12. При каких  $\alpha, \beta, \gamma > 0$  является компактным в пространстве  $C[0, 1]$  оператор  $(Ax)(t) = \int_0^1 t^\alpha s^\beta x(s^\gamma) ds?$
13. При каких  $\alpha, \beta, \gamma$  является компактным в пространстве  $L^2(0, 1)$  оператор  $(Ax)(t) = \int_0^1 t^\alpha s^\beta x(s^\gamma) ds?$